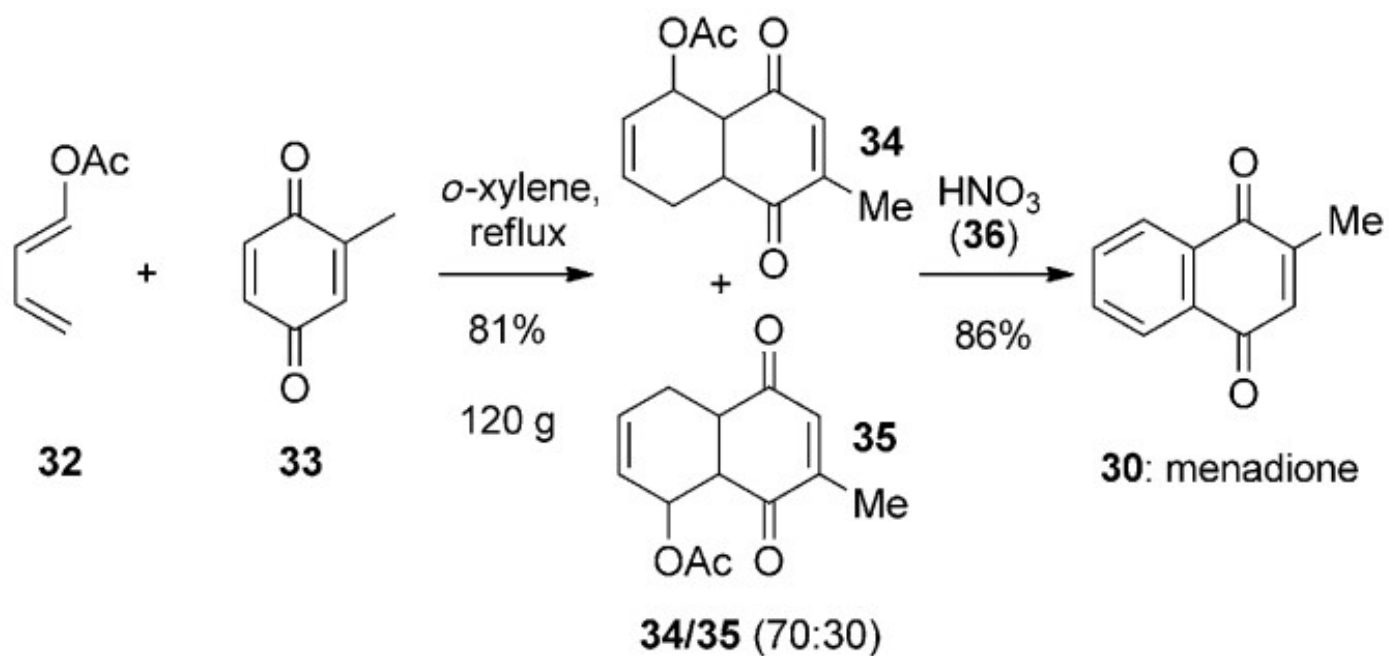
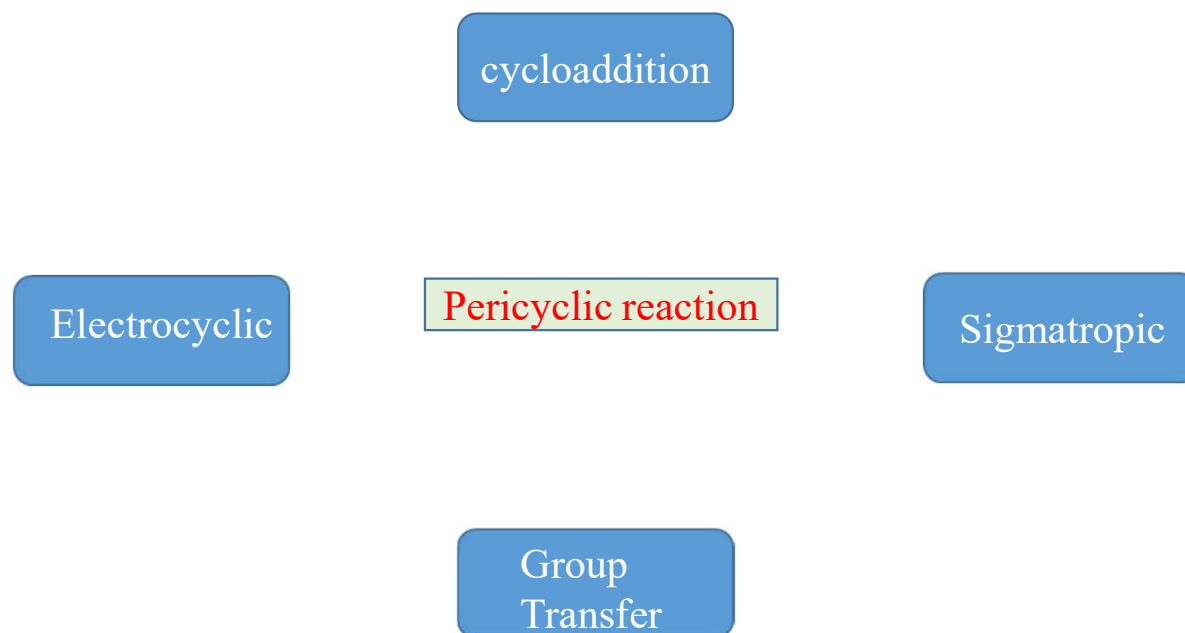




Preparation of provitamin K₃

PERICYCLIC REACTION

- Pericyclic reaction involves a cyclic redistribution of bonding electron through a concerted process.
- Concerted process means bond breaking and making occurs in a single step.

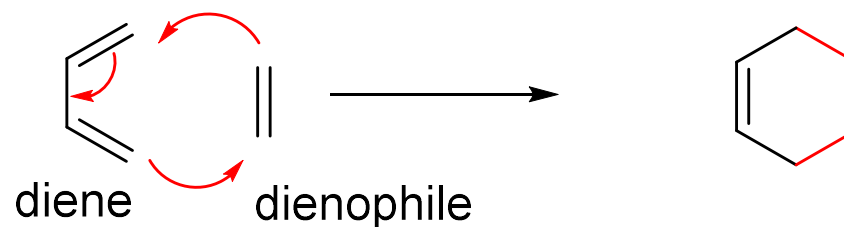


1. Cycloaddition Reaction

- Formation of a new ring.
- More than one bond form simultaneously.
- No intermediate formation.
- Three important classification of cycloaddition reaction
 - I. [4+2]-cycloaddition/Diels-Alder reaction
 - II. [1,3]-Dipolar cycloaddition
 - III. [2+2] cycloaddition

Diels-Alder reaction

- Reaction between a conjugated diene and dienophile.
- Two π bonds disappear and two new σ bond formed.
- This is a [4+2] cycloaddition.
- Diels Alder reaction takes place under thermal condition.

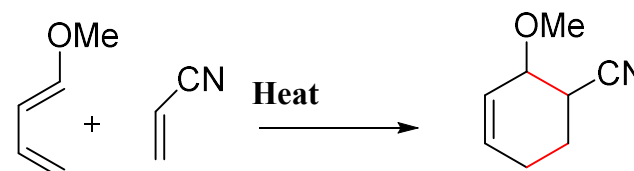


Classification of Diels-Alder reaction:

➤ Normal [4+2]

Diene is electron-rich

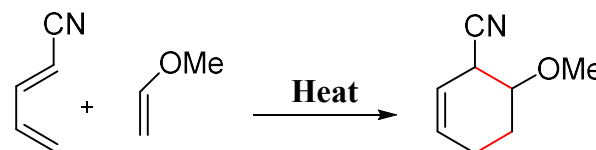
Dienophile is electron-poor



➤ Inverse electron-demand [4+2]

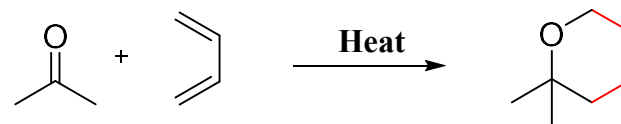
Diene is electron-poor

Dienophile is electron-rich



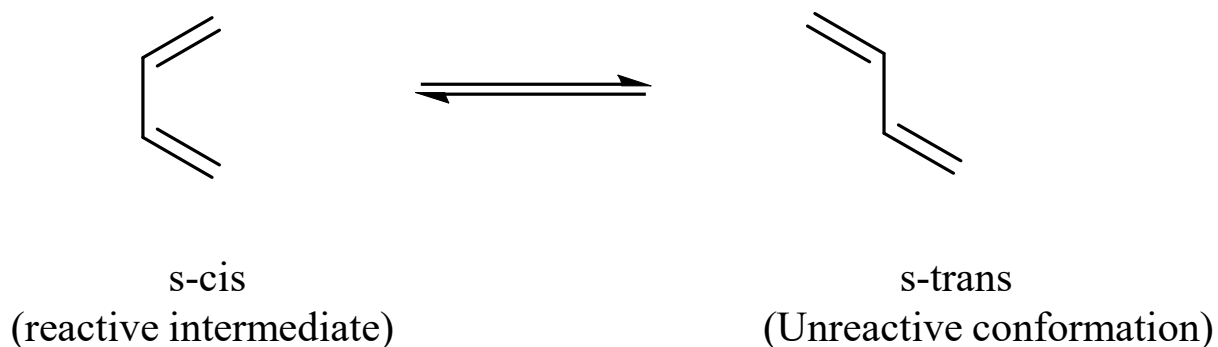
➤ Hetero[4+2]

Hetero atom can be a part of diene or Dienophile or both



Requirements of Diene:

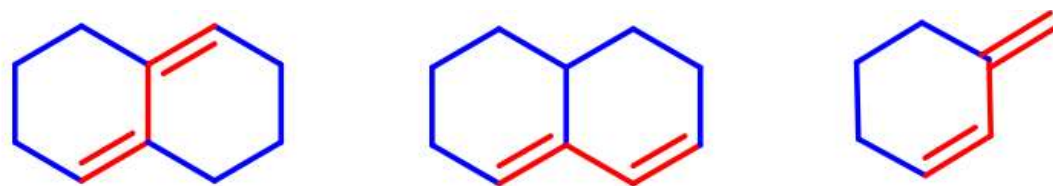
- Diene can be open chain or cyclic.
- Diene should be electron rich and reactivity can be enhanced by electron donating group.
- Acyclic Diene must be s-cis conformation to be reactive.



- Cyclic diene which adopt s-cis conformation to be very reactive.

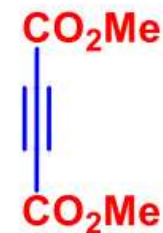
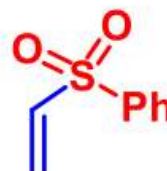
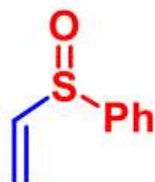
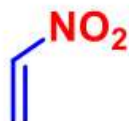
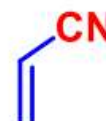
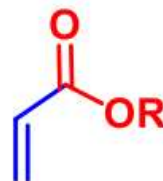
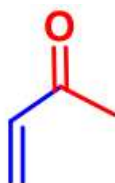
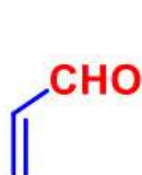


- Cyclic diene which are s-trans conformation are unreactive in Diels-Alder reaction.



Requirements of Dienophile:

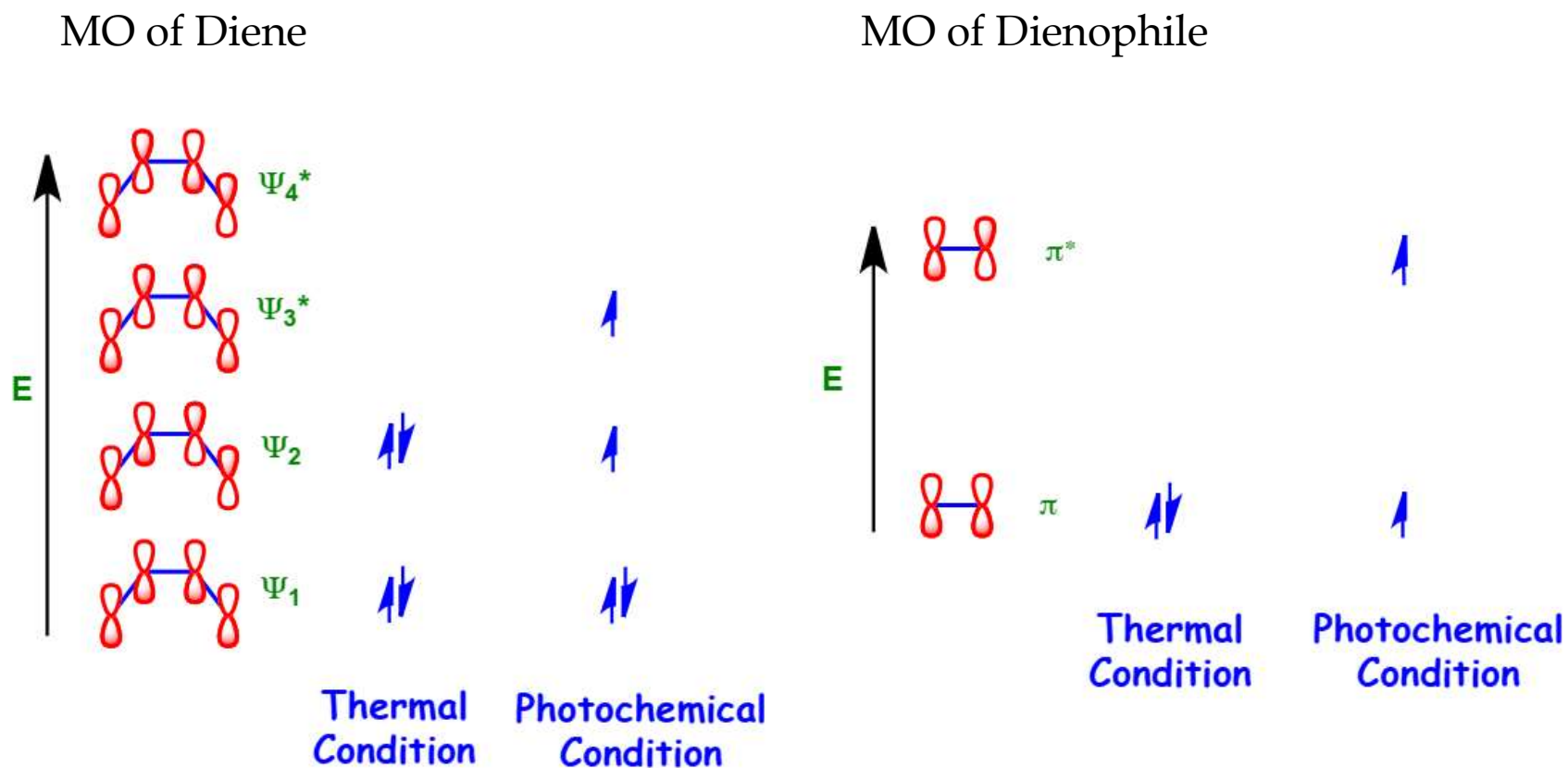
- Dienophile can be open or cyclic.
- Dienophile should be electron deficient and reactivity enhanced by electron withdrawing substituents



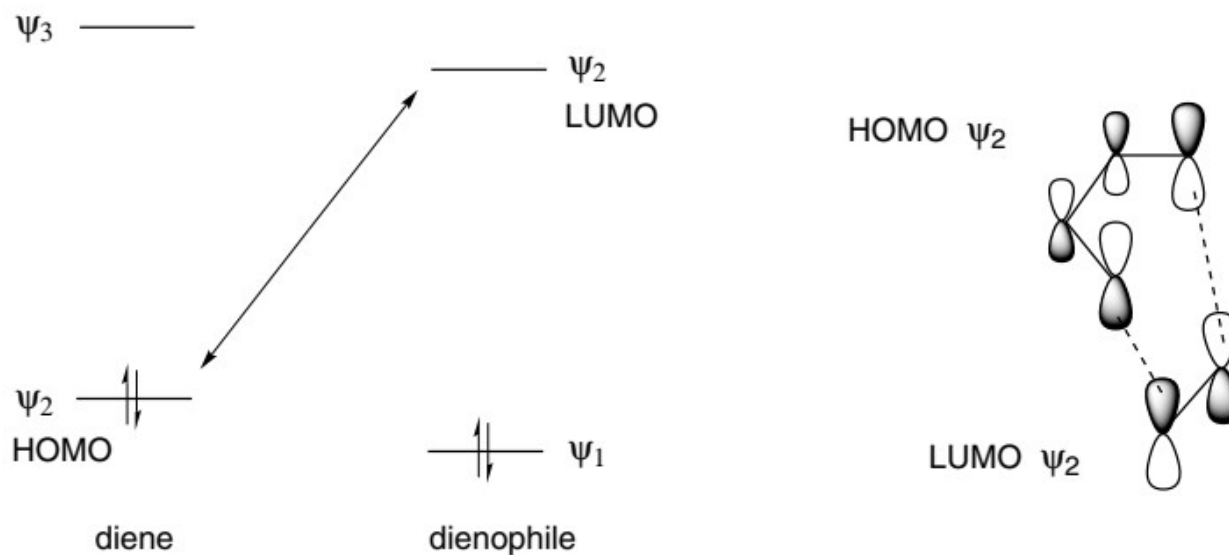
Frontier Molecular Orbital (FMO) Approach

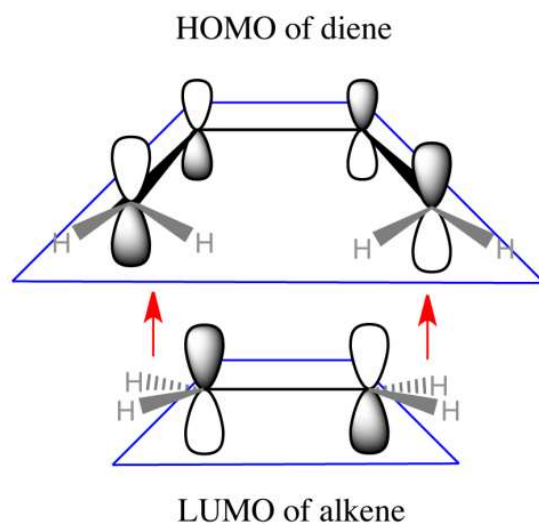
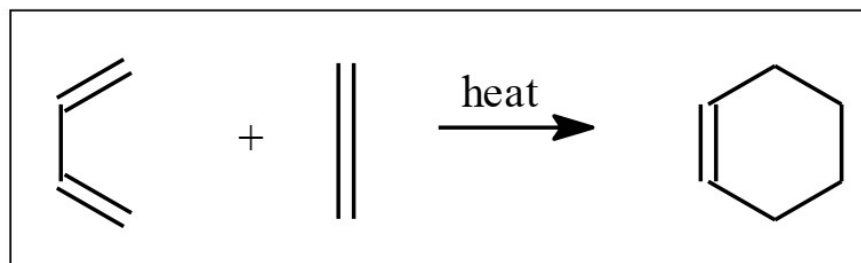
- Highest occupied Molecular Orbital (HOMO) & Lowest Unoccupied Orbital (LUMO) are known as Frontier Molecular Orbitals.
- Occupied orbitals repel each other.
- HOMO and LUMO interact with each other causing attraction.
- FMO simplifies the reactivity by interaction between HOMO of one & LUMO of other.

Frontier Molecular Orbital (FMO) approach:

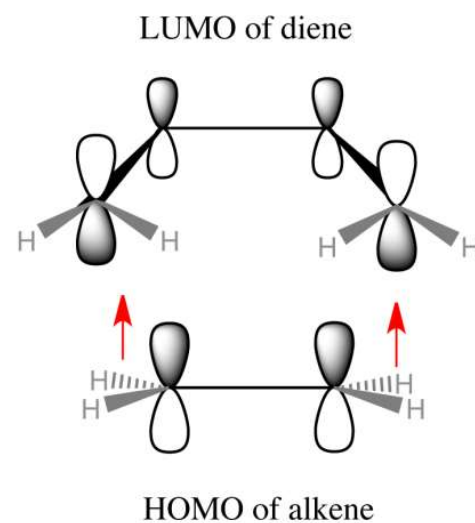


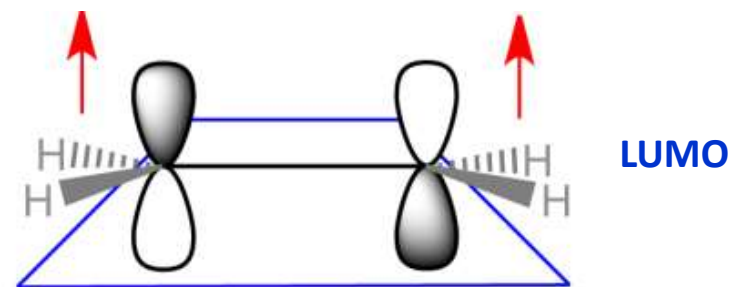
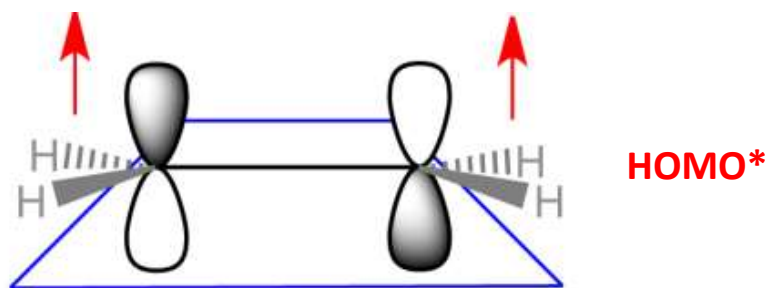
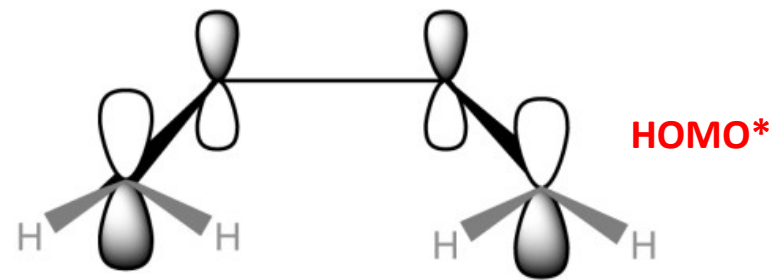
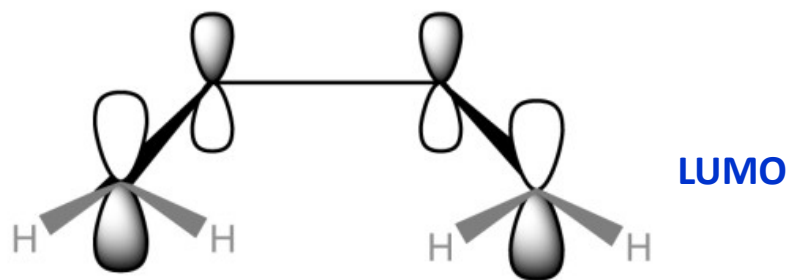
- A normal Diels Alder reaction involves an electron diene and electron deficient dienophile.
- The smaller the energy gap between these frontier orbitals, the better they interact and more readily the reaction occurs.





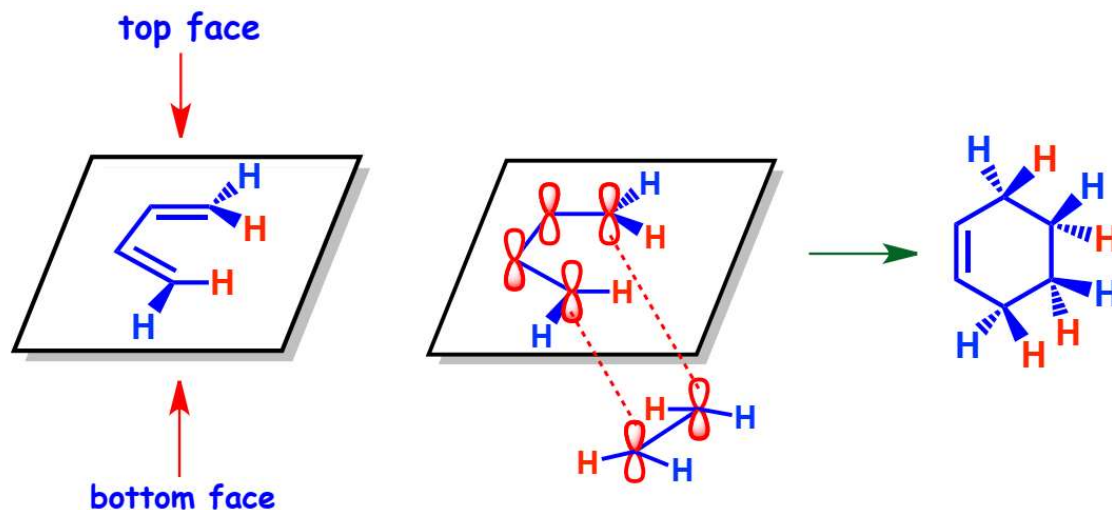
moving up from
the bottom





Stereochemistry of Diels – Alder Reaction

- Diels-Alder reaction is stereospecific.
- Stereochemistry of both diene and dienophile retained in the product.
- Reaction controls the relative stereochemistry at four contiguous centres.
- Diels-Alder reaction occurs due to the overlapping of p-orbitals of diene and dienophile lying perpendicular to the plane of carbon atoms.
- Dienophile can approach either of the faces and lead to racemic mixtures.



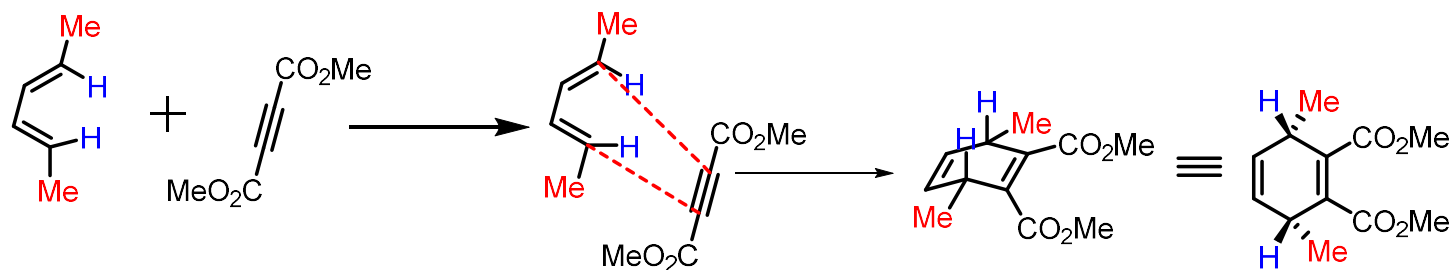
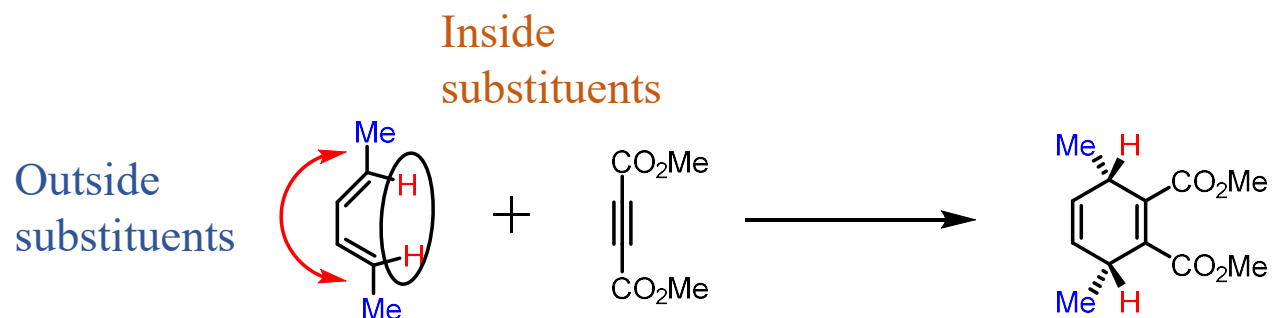
Stereochemistry of Diene

Rule of Thumb-

➤ Given the bottom face approach of dienophile:

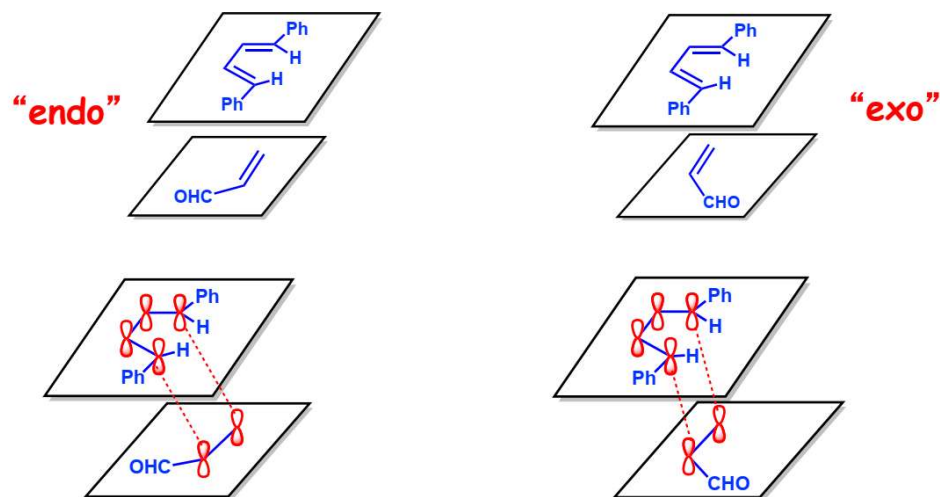
-Inside substituents of diene will become β -configuration in the product.

-Outside substituents of diene will become α -configuration in the product.

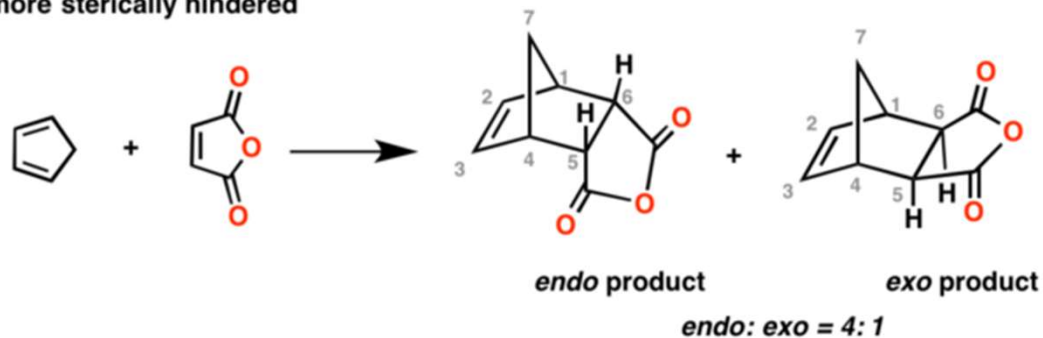


Stereochemistry of Dienophile

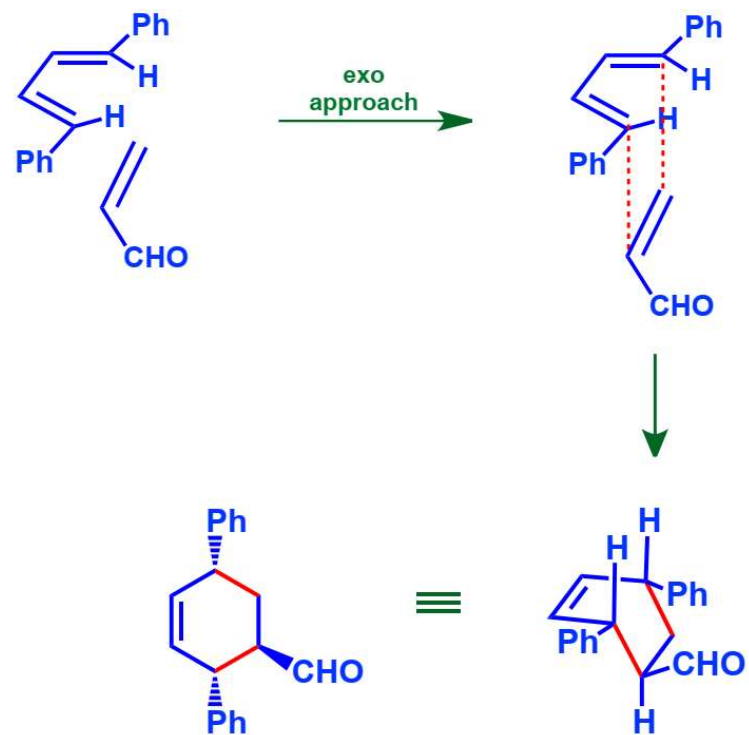
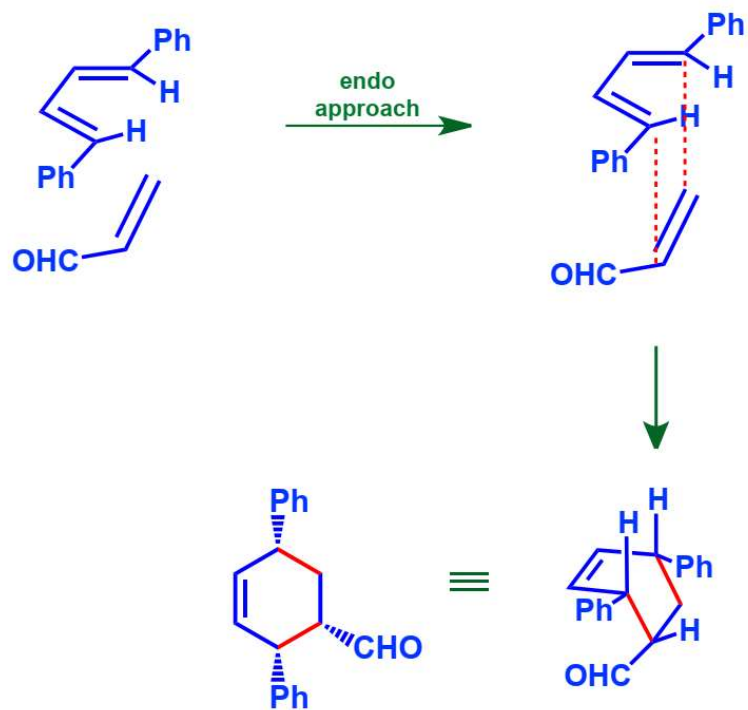
- Given the bottom face approach of dienophile, substituents on dienophile can take different orientations with respect to the plane of the diene called **endo** or **exo**.



endo products tend to be favored in the Diels-Alder, even though they are more sterically hindered

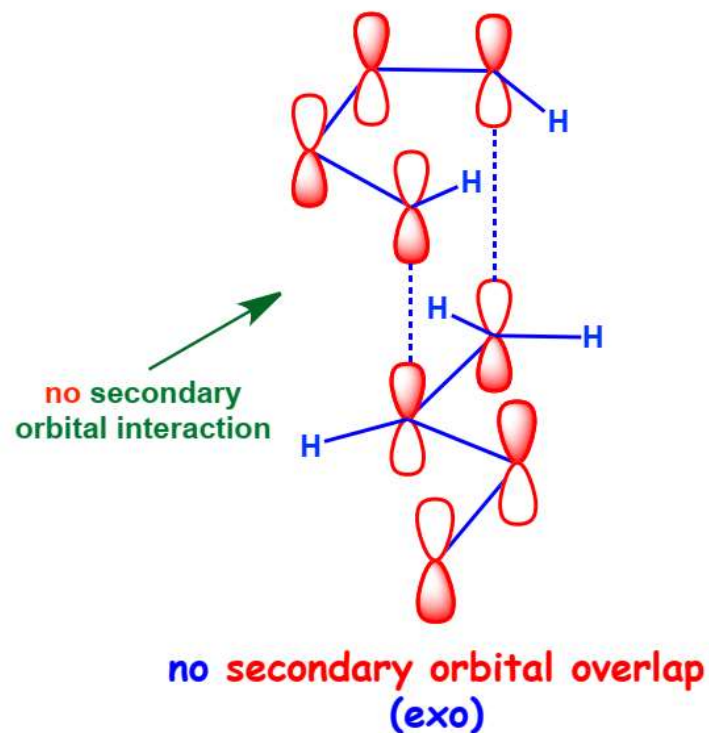
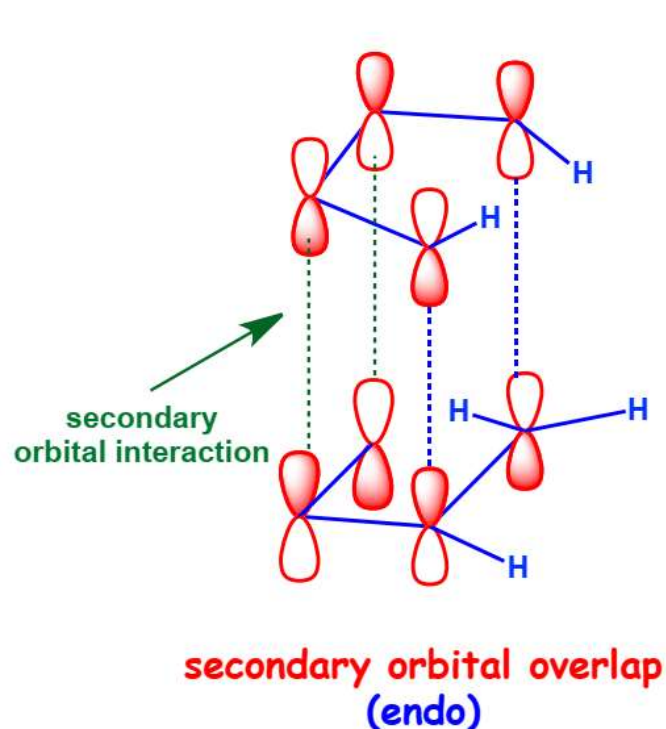


Example-

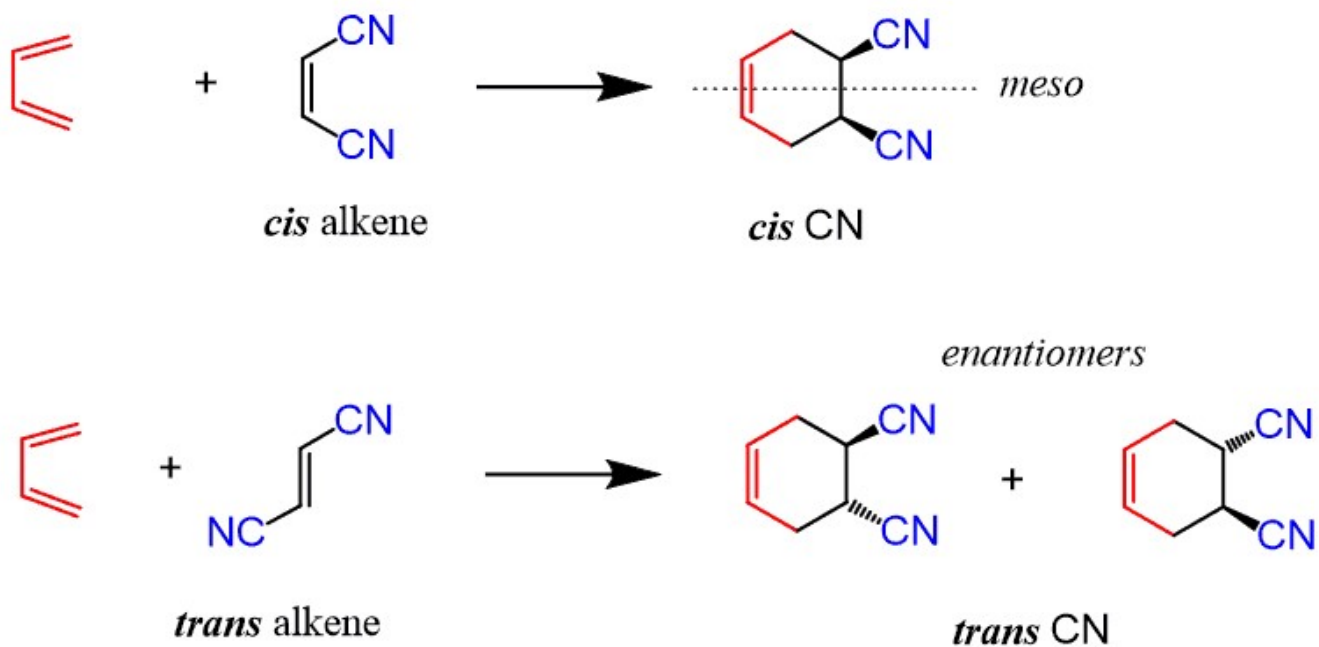


Alder's endo Rule

- **Endo** mode of activation is usually preferred to **exo** addition due to secondary orbital overlap between the dienophile's activating substituent and the diene.



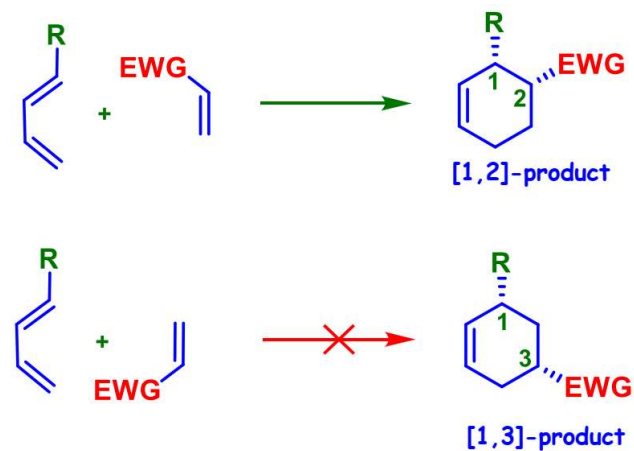
Stereospecific reaction - *cis* produces *cis*, *trans* produces *trans*





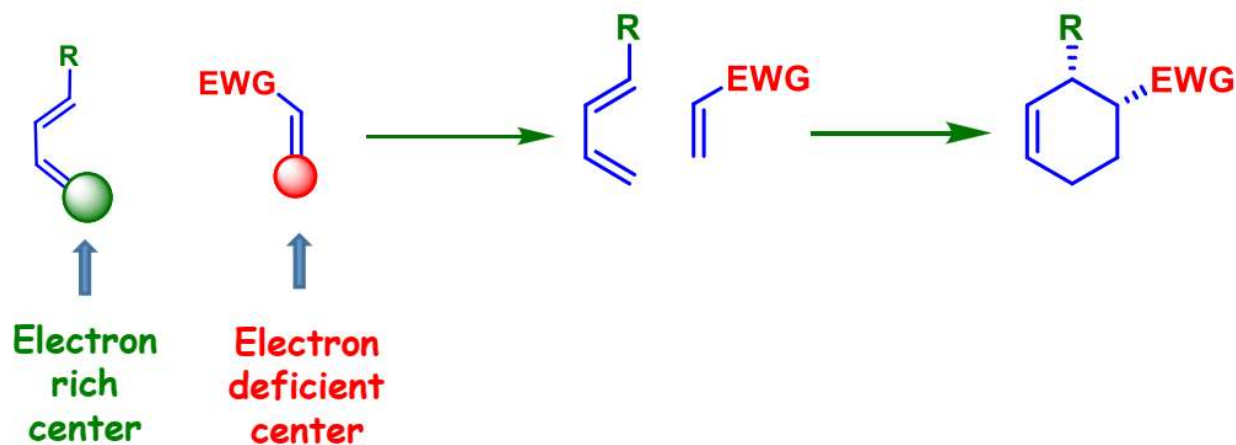
Regiochemistry of Diels – Alder Reaction

- For a mono- substituted diene, outcome of the product depends on carbon atom where it is substituted.
- Consider two classical cases, where the electron donating substituent could be present at carbon **1** and **2**.

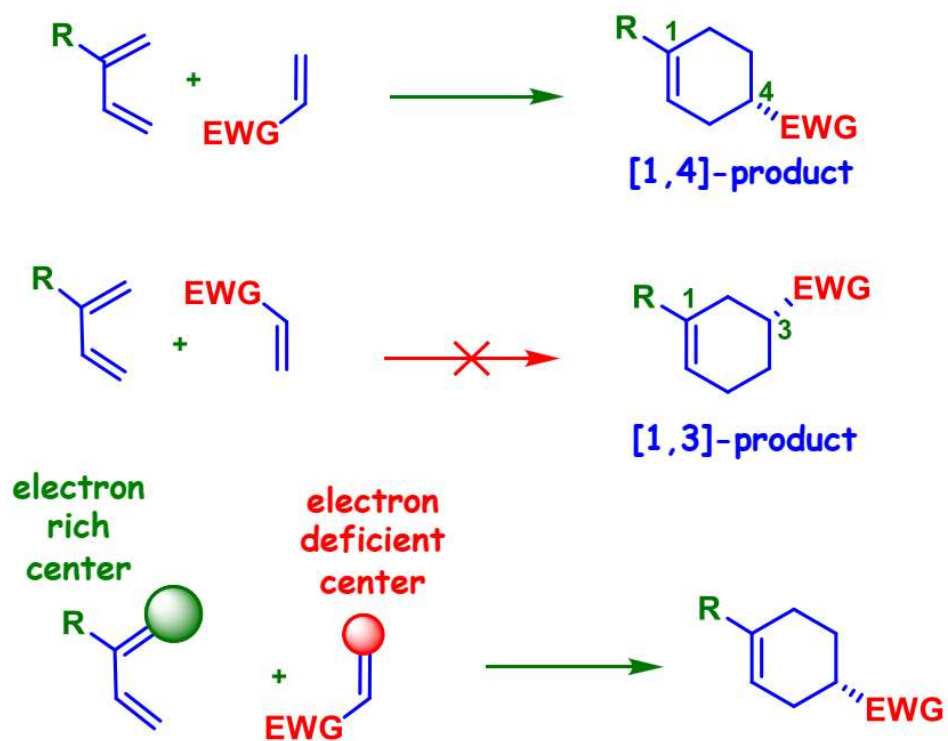


- 1- substituted diene react to give mainly [1,2]-product.

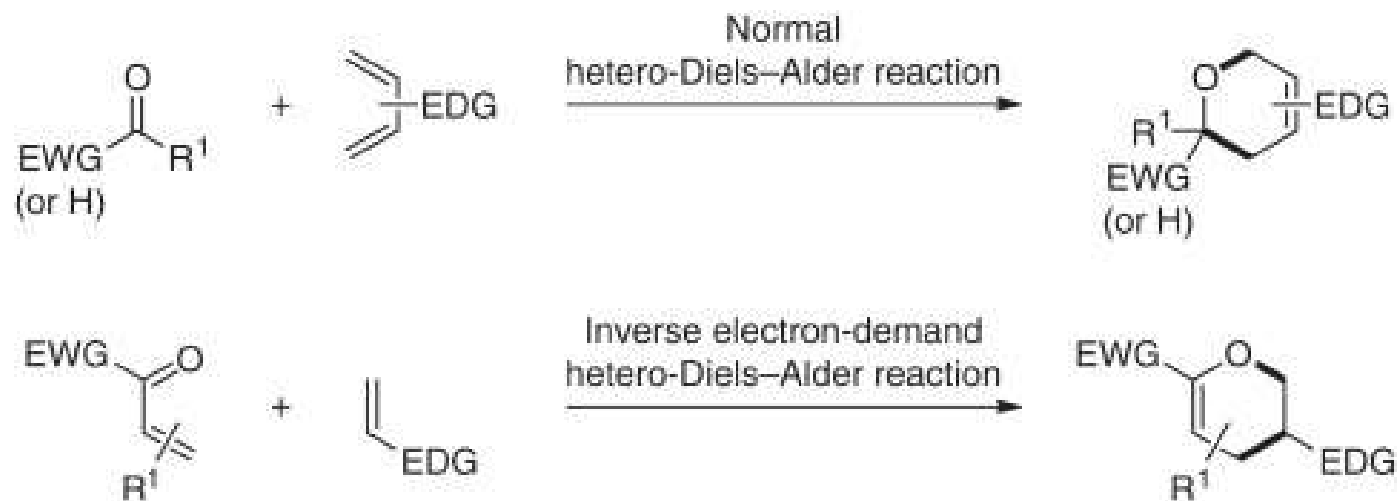
- Regioselectivity for the preference of [1,2]- product can be explained by considering the polarization of diene and dienophile.
- Connect the **electron rich** centre of diene to **electron deficient** centre of dienophile.



➤ 2- substituted diene react to give mainly [1,4]- substituted product.

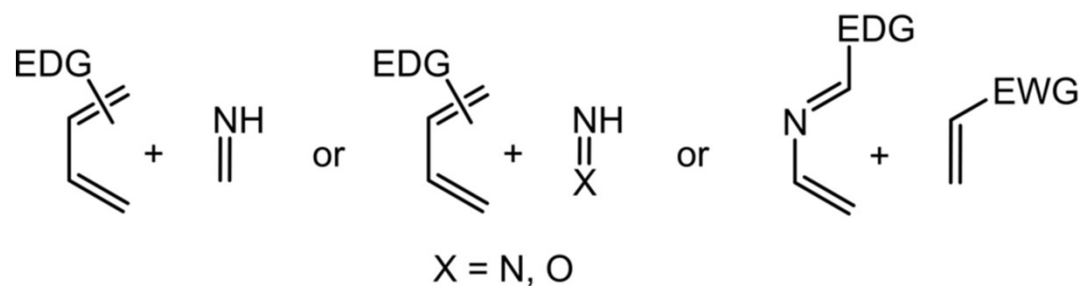


hetero-Diels-Alder reaction

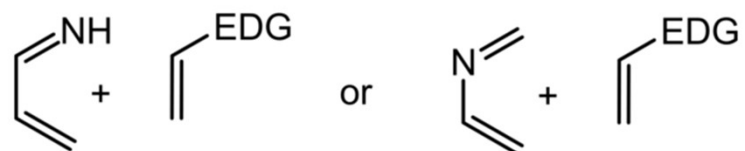


EDG, electron-donating group; EWG, electron-withdrawing group

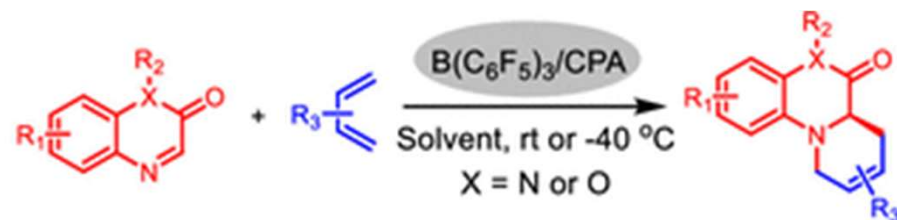
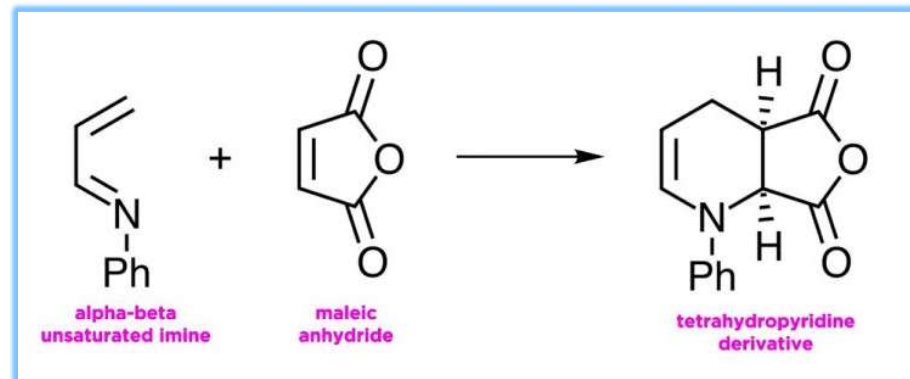
Normal electron demand ($\text{HOMO}_{\text{diene}}$ -controlled) aza-DAR



Inverse electron demand ($\text{LUMO}_{\text{diene}}$ -controlled) aza-DAR



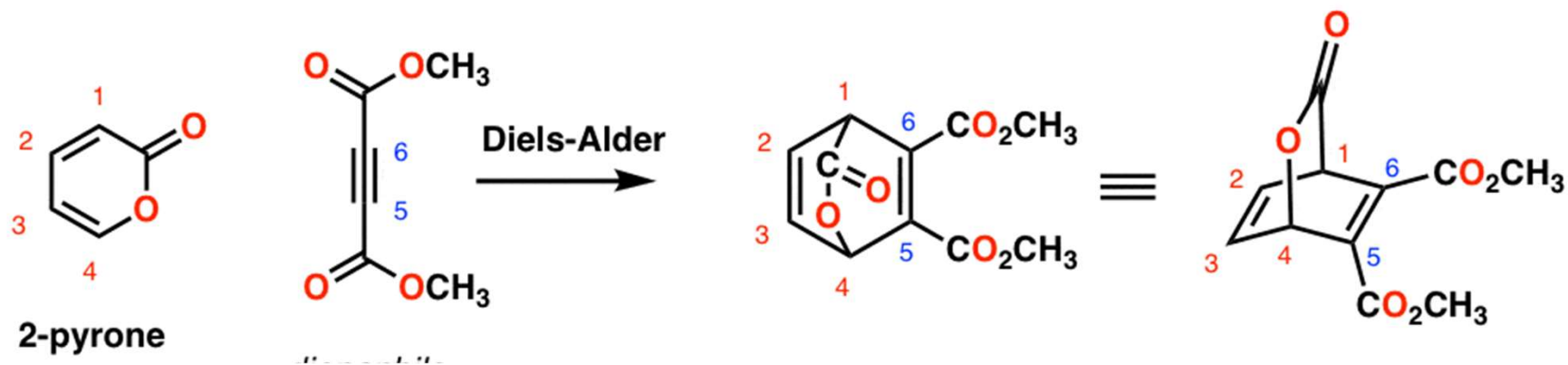
EDG = electron donating group
EWG = electron withdrawing group



Retro Diels-Alder Reaction

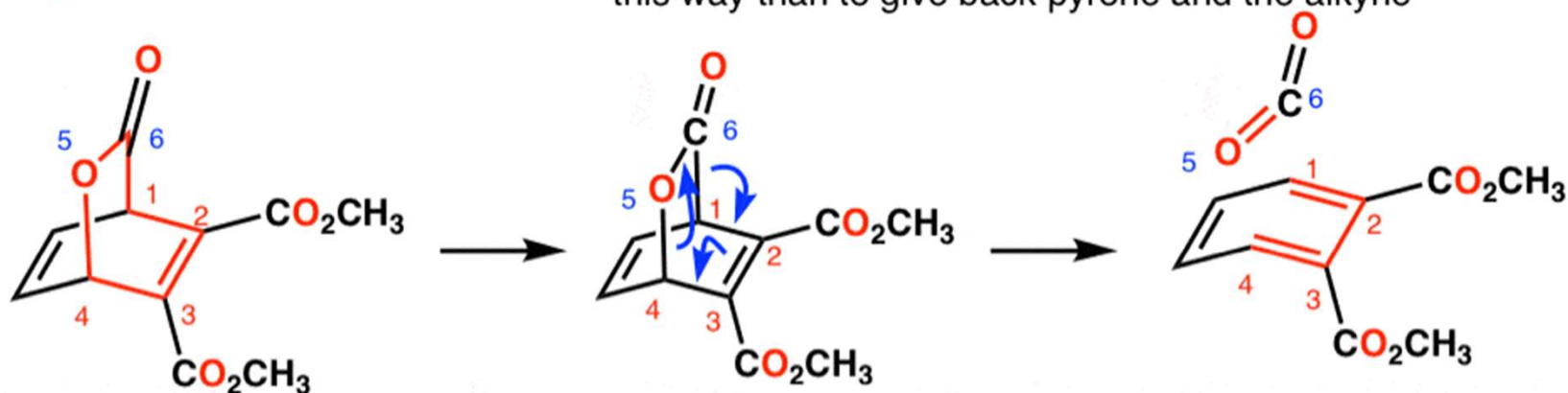
Diels-Alder product can undergo **more than one** possible *retro*-Diels Alder reaction

2-Pyrone as a diene in the Diels-Alder with an electron-poor alkyne dienophile:



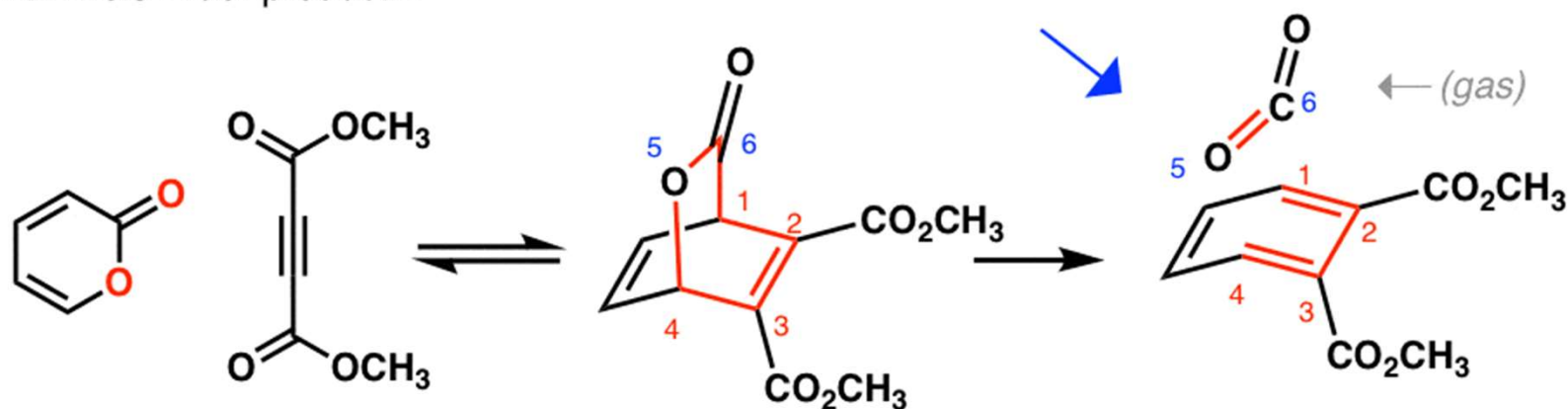
The 6 membered ring highlighted in red undergoes retro Diels-Alder fragmentation

- The products of this retro-Diels alder are CO_2 gas and a (very stable) aromatic ring
- More favorable (and irreversible) to fragment this way than to give back pyrone and the alkyne



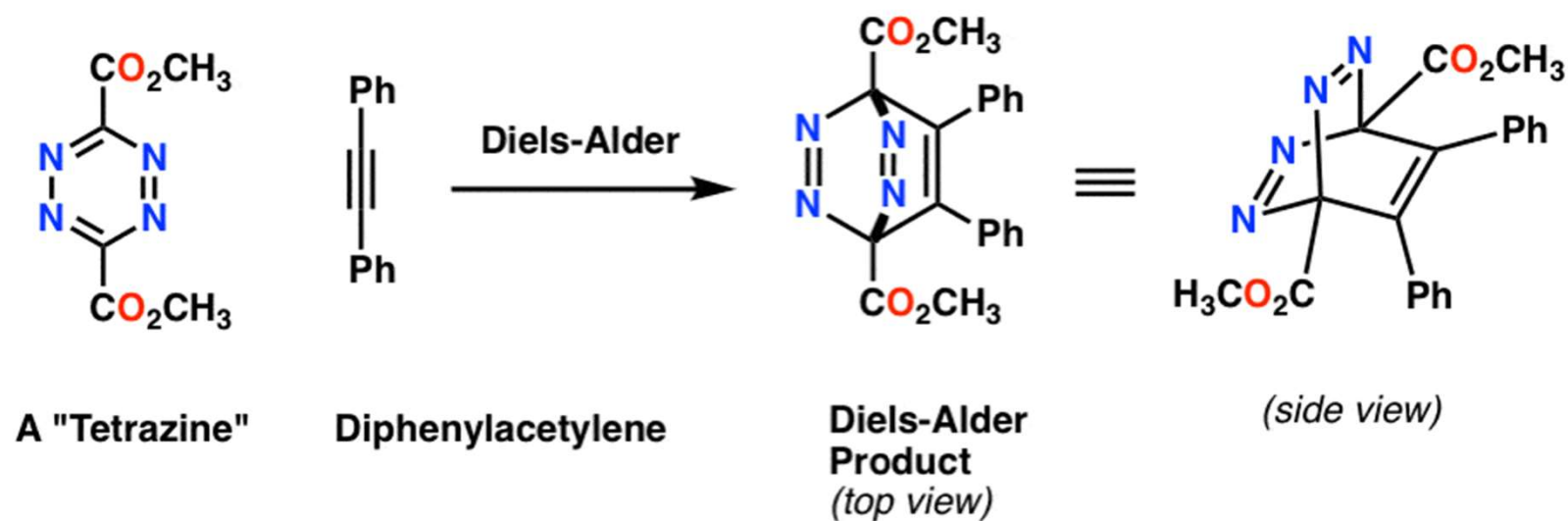
Pyrone and alkyne: in equilibrium with Diels-Alder product...

...but this *retro*-Diels Alder is **irreversible**

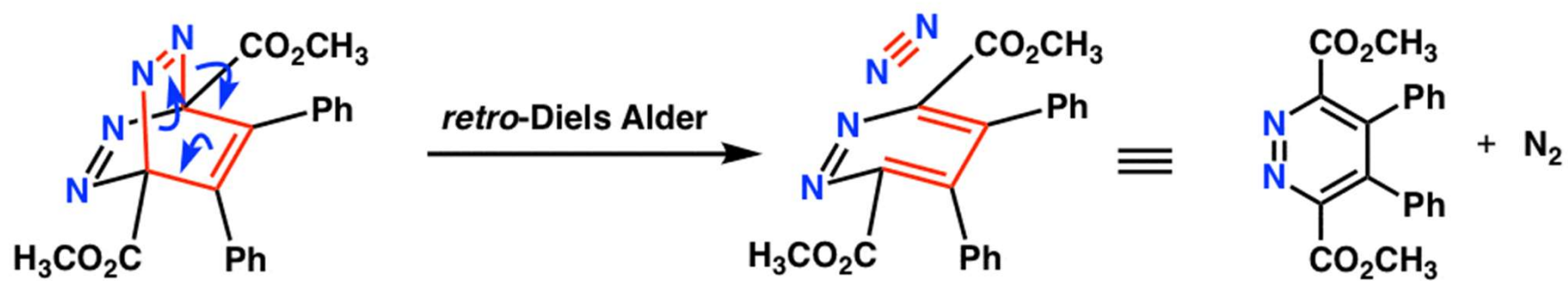


Even if the *retro*-Diels Alder to regenerate the pyrone and alkyne does occur, it just sets up an equilibrium.

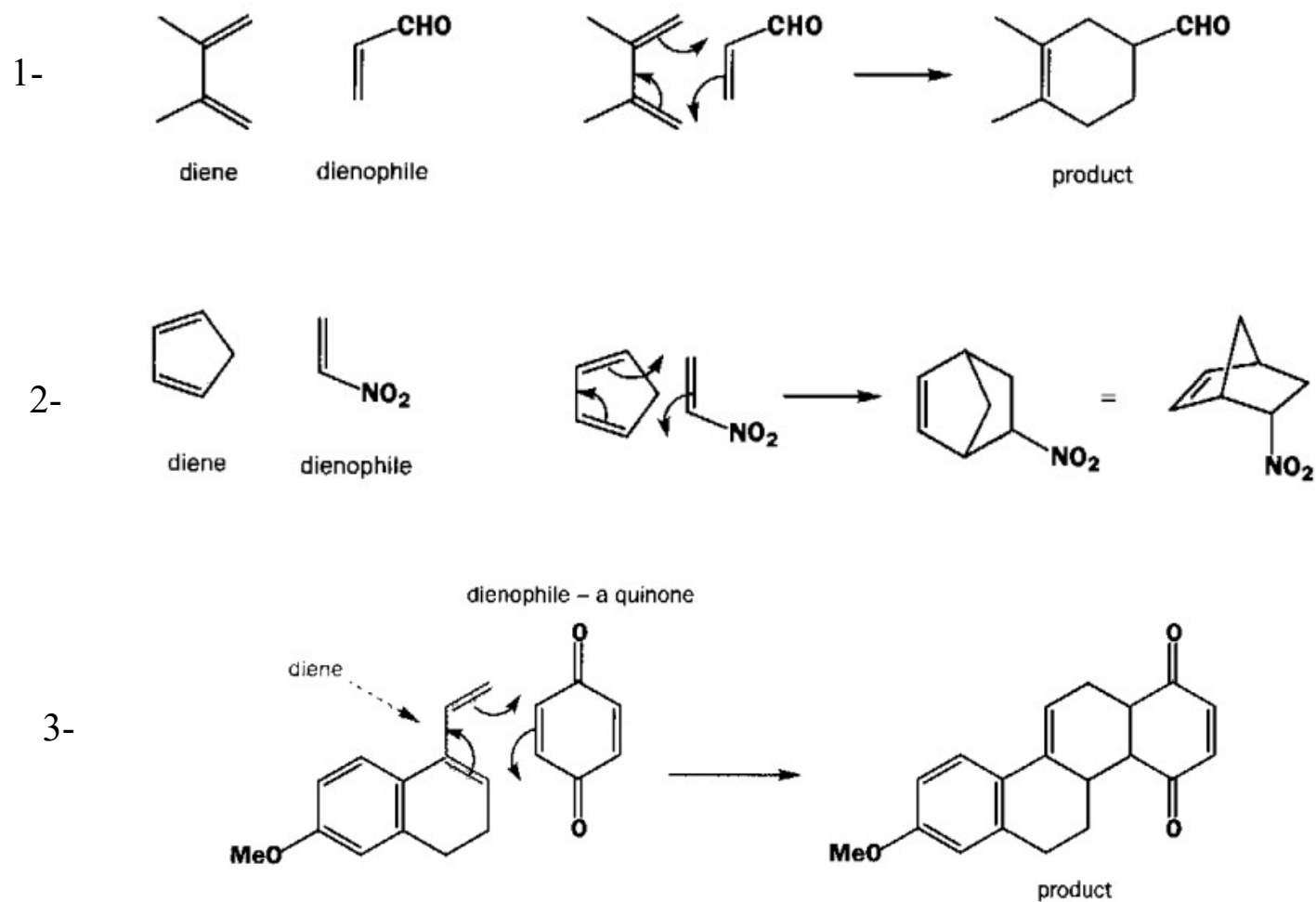
Another Diels-Alder / Retro-Diels Alder Sequence



Heating Leads to A Retro-Diels Alder With Irreversible Loss Of N₂

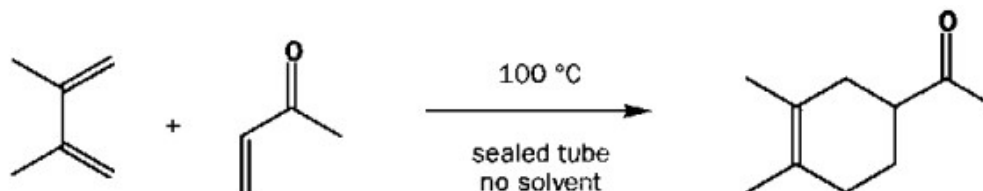


Example:

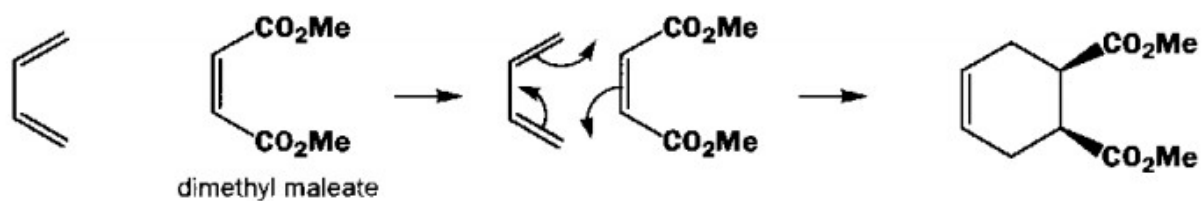


Example:

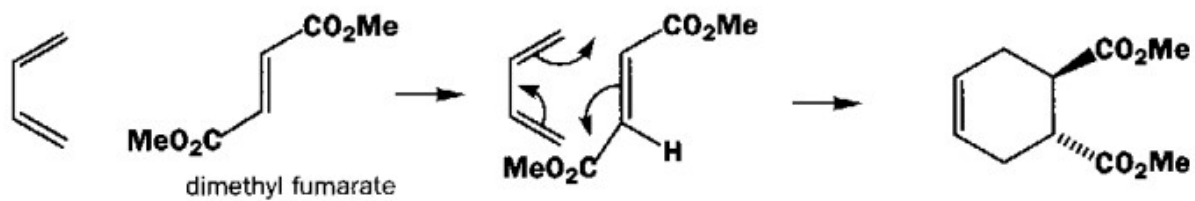
4-



5-

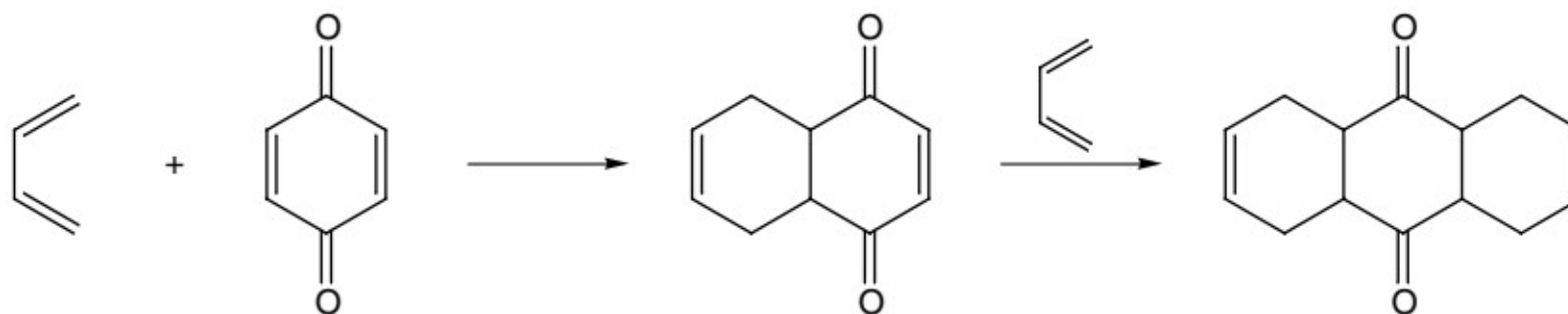


6-

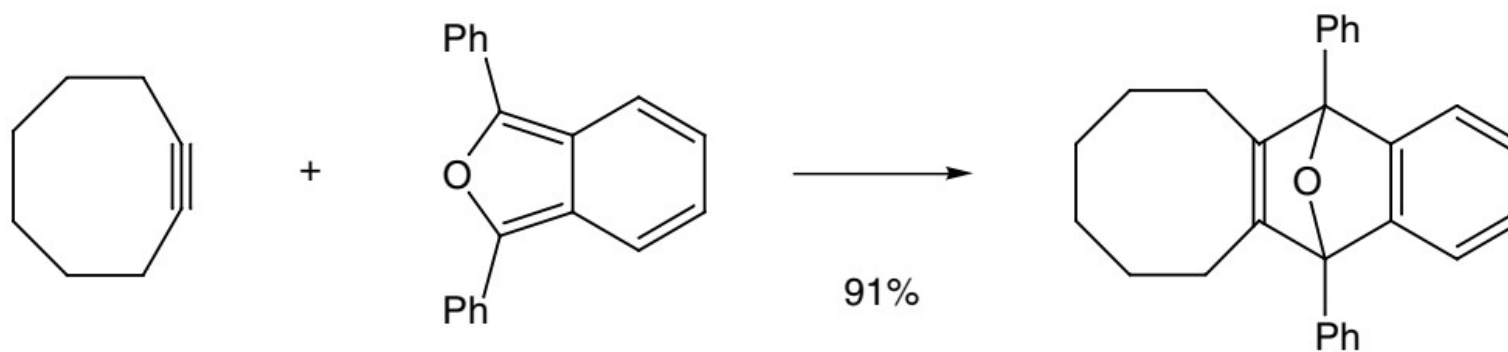


Example:

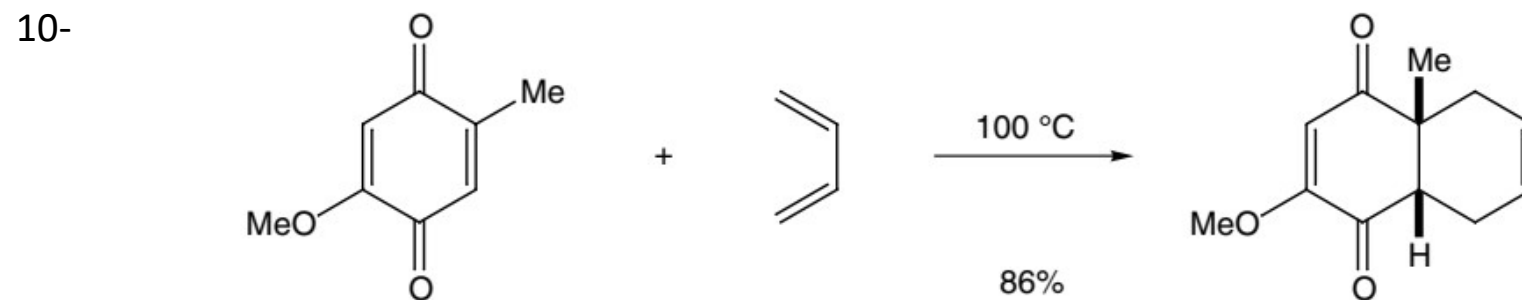
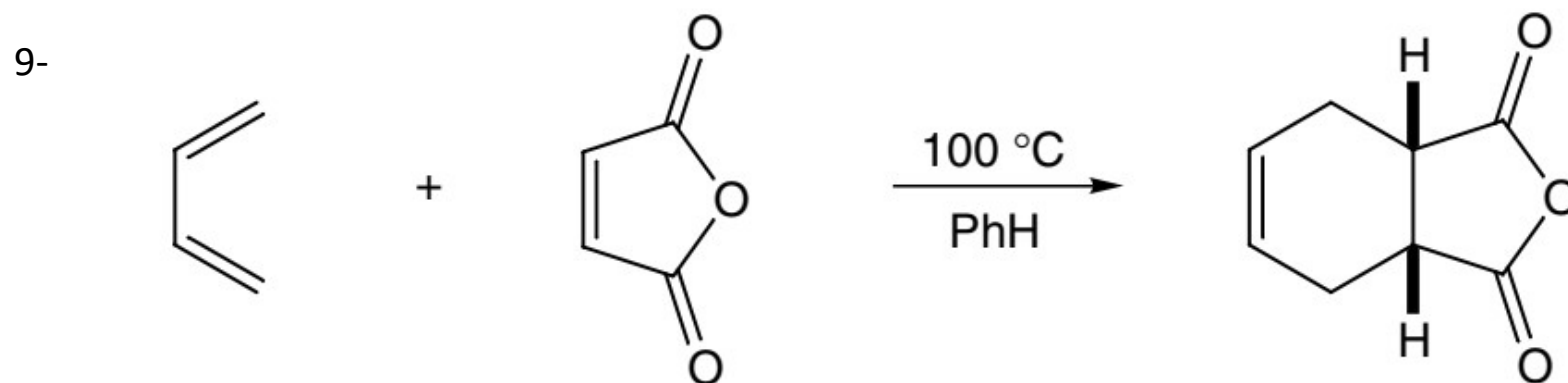
7-



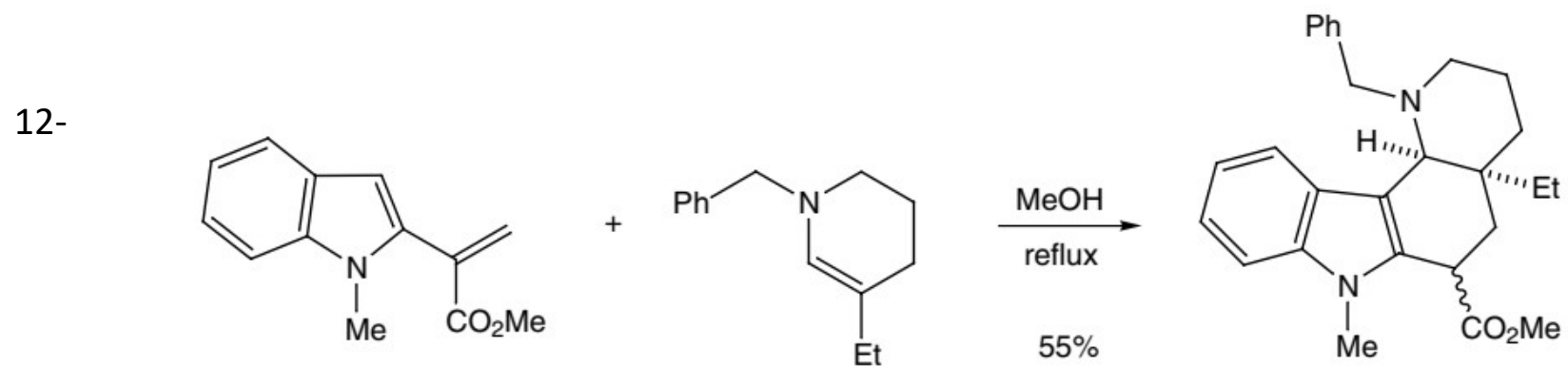
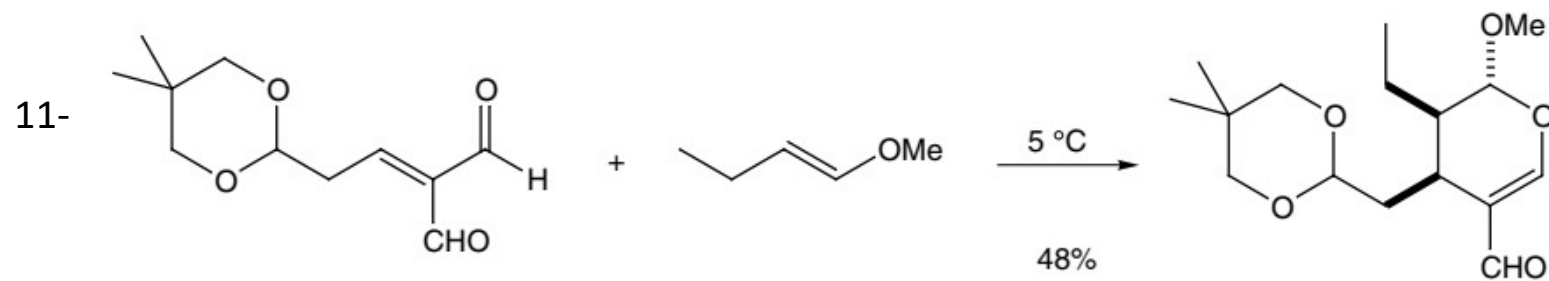
8-



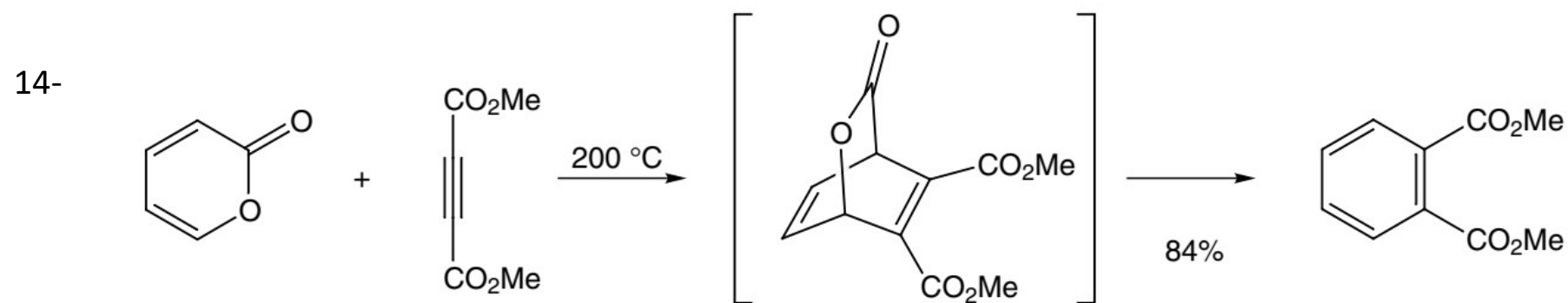
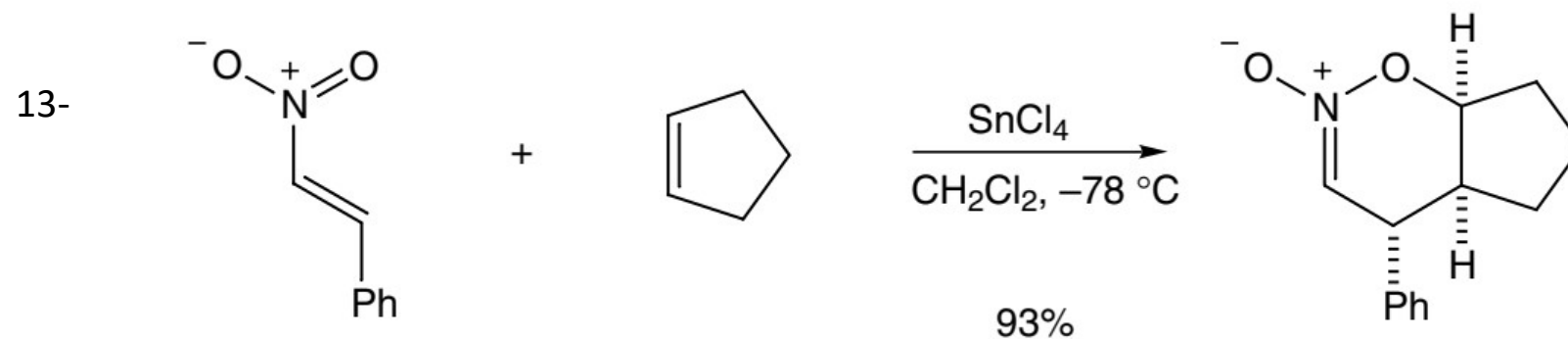
Example:



Example:

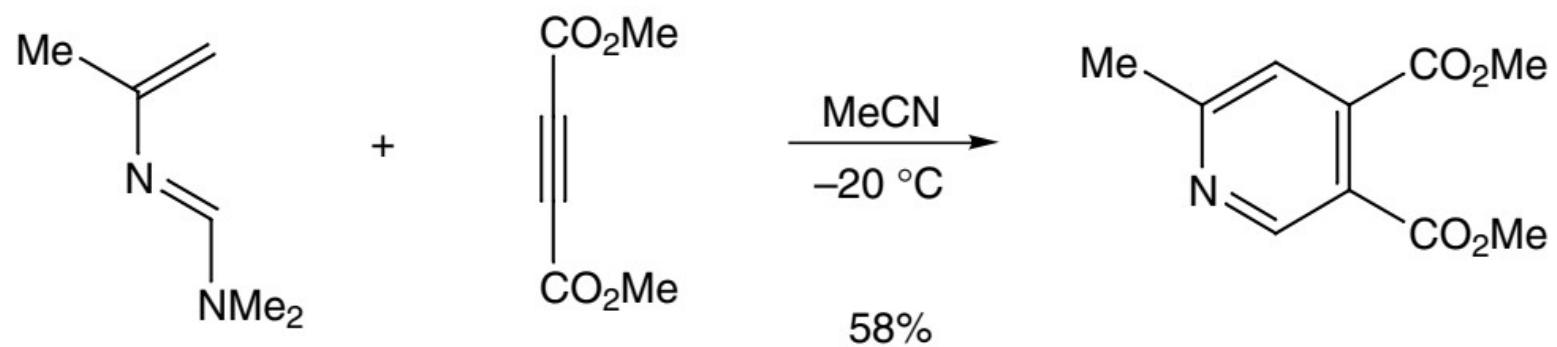


Example:

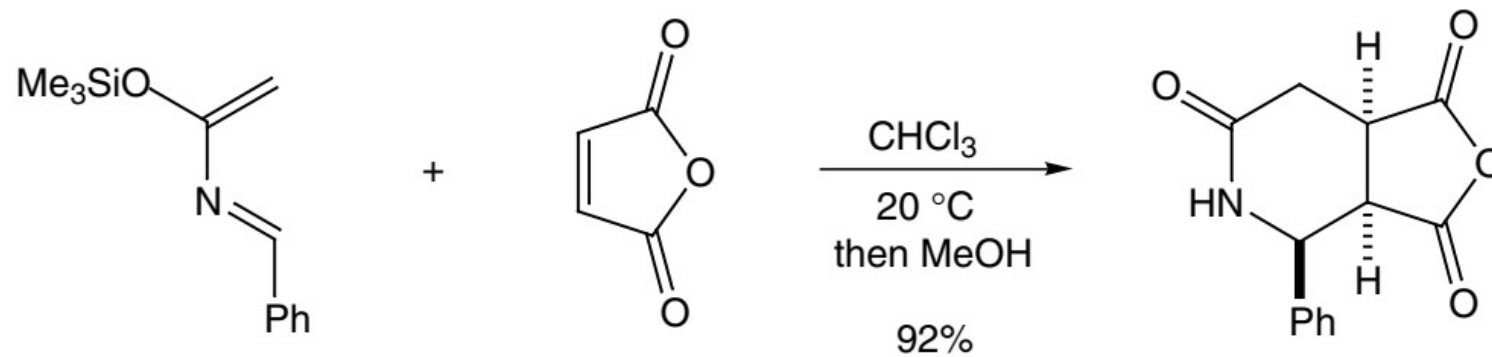


Example:

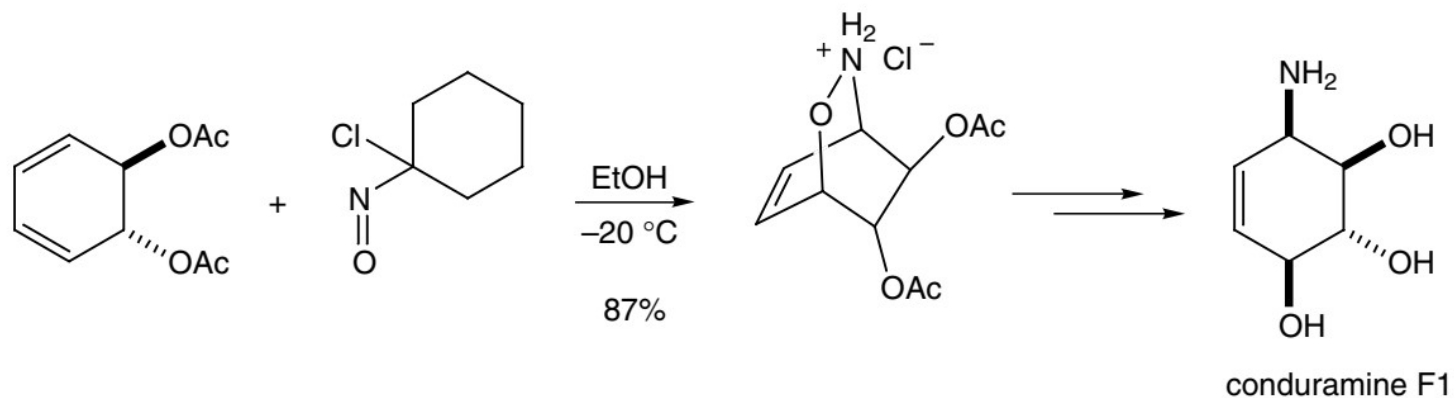
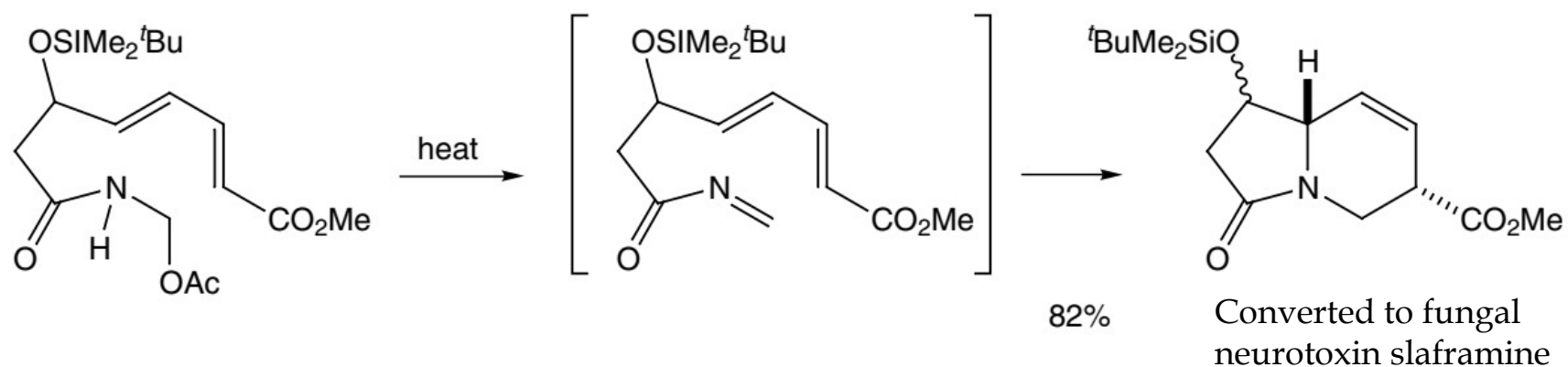
15-

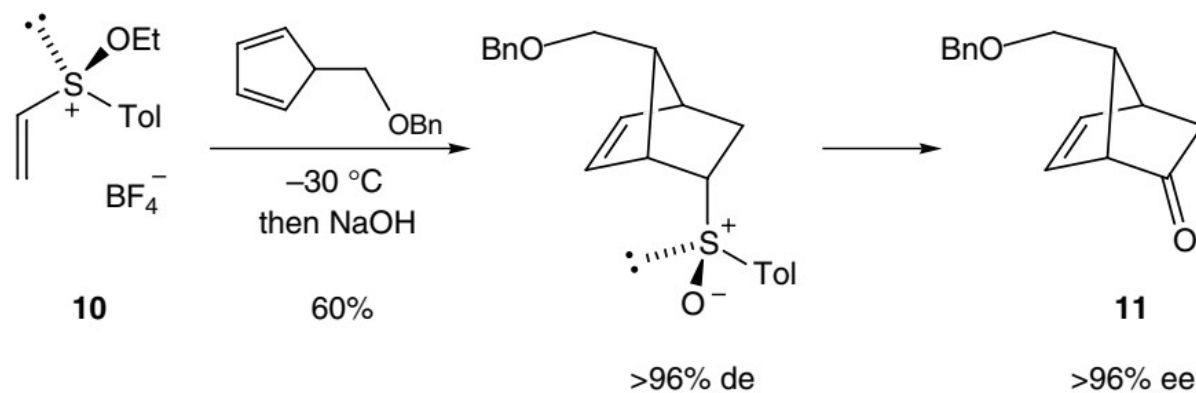


16-

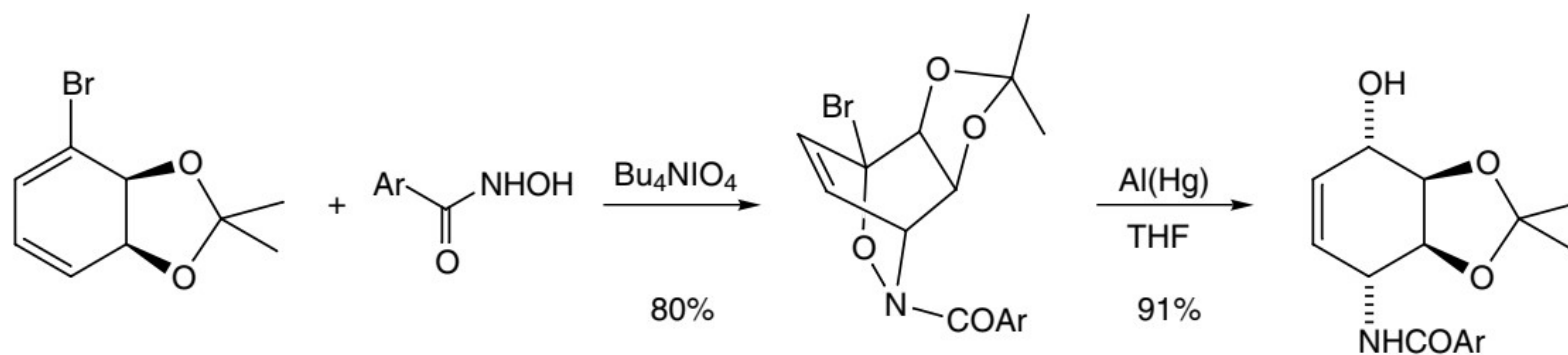


Application of Diels Alder reaction in synthesis of drugs

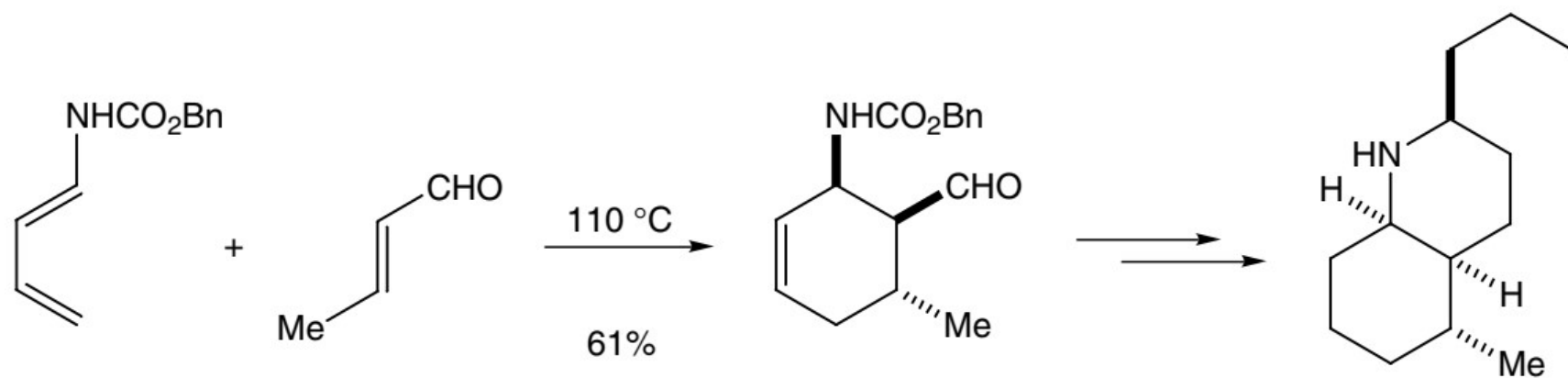




Used in the synthesis of
prostaglandins



Used in the synthesis of
lycoridine



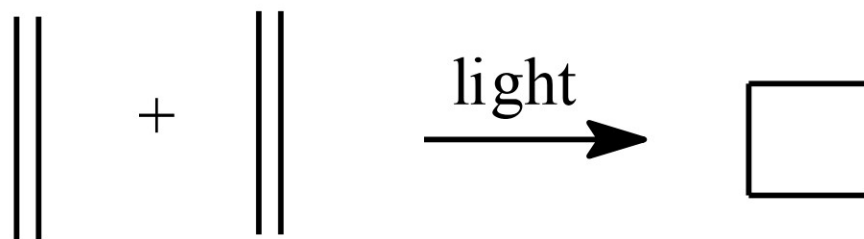
Pumilitotoxine (Poison)

1. Cycloaddition Reaction

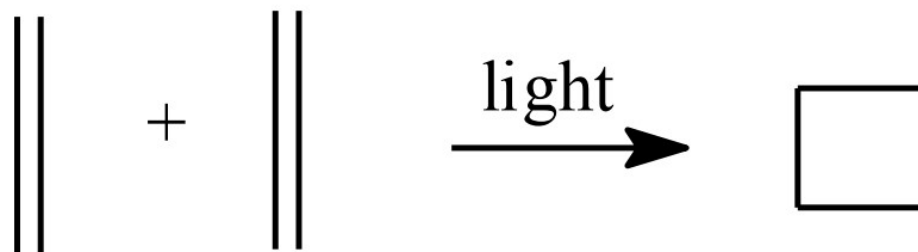
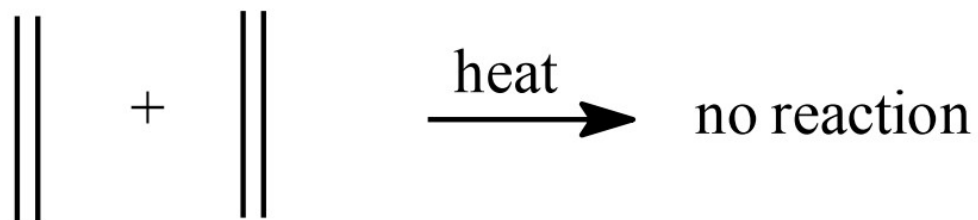
- Formation of a new ring.
- More than one bond form simultaneously.
- No intermediate formation.
- Three important classification of cycloaddition reaction
 - I. Diels-Alder reaction
 - II. [1,3]-Dipolar cycloaddition
 - III. [2+2] cycloaddition**

[2+2] Cycloaddition

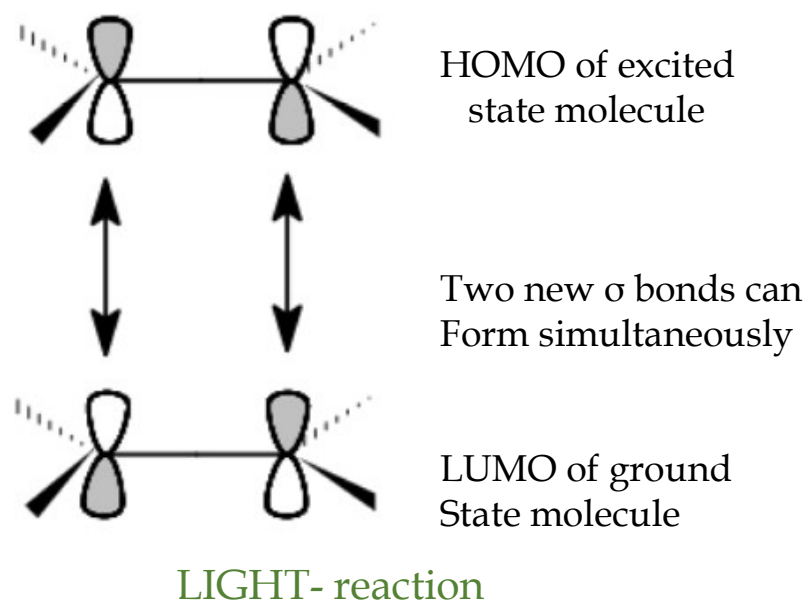
- These reaction takes place under photochemical conditions.
- Formation of compounds containing a four membered ring.



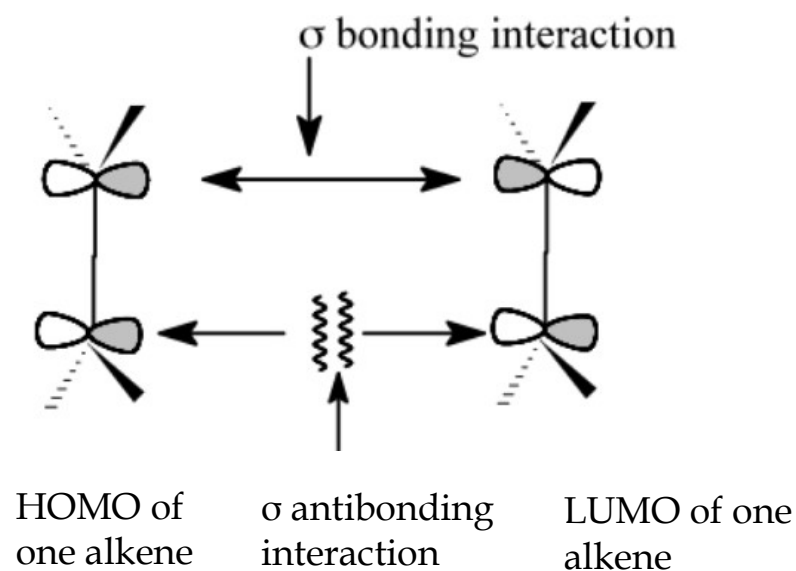
WHY?



Reason:



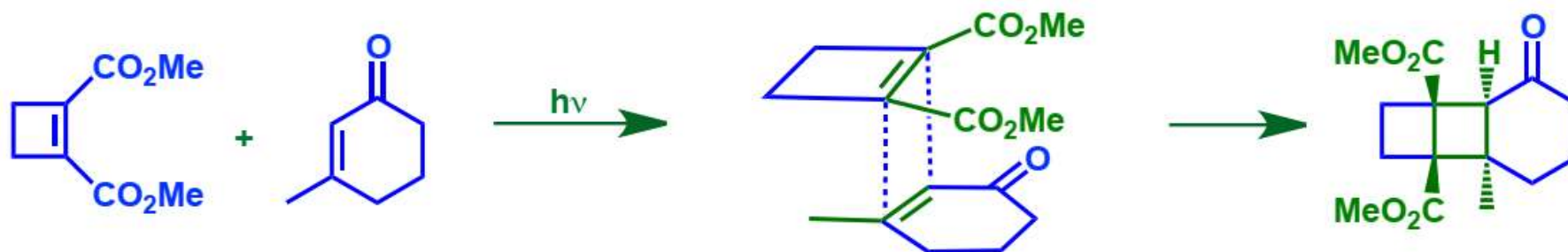
Irradiation *EXCITES* an electron
From the HOMO to the LUMO
So, the excited state HOMO is now
the same symmetry as the ground state LUMO



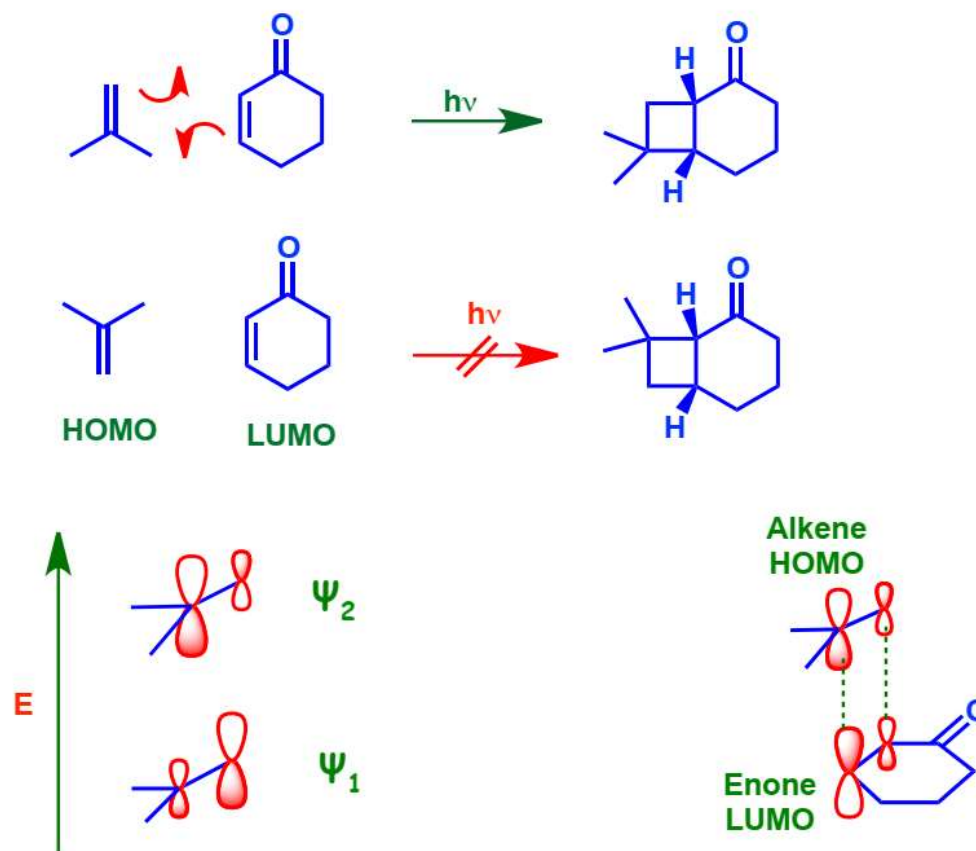
HEAT- no reaction

[2+2] Cycloaddition is stereospecific.

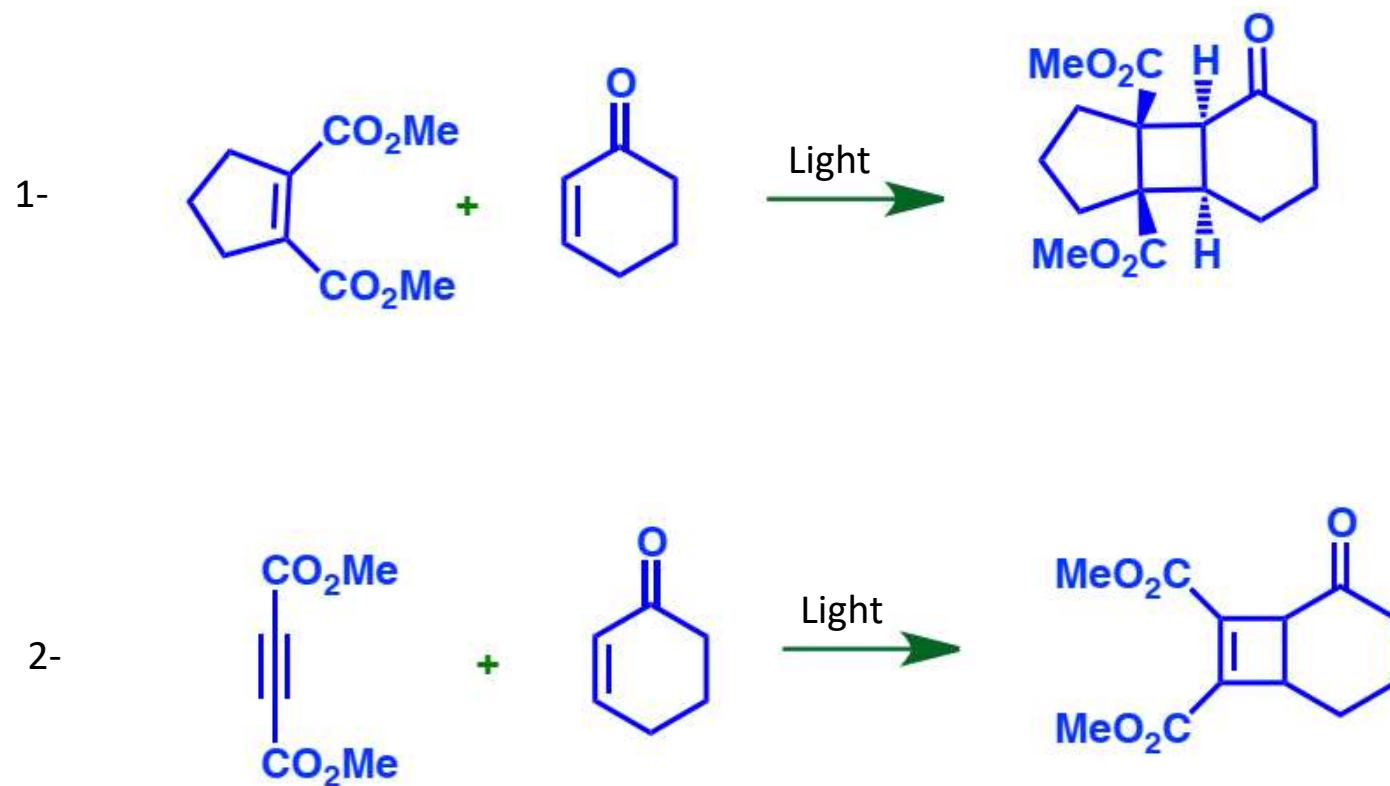
- Less hindered transition is observed



[2+2] Cycloaddition is regioselective

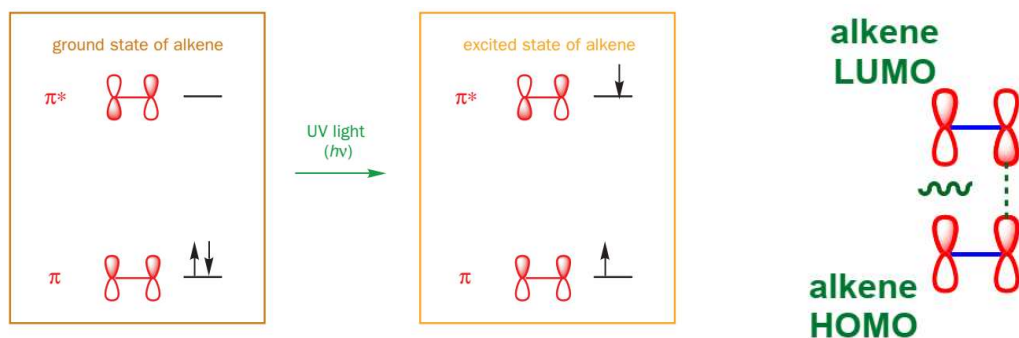


Example:

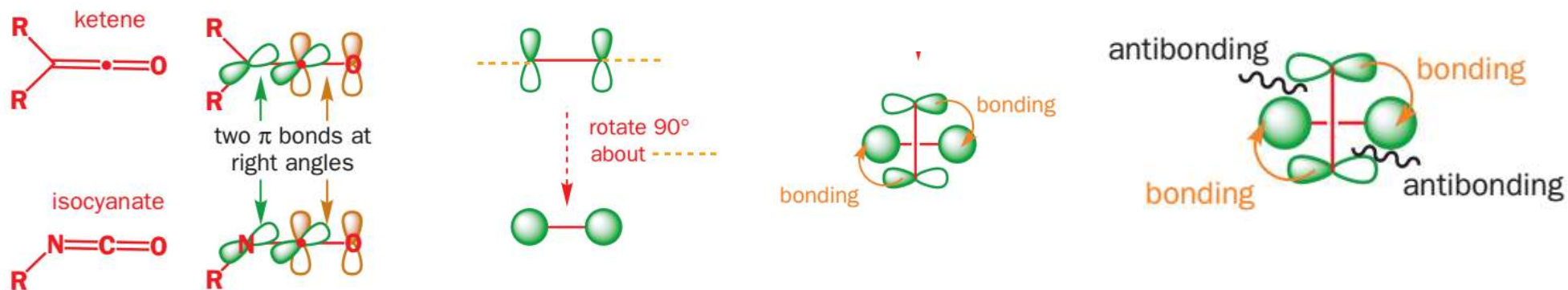


Thermal [2+2] cycloaddition

- HOMO/LUMO combination is antibonding at one end

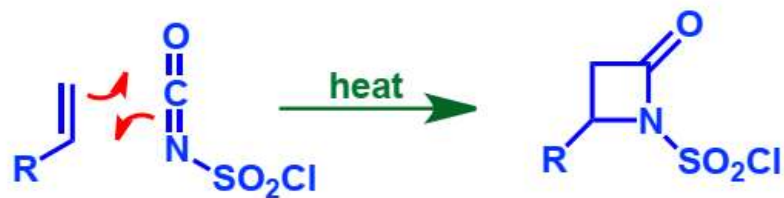


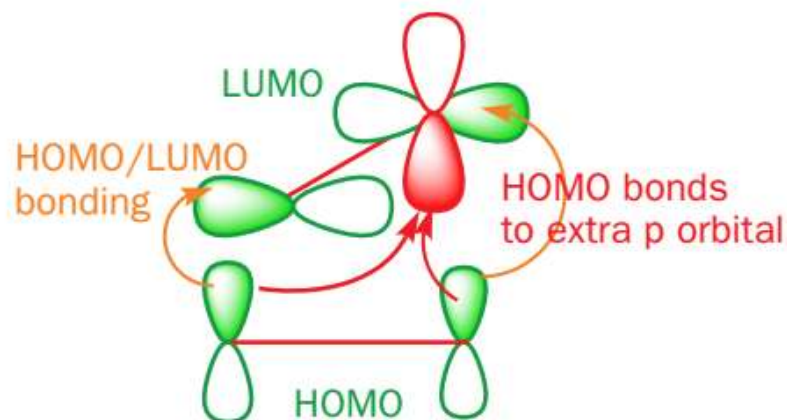
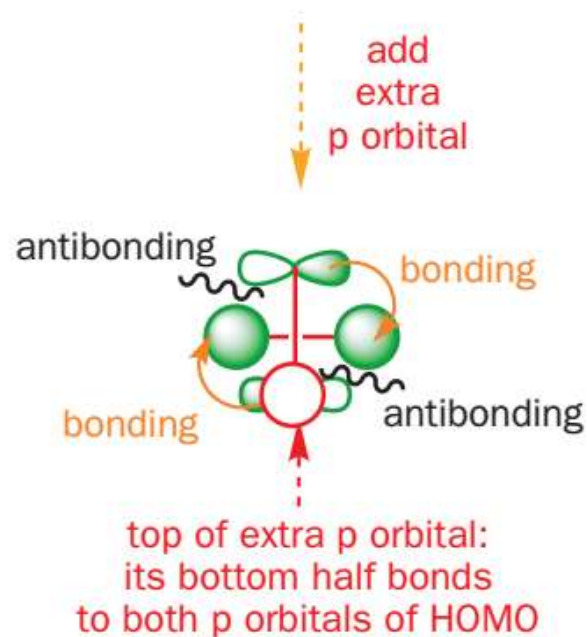
- If HOMO & LUMO approaches perpendicular to each other



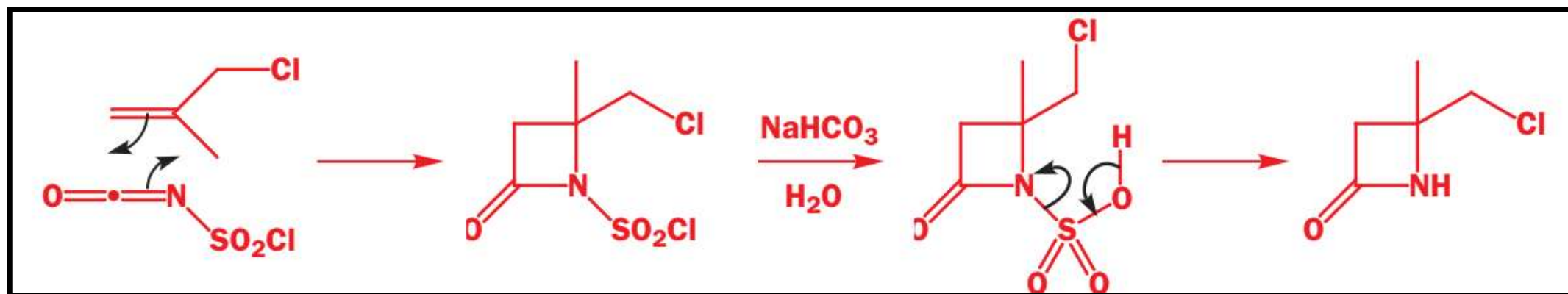
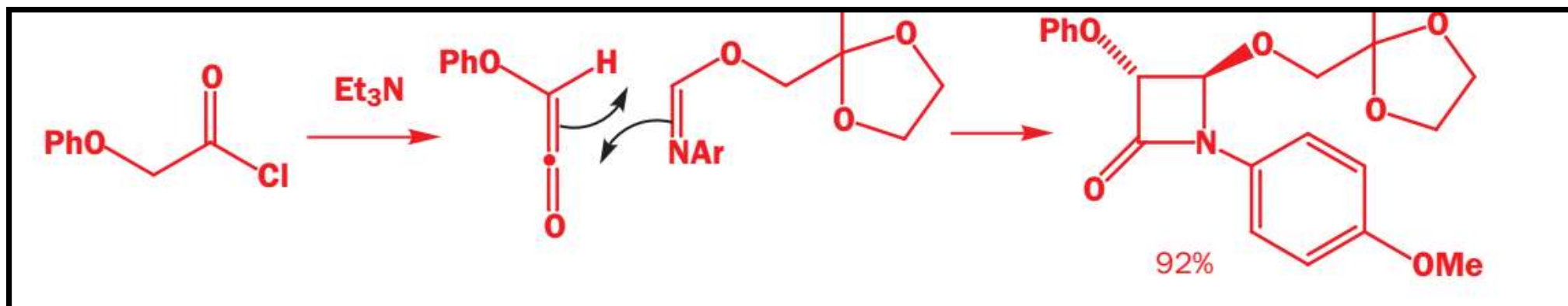
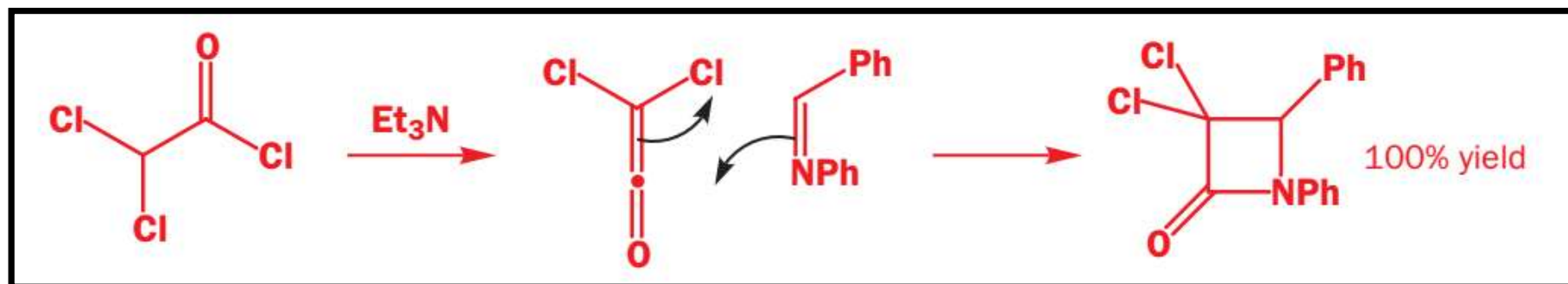
Condition for Thermal [2+2] cycloaddition

- Two double bond at single carbon.





Ketenes have a central sp carbon atom with an extra π bond (the C=O) at right angles to the first alkene – perfect for thermal [2 + 2] cycloadditions. They are also electrophilic and so have suitable low-energy LUMOs.



Acids & Bases in Organic Chemistry

Common definitions of acids and bases:

Arrhenius: Acids produce H^+ & bases produce OH^- ion in aqueous solution.

introduced in 1887

by the Swedish scientist [Svante Arrhenius](#)

Arrhenius acid increases the concentration of hydrogen ions in an aqueous solution



Arrhenius base increases the concentration of hydroxide ions in an aqueous solution

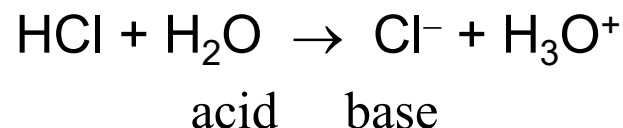


Brønsted-Lowry: Acids are H^+ donors & bases are proton acceptors.

developed by [Johannes Nicolaus Brønsted](#)

and [Thomas Martin Lowry](#)

independently in 1923



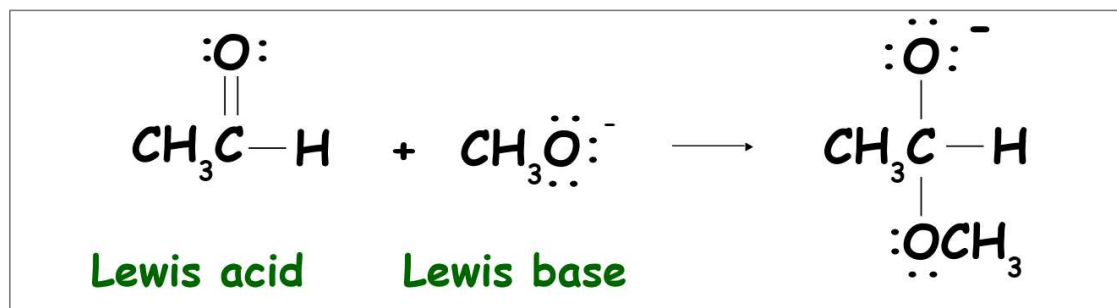
- Many of the organic reactions you will study involve acid-base reactions.

Understanding these reactions will require you to:

- Recognize organic compounds that can serve as acids or bases.
- Throw away the idea that anything with an OH is a base and anything with an H is an acid!!!

Acids & Bases in Organic Chemistry

The following reaction does not look like a “classic” acid-base reaction...neither reactant gains or loses an H^+ .



It is, however, a Lewis acid-base reaction.

Lewis acid-base concept

Gilbert N. Lewis

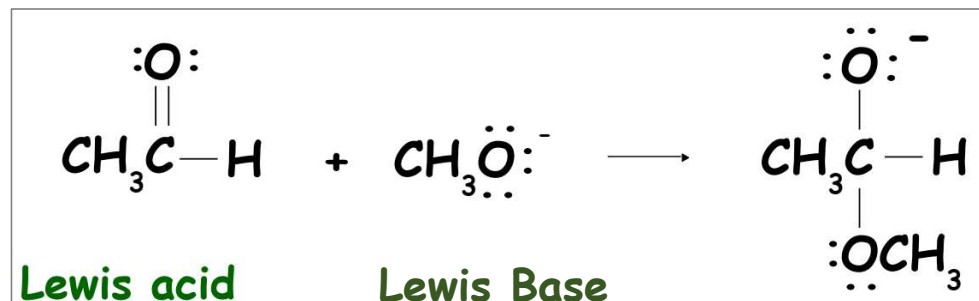
The Lewis acid-base definition is the broadest definition of acids and bases.

Lewis acid:

- an electron pair acceptor
- an **electrophile**
- “electron lover”
- a substance that accepts a pair of electrons to form a new bond

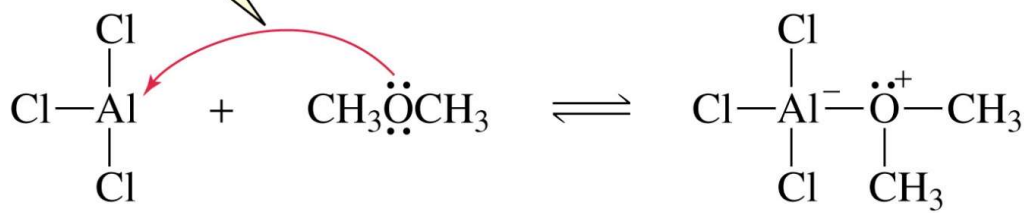
Lewis Base:

- an electron pair donor
- a **nucleophile**
- “nuclei lover”
- a substance with a pair of electrons that can be donated to another nucleus to form a new bond

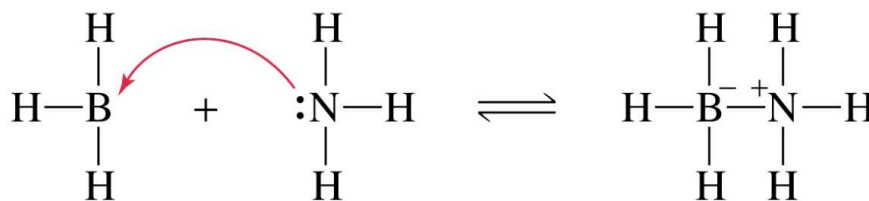


Example of Lewis acid-base

the curved arrow indicates
where the pair of electrons
starts from and where it ends up



aluminum trichloride dimethyl ether
a Lewis acid a Lewis base

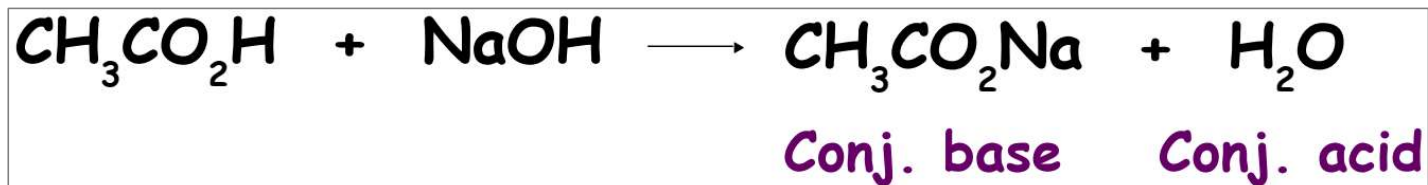


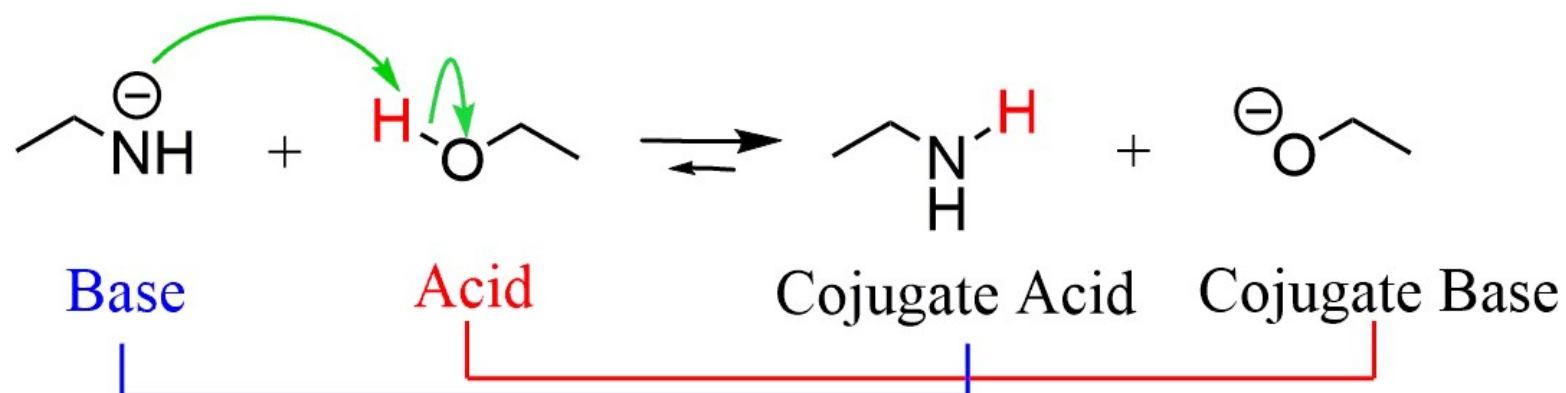
borane ammonia
a Lewis acid a Lewis base

Acid-base reactions

Acid-base reactions always produce a new acid and a new base.

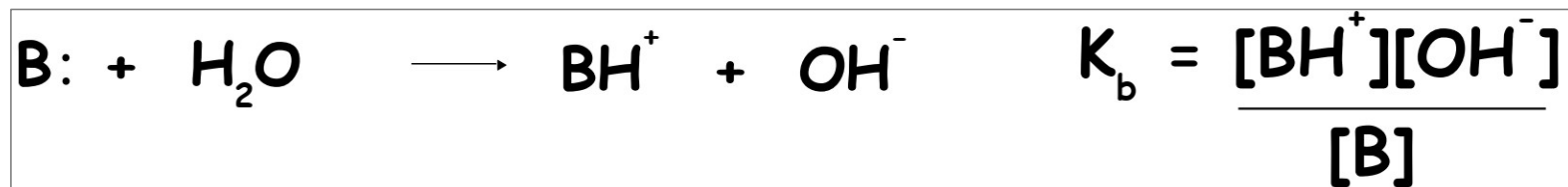
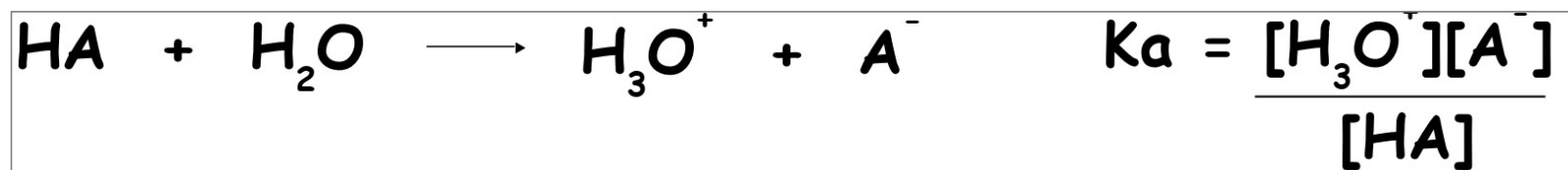
- **Conjugate acid:**
 - The new acid formed when the base gains a proton.
 - always found on the product side.
- **Conjugate base:**
 - The new base formed by removing a proton from an acid.
 - always found on the product side.





Strength of an acid or base

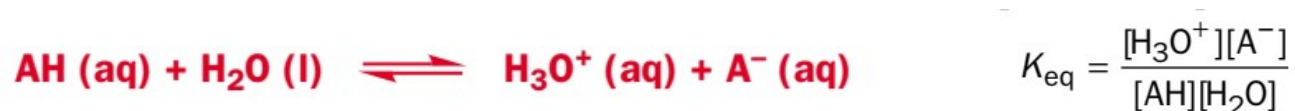
The relative strength of an acid can be determined using the magnitude of K_a (or **pKa**)



- K_a = acid dissociation constant
- K_b = base dissociation constant

The definition of pKa

Now we need to be clearer about ‘strong’ and ‘weak’ acids. In order to measure the strength of an acid relative to water and find out how effective a proton donor it is, we must look at the equilibrium constant for the reaction.



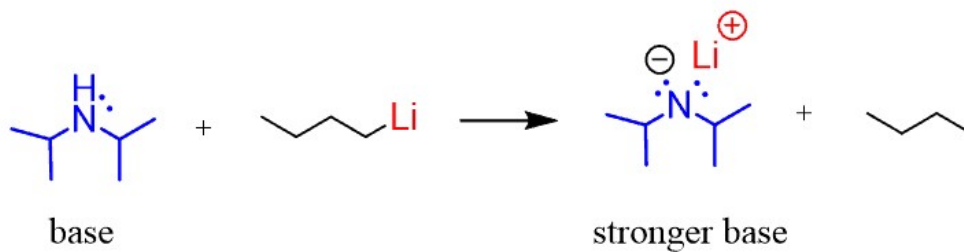
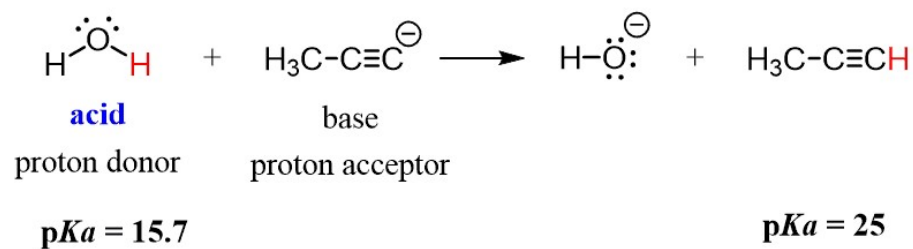
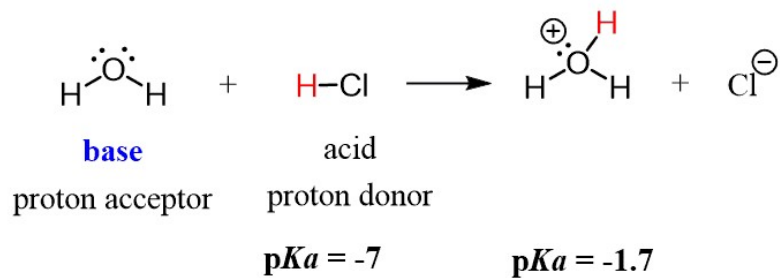
The concentration of water remains constant (at 55.56 mol dm⁻³) with dilute solutions of acids wherever the equilibrium may be and a new equilibrium constant, K_a , is defined and called the **acidity constant**.

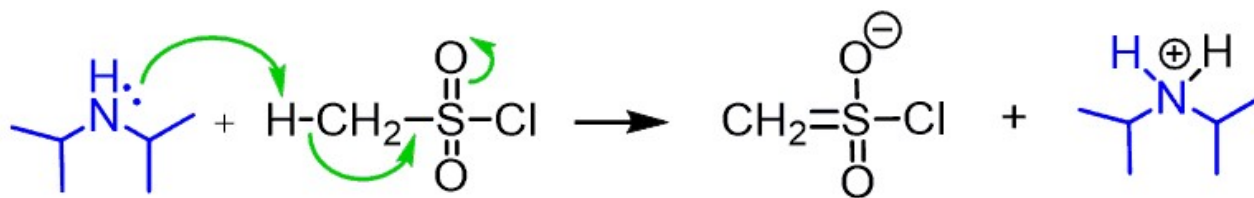
$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{AH}]}$$

$$\text{p}K_a = -\log K_a$$

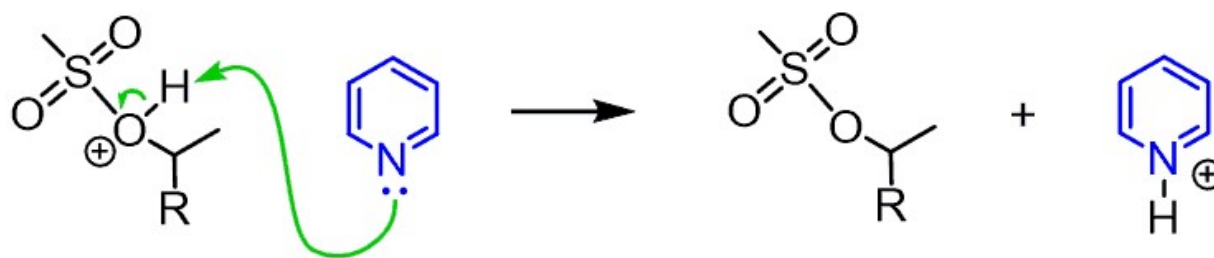
Because of the minus sign in this definition, the lower the $\text{p}K_a$, the larger the equilibrium constant, K_a , is and hence the stronger the acid.

Acid-Base role changes depending on the pKa





diisopropylamine
base



Pyridine - base

The pKa value of some compounds

	Functional group	Example	pKa	Conjugate base	
Stronger acid	Hydrochloric acid	HCl	-7	$\text{:}\ddot{\text{Cl}}\text{:}^-$	Weaker base
	Carboxylic acid	<chem>CC(=O)O</chem>	5	<chem>CC(=O)[O-]</chem>	
	Phenol	<chem>c1ccccc1O</chem>	10	<chem>c1ccccc1[O-]</chem>	
	Alcohol	<chem>CCO</chem>	16	<chem>CC[O-]</chem>	
	Ketone	<chem>CC(=O)C</chem>	19	<chem>CC(=O)[CH2-]</chem>	
	Alkyne	<chem>C#CC</chem>	25	<chem>C#C[CH2-]</chem>	
	Alkene	<chem>C=CC</chem>	44	<chem>C=CC[CH2-]</chem>	
Weaker acid	Alkane	<chem>CCC</chem>	51	<chem>CCC[CH2-]</chem>	Stronger base

Acid	pK _a	Conjugate base
HI	ca. -10	I ⁻
HCl	ca. -7	Cl ⁻
H ₂ SO ₄	ca. -3	HSO ₄ ⁻
HSO ₄ ⁻	2.0	SO ₄ ²⁻
CH ₃ COOH	4.8	CH ₃ COO ⁻
H ₂ S	7.0	HS ⁻
NH ₄ ⁺	9.2	NH ₃
C ₆ H ₅ OH	10.0	C ₆ H ₅ O ⁻
CH ₃ OH	15.5	CH ₃ O ⁻
<chem>CC(=O)C</chem>	20.0	<chem>CC(=O)[CH2-]</chem>
<chem>C#CC</chem>	25	<chem>C#C[CH2-]</chem>
NH ₃	33	NH ₂ ⁻
C ₆ H ₆	ca. 43	C ₆ H ₅ ⁻
CH ₄	ca. 48	CH ₃ ⁻

The relative strength of an acid can be determined using:

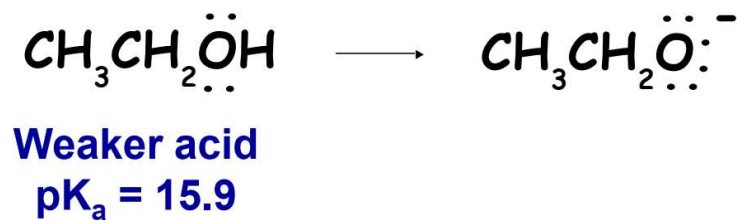
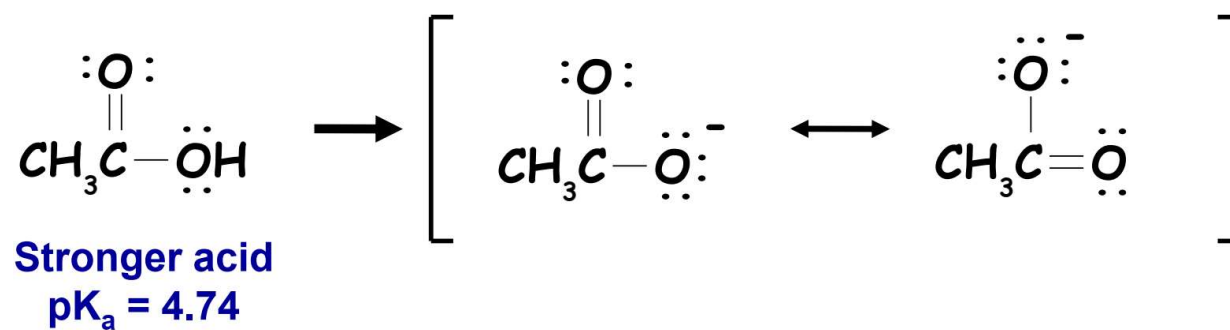
- The magnitude of K_a (or pK_a)
 - As K_a increases, the strength of the acid increases.
 - As pK_a decreases, the strength of the acid increases.
- Structural trend: The strength of an acid, HX, depends on
 - the electronegativity of the atom containing the acidic hydrogen.
 - the stability of the conjugate base, X^-



- Within a group, the strength of an acid increases moving down the group
 - negative charge is more stable when spread out over a larger region.



- Acidity increases when the charge on the conjugate base can be delocalized over two or more atoms through resonance



- The strength of a base is inversely related to its conjugate acid.
- Strong acids form conjugate bases with negligible basicity.



- Weak acids form stronger conjugate bases.

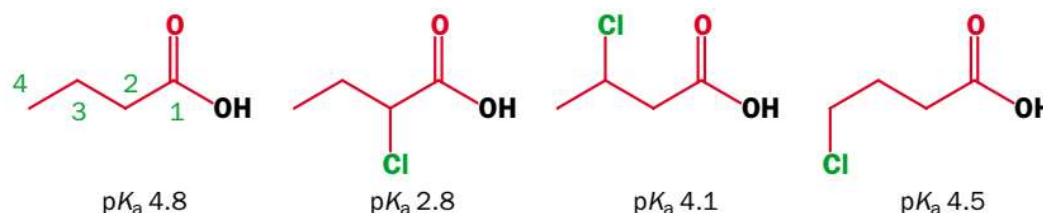


- Substances with negligible acidity form very strong conjugate bases.



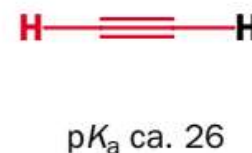
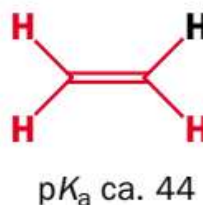
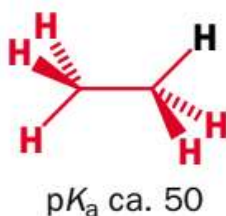
Effect that can affects the pK_a value of some compounds.

Inductive effects:

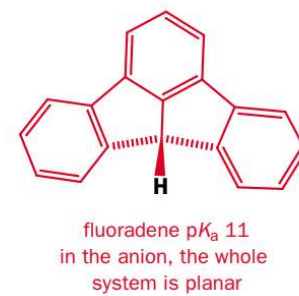
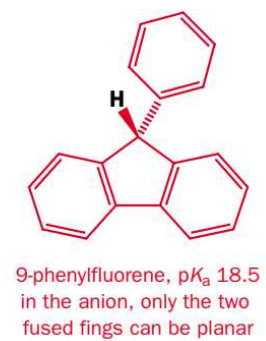
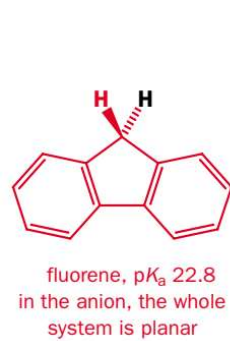
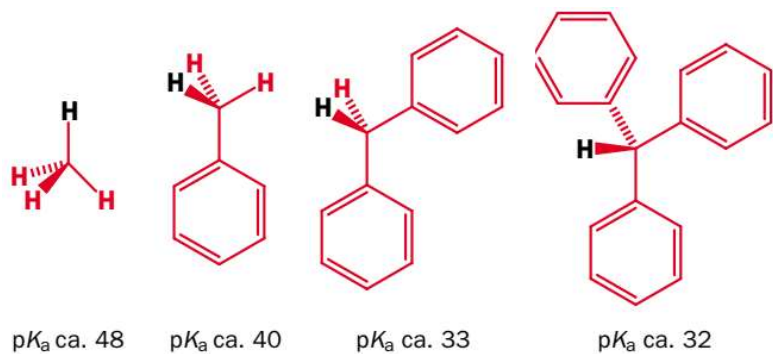


Hybridization:

The hybridization of the orbital from which the proton is removed also affects the pK_a . Since s orbitals are held closer to the nucleus than are p orbitals, so the more s character an orbital has, the more tightly held are the electrons .



Highly conjugated carbon acids:



Aromaticity:



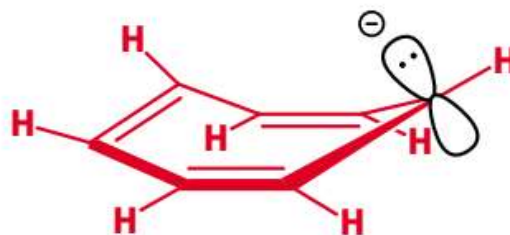
cyclopentadiene
 pK_a 15.5



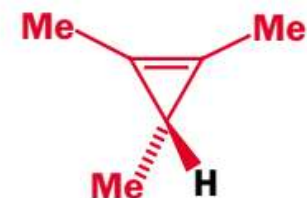
anion is planar
 6π electrons
make it aromatic



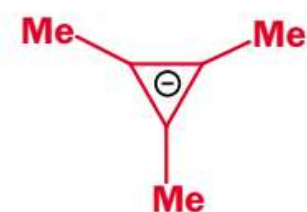
cycloheptatriene
 pK_a ca. 36



anion not planar
neither aromatic or anti-aromatic

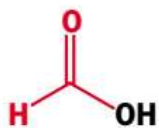


trimethylcyclopropene
 pK_a ca. 62

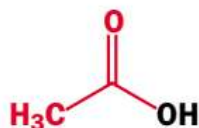


anion is planar
 4π electrons
make it anti-aromatic

Electron-donating groups decrease acidity:



formic (methanoic) acid
 pK_a 3.7



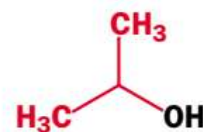
acetic (ethanoic) acid
 pK_a 4.8



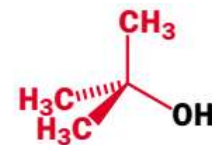
methanol
 pK_a 15.5



ethanol
 pK_a 16.0



isopropyl alcohol
 pK_a 17.1



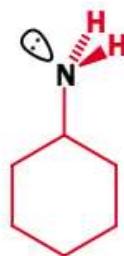
tert-butyl alcohol
 pK_a 19.2

ORGANIC BASES

A **base** is a substance that can accept a proton by donating a pair of electrons.



aniline



Strong bases commonly used in organic chemistry:

potassium *tert*-butoxide: $\text{K}^+\text{OC}(\text{CH}_3)_3$

sodium amide: Na^+NH_2^-

sodium hydride: Na^+H^-

Basicity

How can we measure a base's strength?



ammonia

pK_a ca. 33



ammonium

pK_b 9.24

pK_{aH} of ammonia is **9.24**

we must look up the pK_a of its conjugate acid,
The ammonium cation, NH_4^+

Scales for basicity— pK_b and pK_{aH}



$$K_B = \frac{[\text{OH}^-][\text{BH}^+]}{[\text{B}]}$$

Hence

$$pK_B = -\log(K_B)$$

lower the pK_b , stronger the base

pK_b or pK_{aH} ??

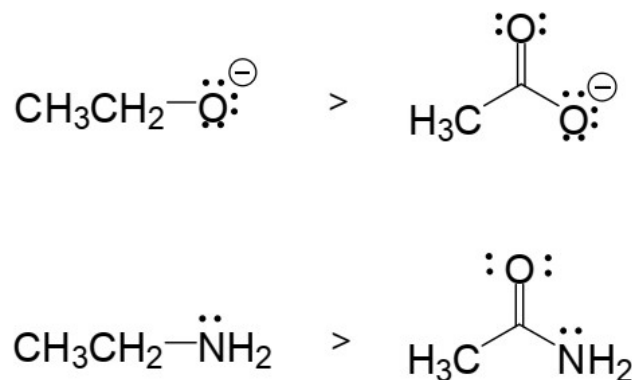
higher the pK_{aH} , stronger the base

What factors affect how basic a compound is?

There are two main factors that determine the strength of a neutral base:

- How accessible is the lone pair.
- What extent can the resultant positive charge formed be stabilized either by delocalization or by the solvent.

Effect of resonance



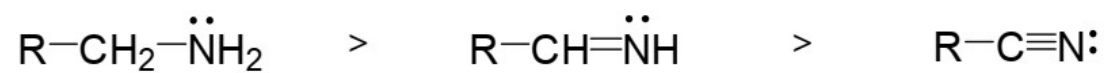
Effect of charge



Effect of electronegativity



Effect of hybridization



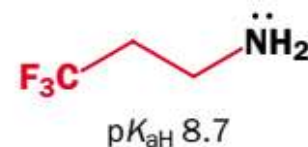
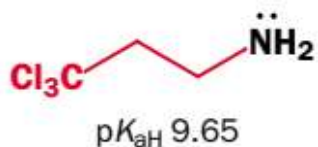
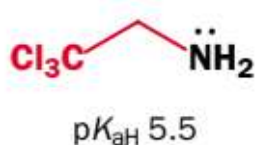
Neutral nitrogen bases

- Ammonia is the simplest nitrogen base and has a pK_aH of 9.24.
- Any substituent that **increases the electron density on the nitrogen** therefore raises the energy of the lone pair thus making it more available for protonation and **increasing the basicity** of the amine .
- Any substituent that **withdraws electron density** from the nitrogen makes it **less basic**.

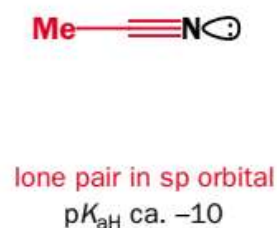
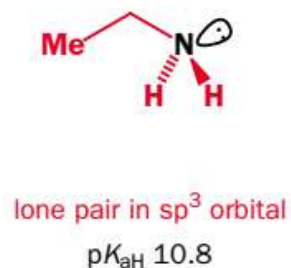
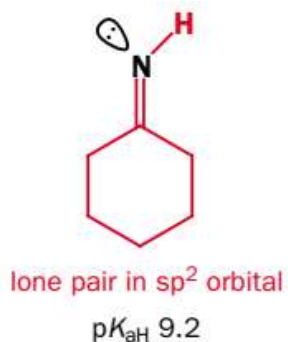
Effects that decrease the electron density on nitrogen

The lone pair on nitrogen will be *less* available for protonation, and the amine *less* basic, if

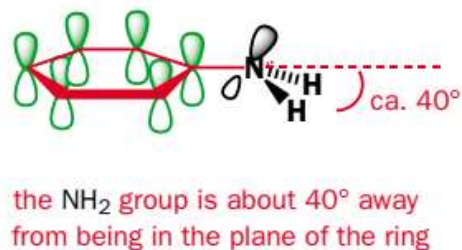
- ✓ The nitrogen atom is attached to an electron-withdrawing group



✓ The lone pair is in an sp or sp_2 hybridized orbital



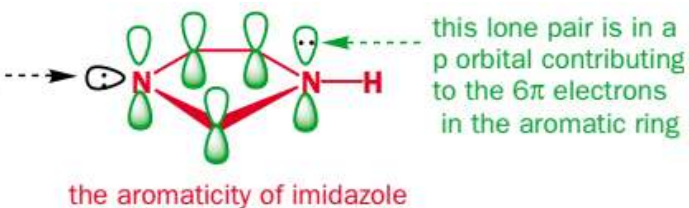
✓ The lone pair is conjugated with an π group



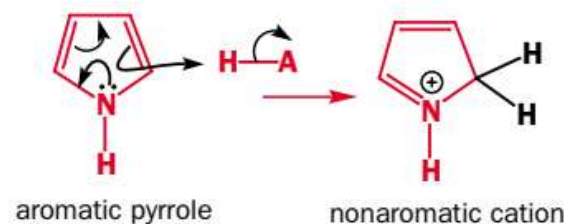
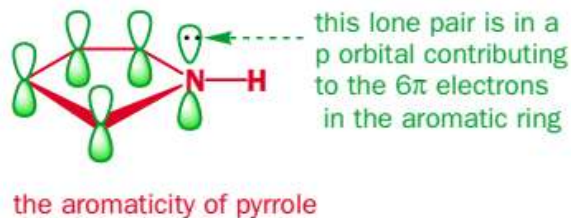
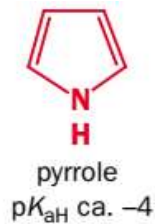
✓ The lone pair is involved in maintaining the aromaticity of the molecule



this lone pair is in an sp^2 orbital and is not involved with the aromaticity of the ring. Protonation occurs here



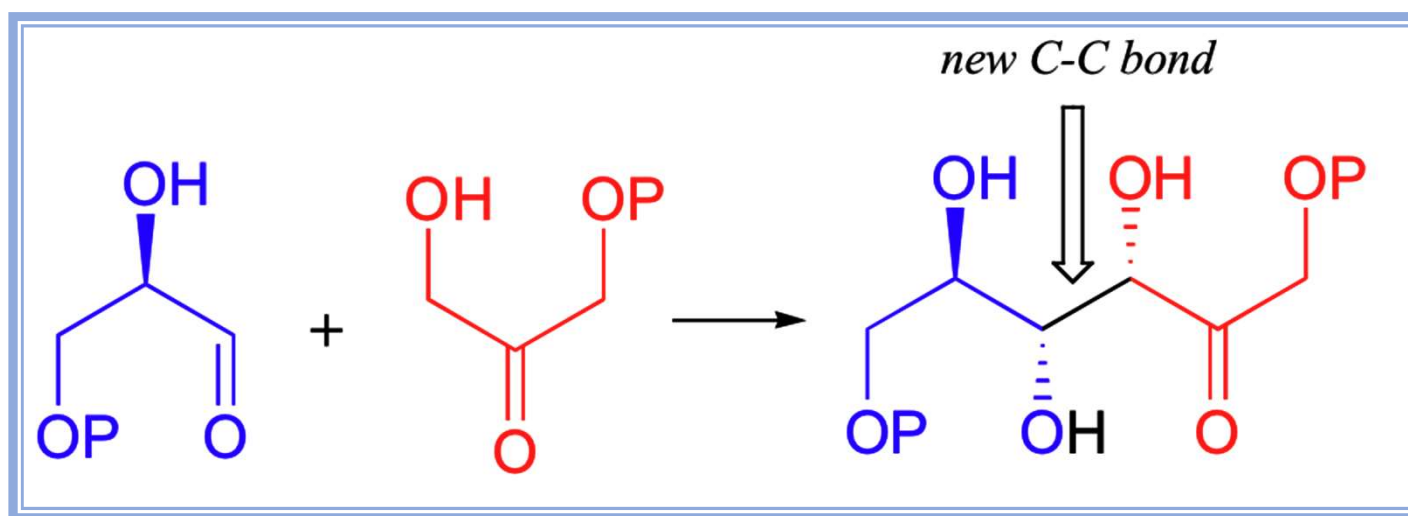
this lone pair is in a p orbital contributing to the 6π electrons in the aromatic ring



BTech Attendance
(21th Oct, 1-1.50
pm



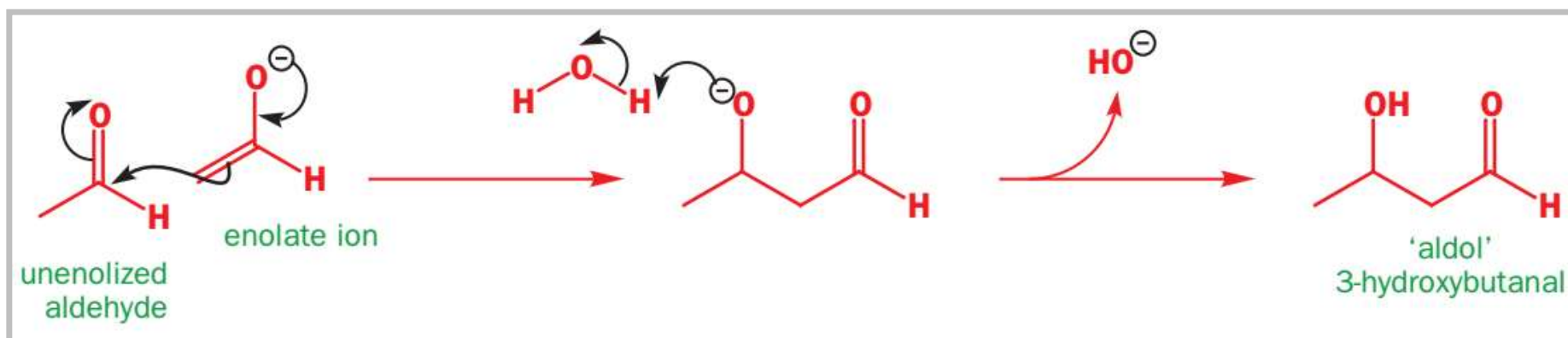
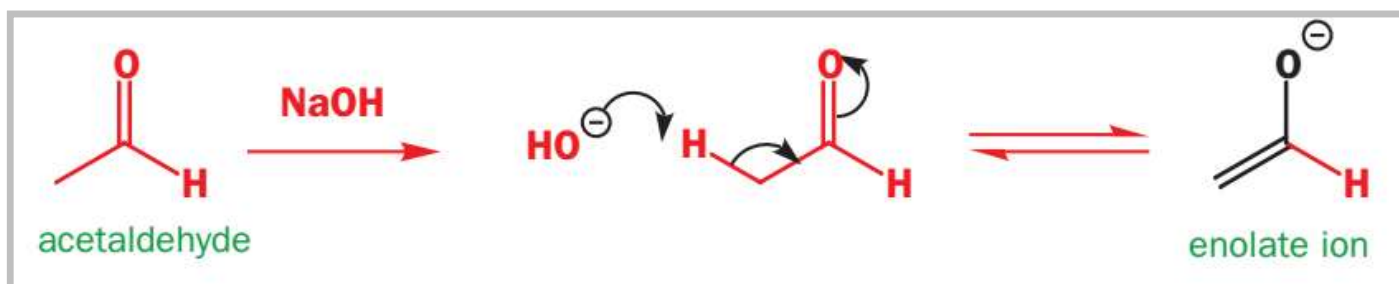
Enolate/enol chemistry

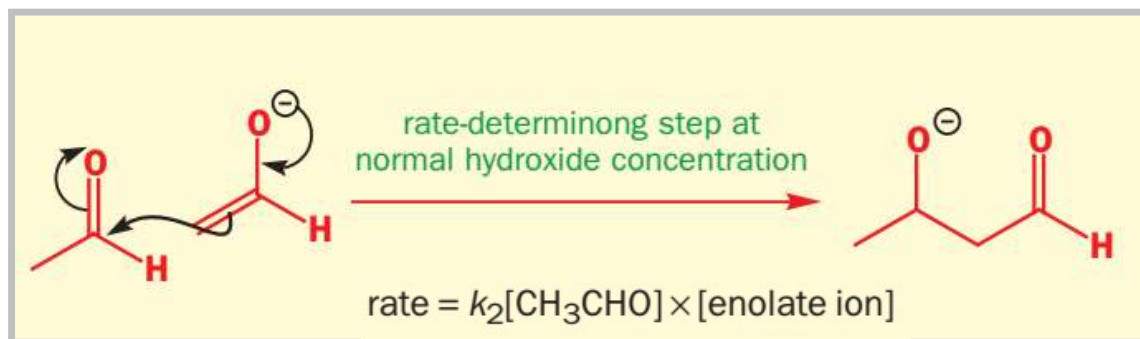
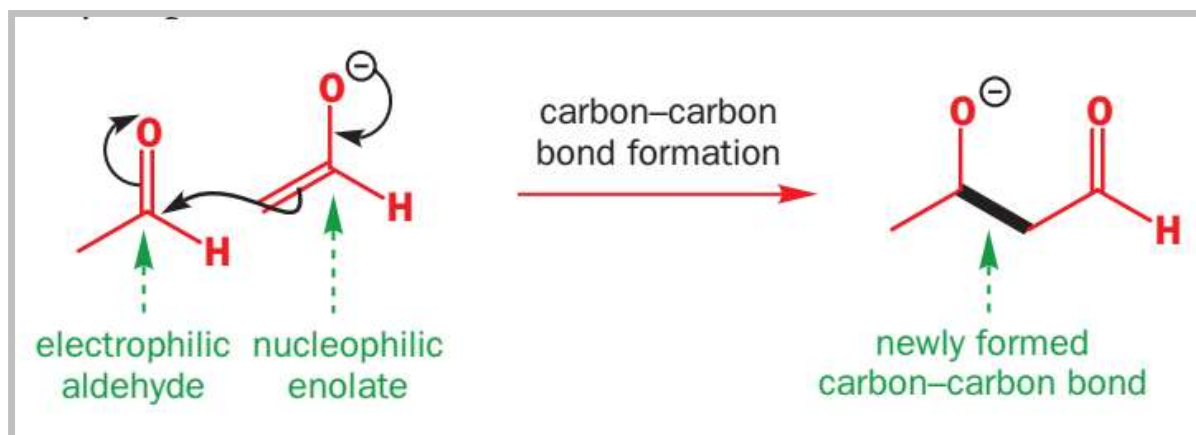


aldol reactions occur in several biological pathways, most commonly in the metabolism of carbohydrates

the aldol reaction

What happens if we add a small amount of **Base** to aldehyde?

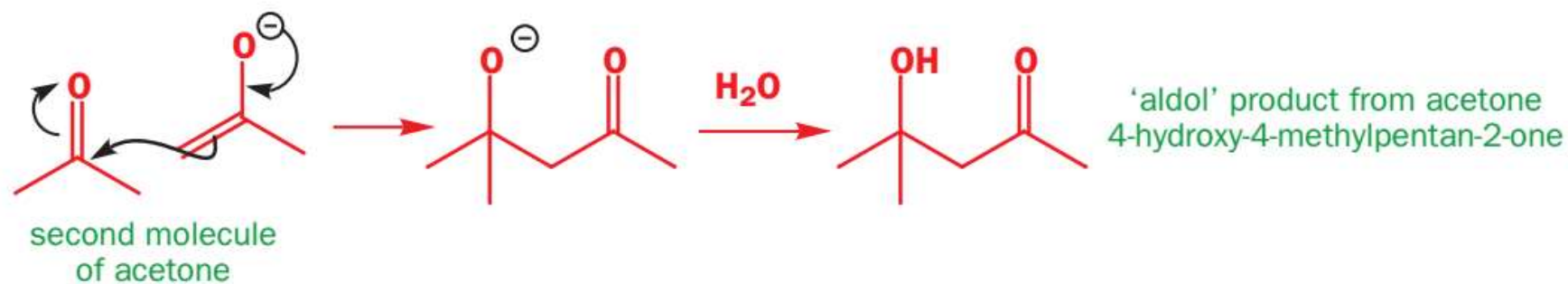




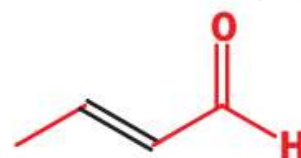
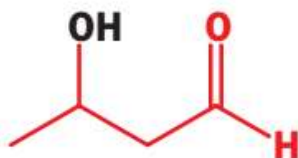
the enolization step



the carbon-carbon bond-forming step

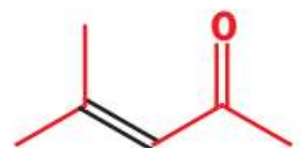
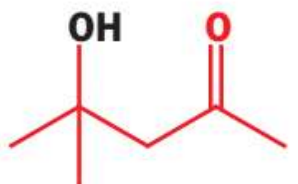


aldol product
from acetaldehyde

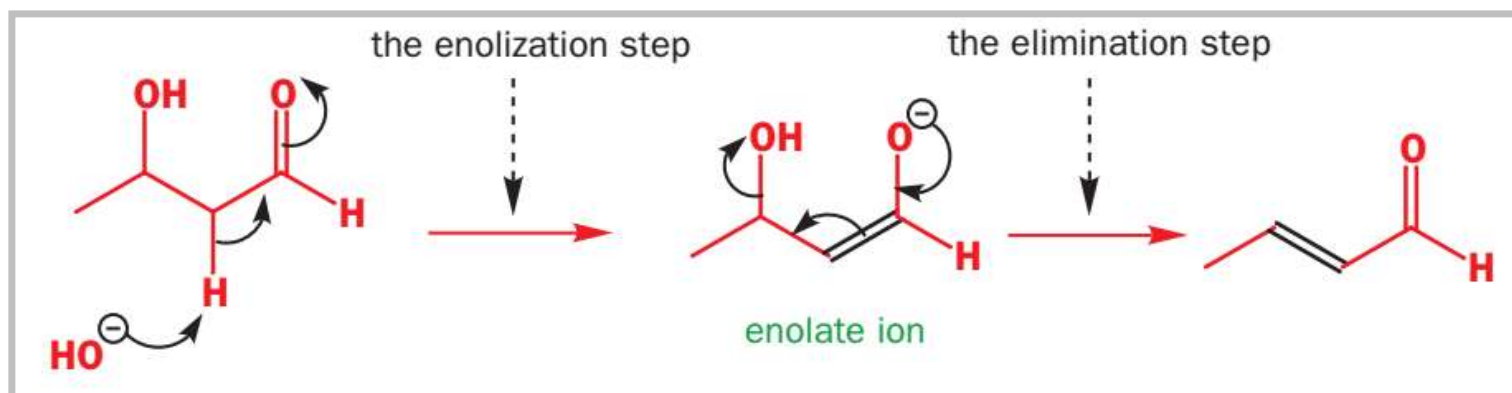


dehydration product
but-2-enal, or
'crotonaldehyde'

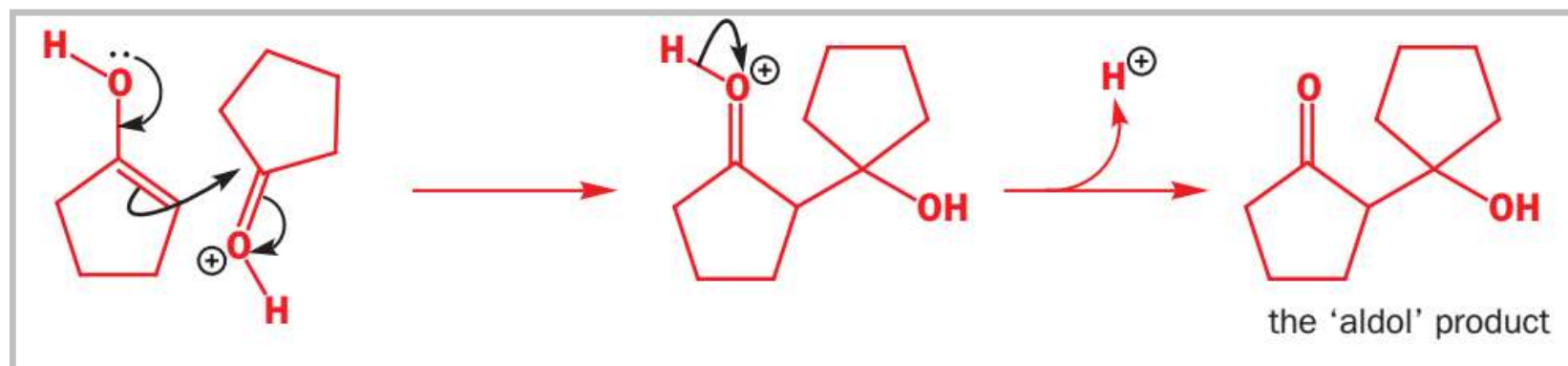
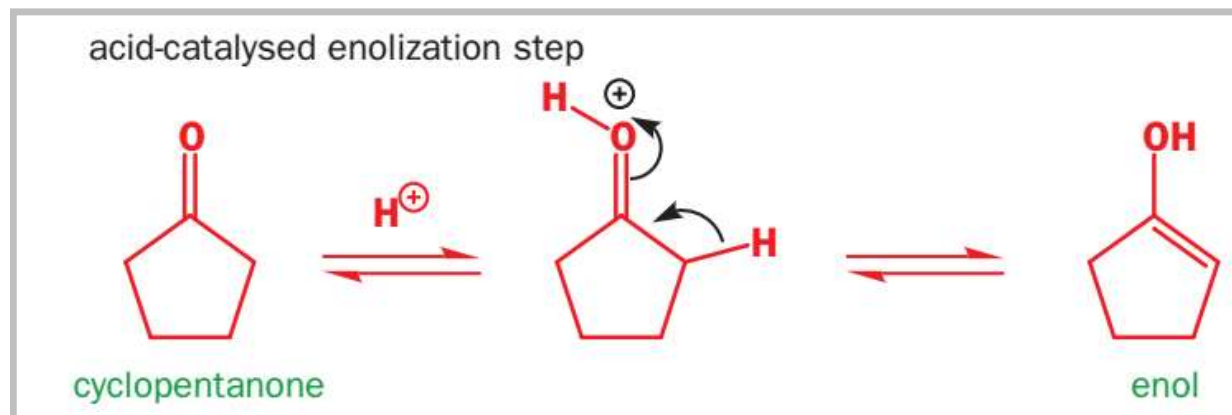
aldol product
from acetone



dehydration product
4-methylpent-3-en-2-one,
or 'mesityl oxide'

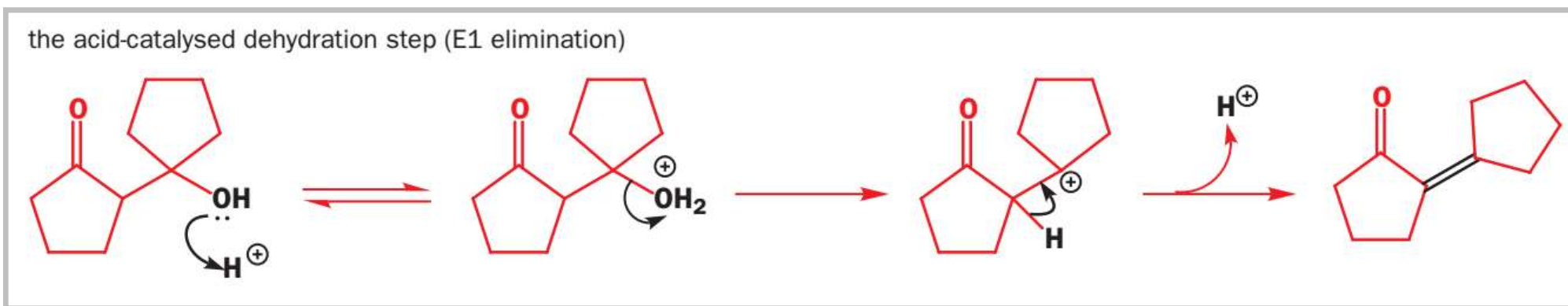


What happens if we add a small amount of **Acid**?



The elimination is even easier in acid solution

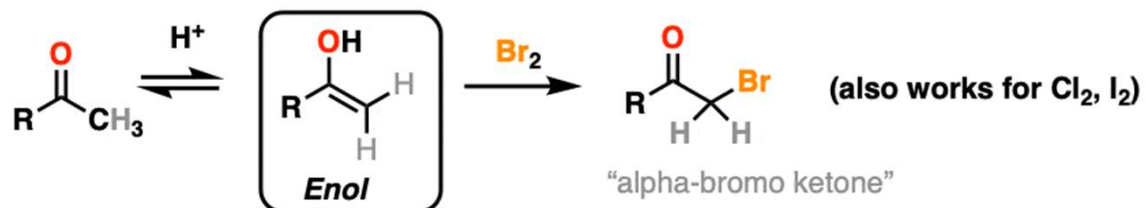
Acid-catalysed aldol reactions commonly give **unsaturated products** instead of **aldols**.



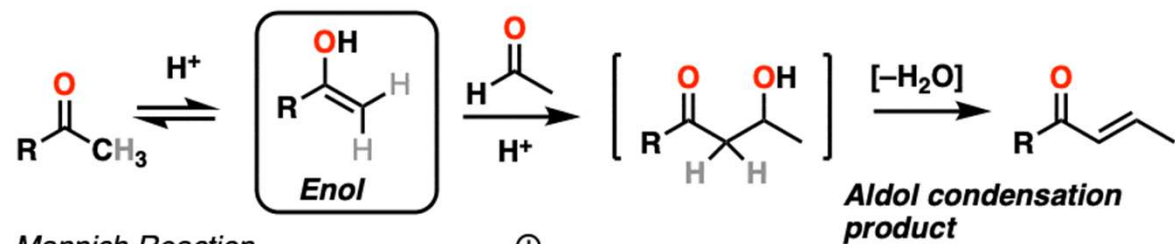
This reaction is called an **aldol condensation**

- Base-catalysed aldol reactions may give the aldol product, or may give the dehydrated enone or enal by an E1cB mechanism
- Acid-catalysed aldol reactions may give the aldol product, but usually give the dehydrated enone or enal by an E1 mechanism

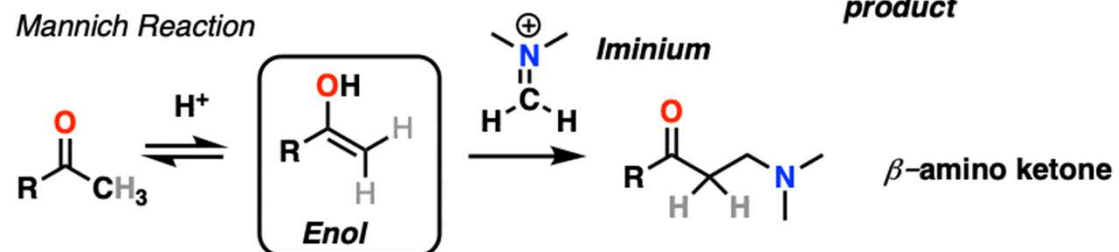
Acid-catalyzed halogenation of ketones



Acid-catalyzed aldol condensation

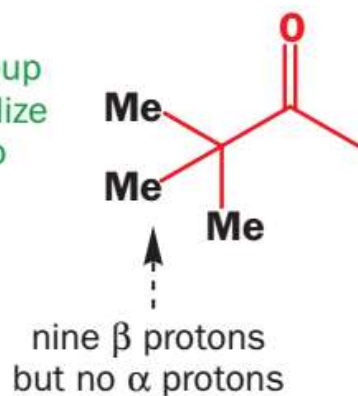


Mannich Reaction

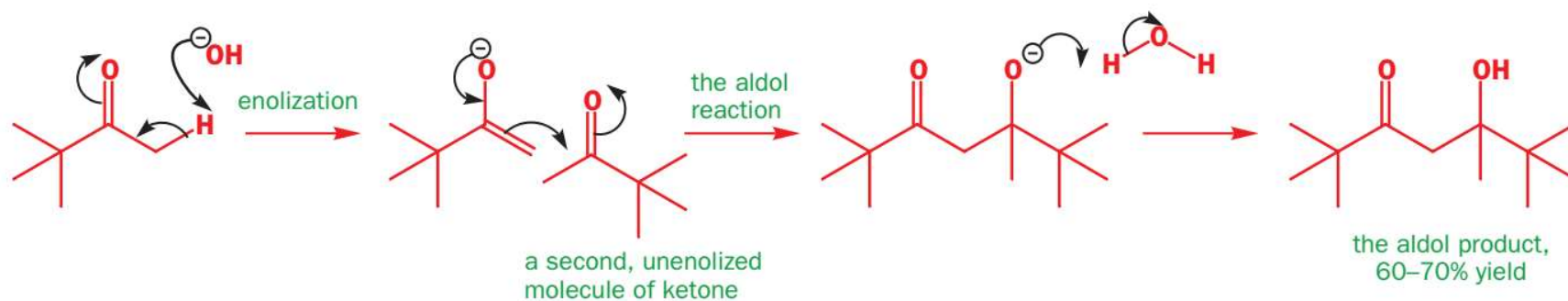
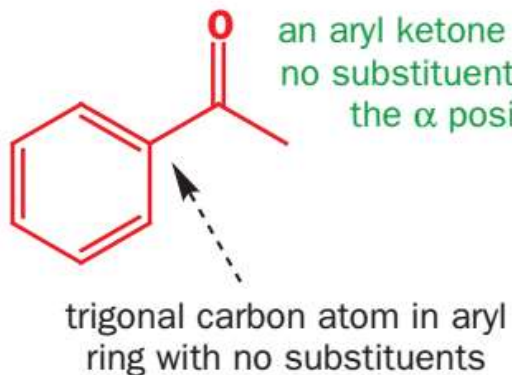


ketones which can enolize only one way:

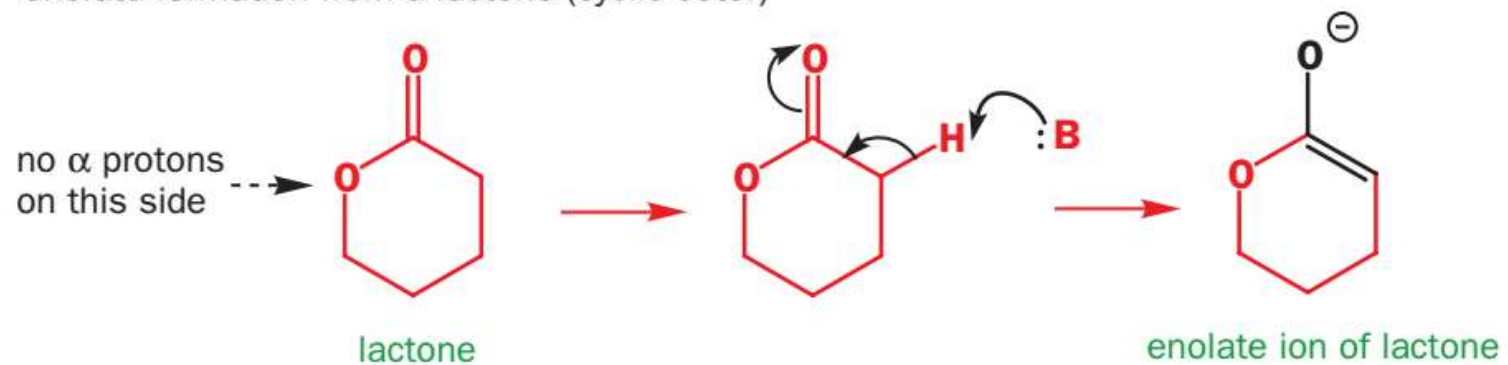
a *t*-alkyl group cannot enolize as it has no α protons



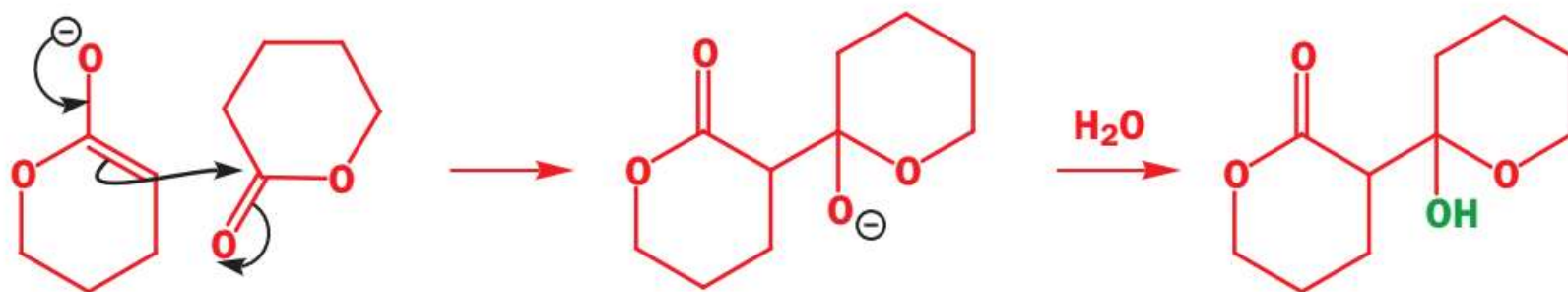
an aryl ketone has no substituents at the α position



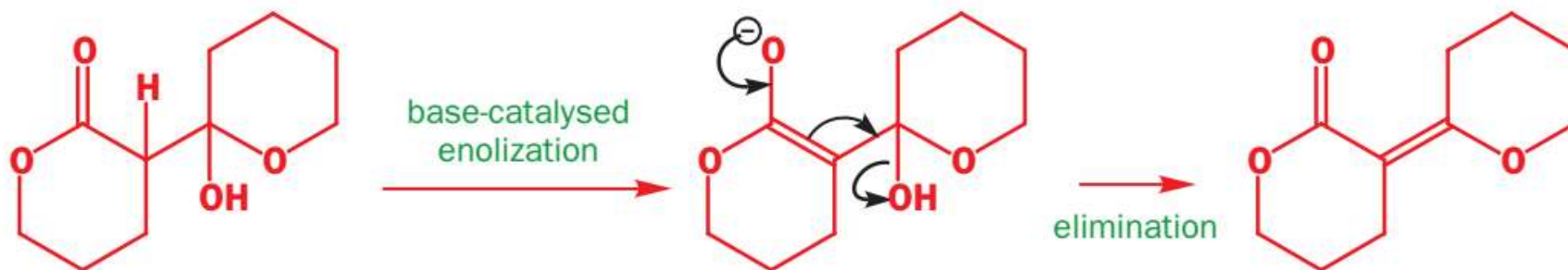
enolate formation from a lactone (cyclic ester)



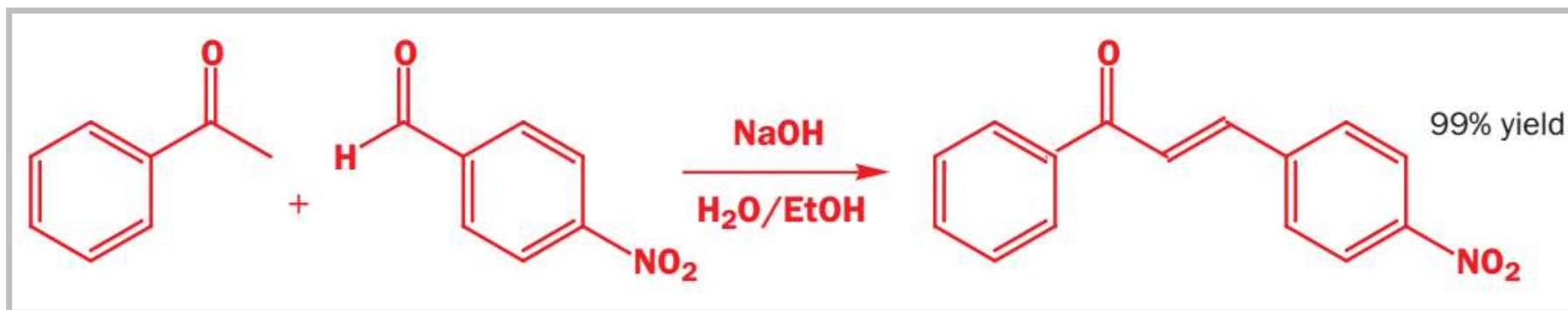
aldol reaction of a lactone (cyclic ester)

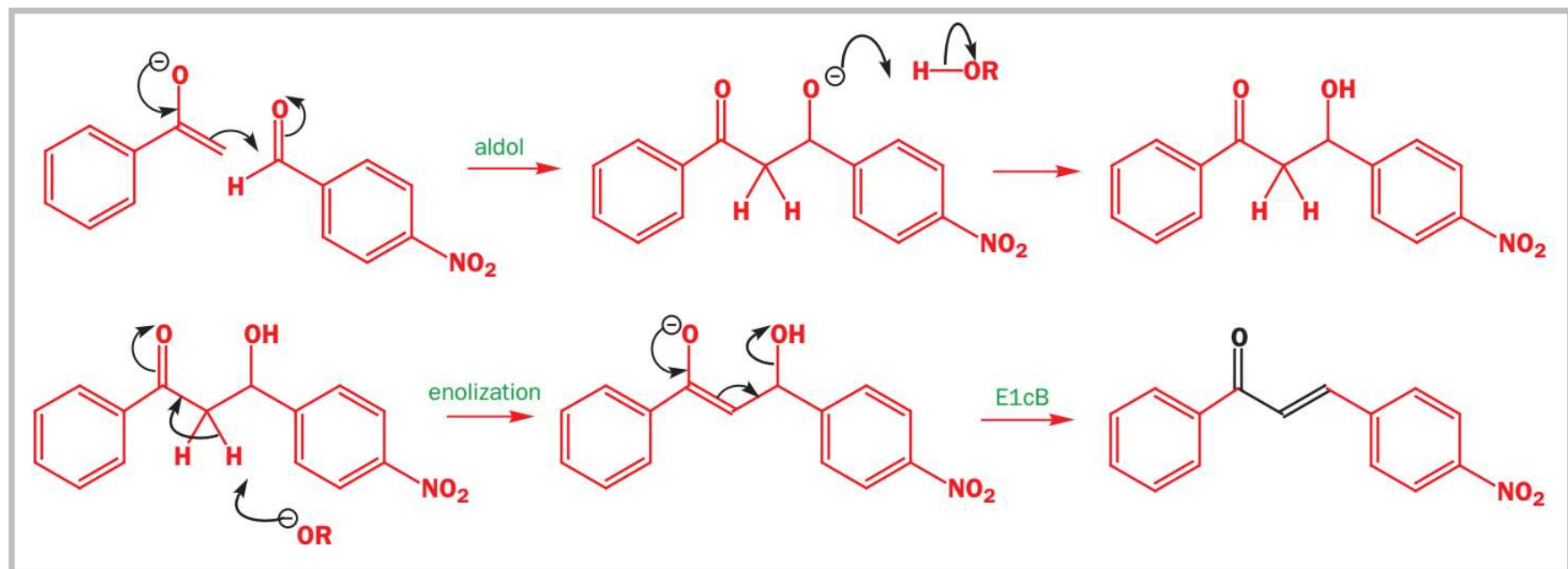
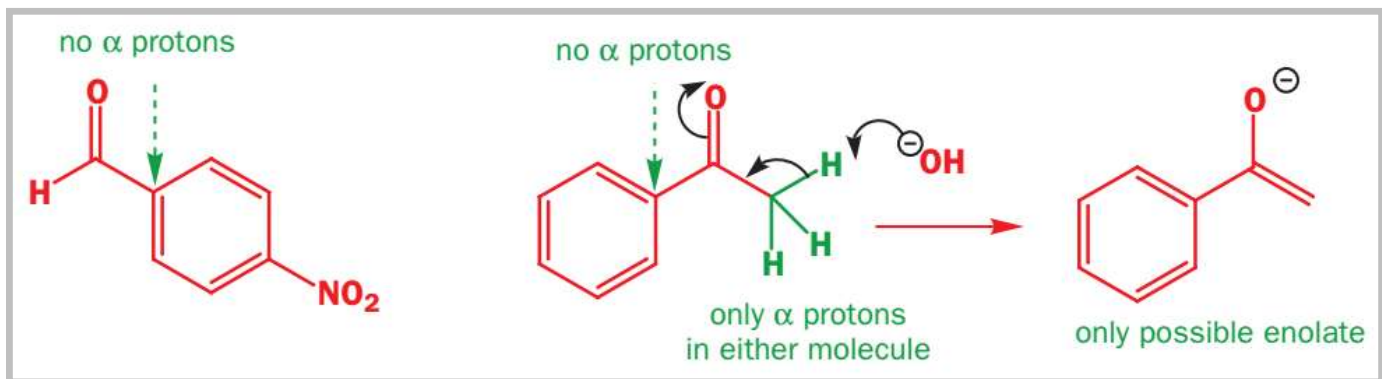


the dehydration step

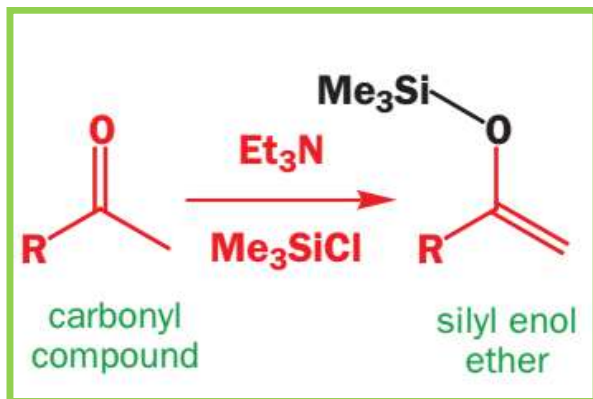


Cross-condensations

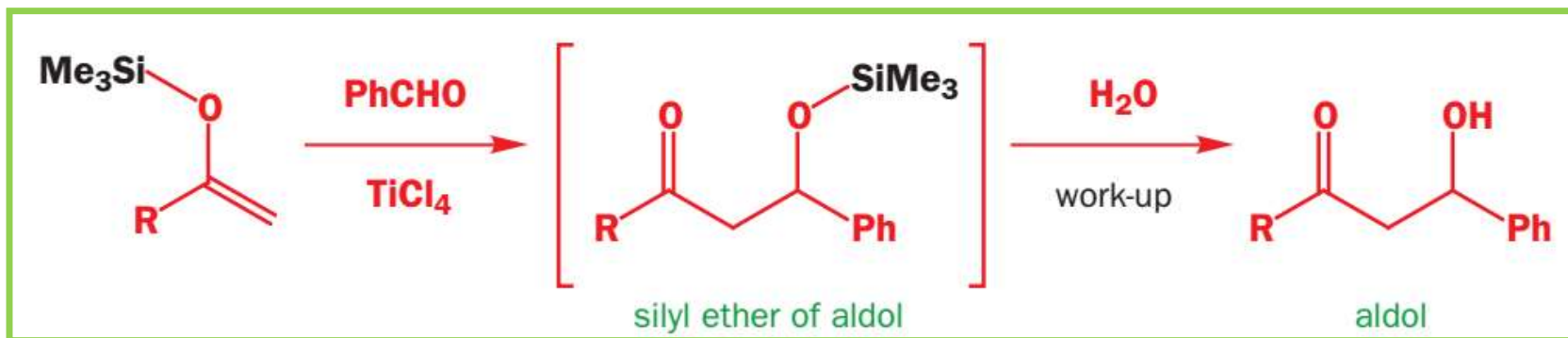




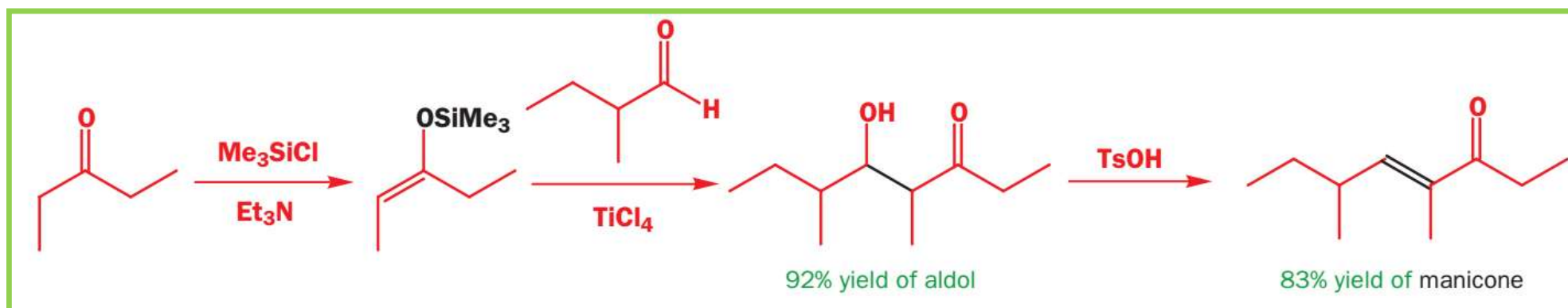
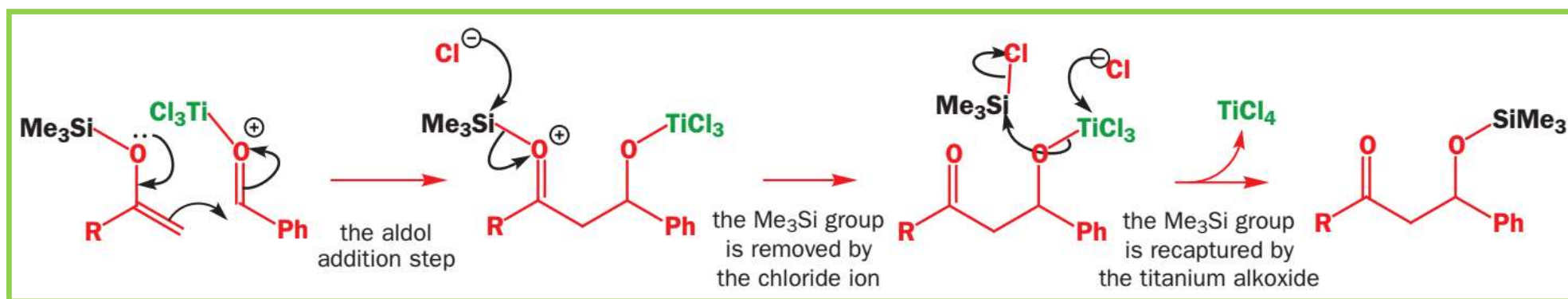
Silyl enol ethers in aldol reactions: **Cross aldol reaction becomes easier**

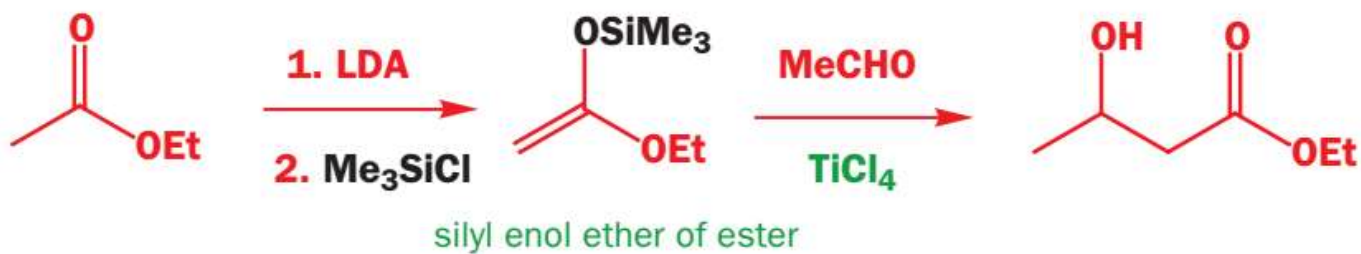
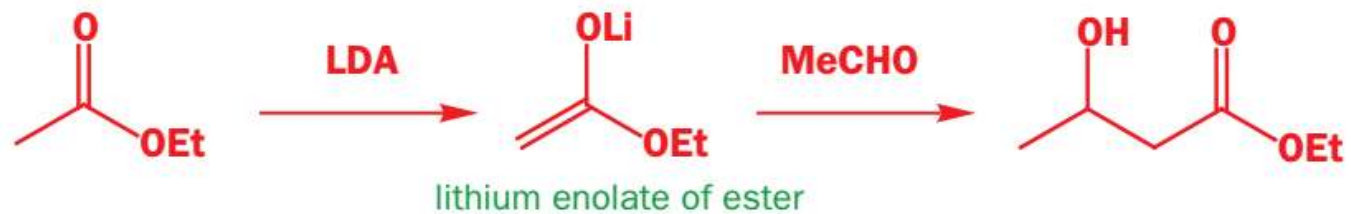


Mukaiyama aldol addition



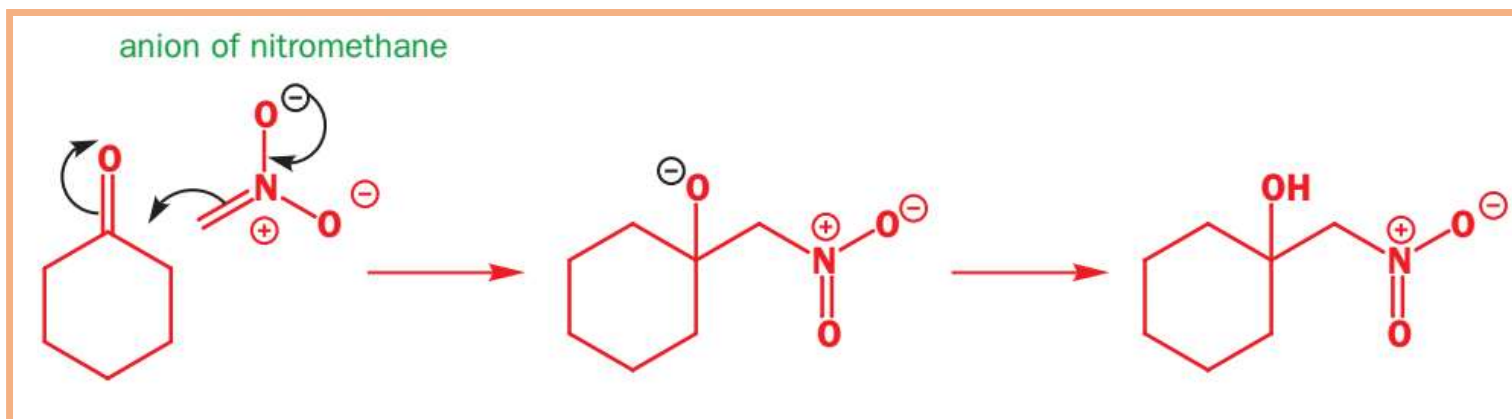
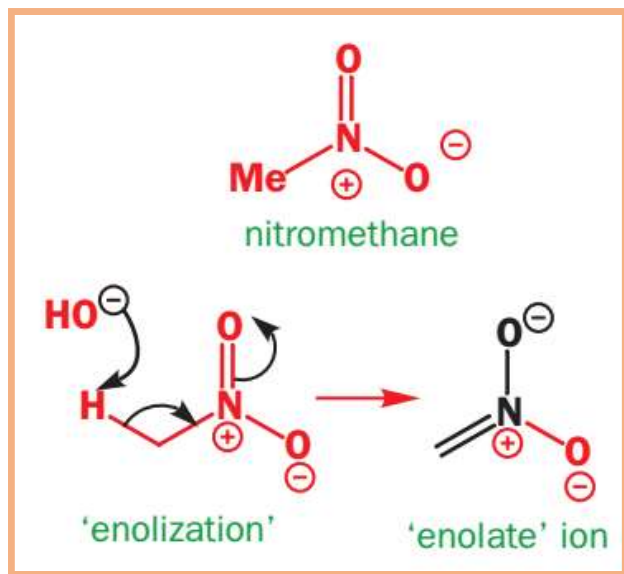
Mechanism: through enol ether

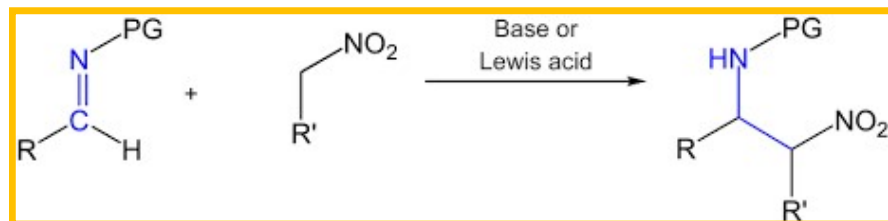




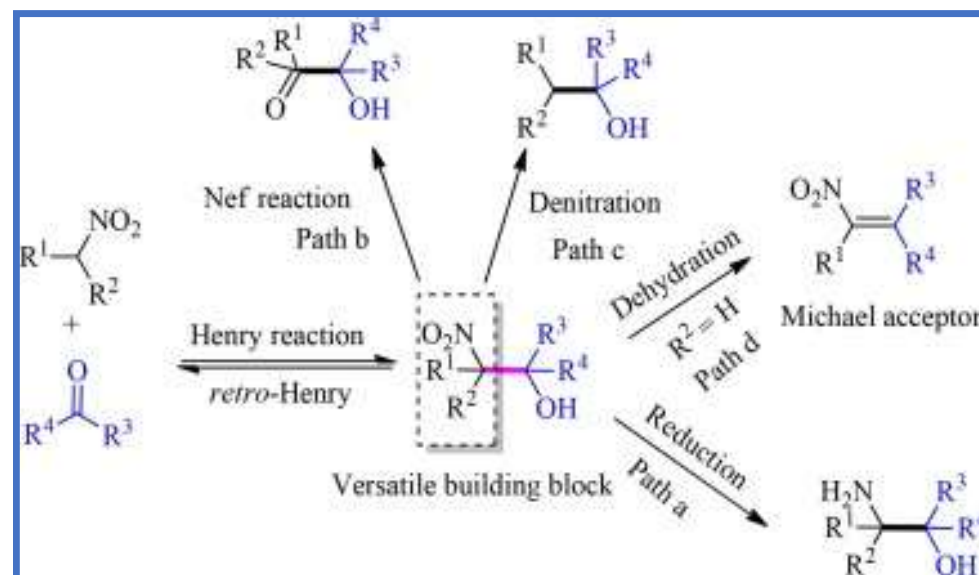
Compounds that can enolize but
that are not electrophilic

Nitro-aldol or Henry Reaction

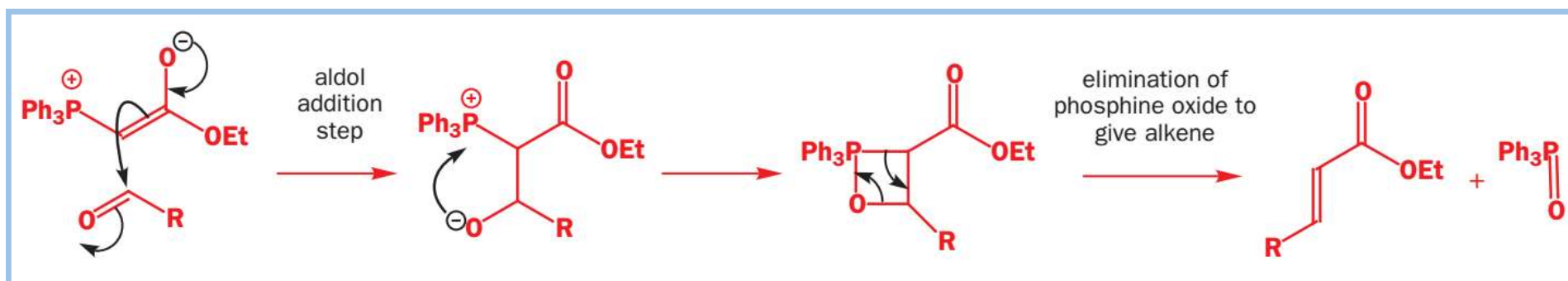
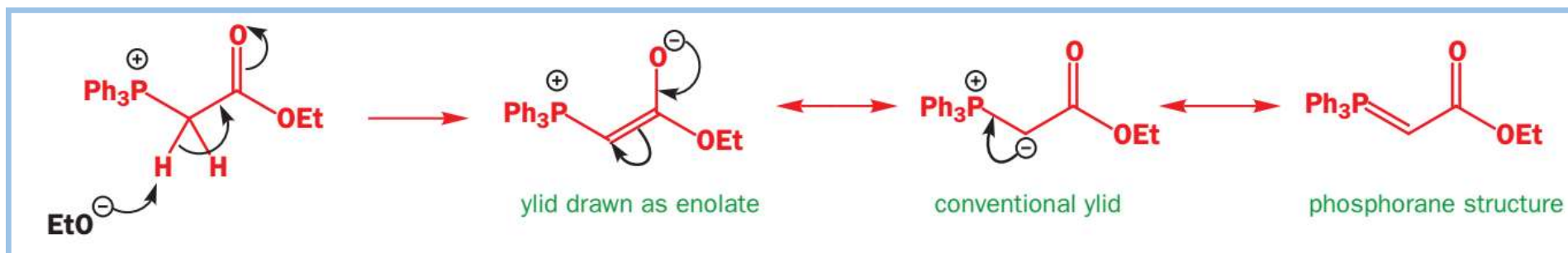
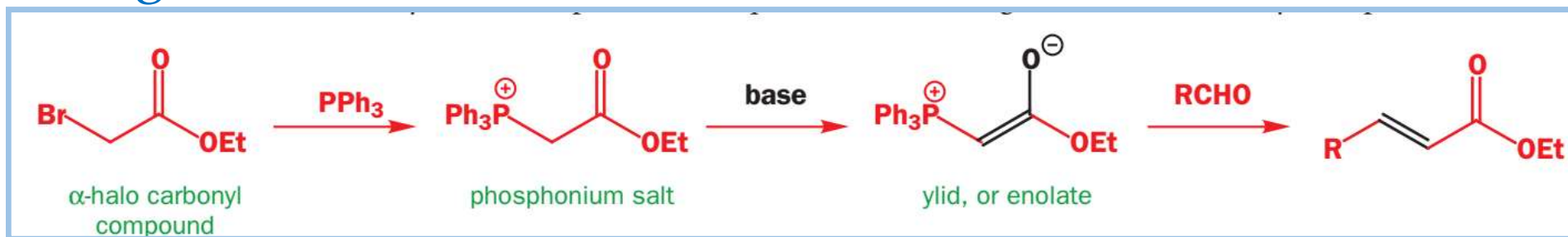


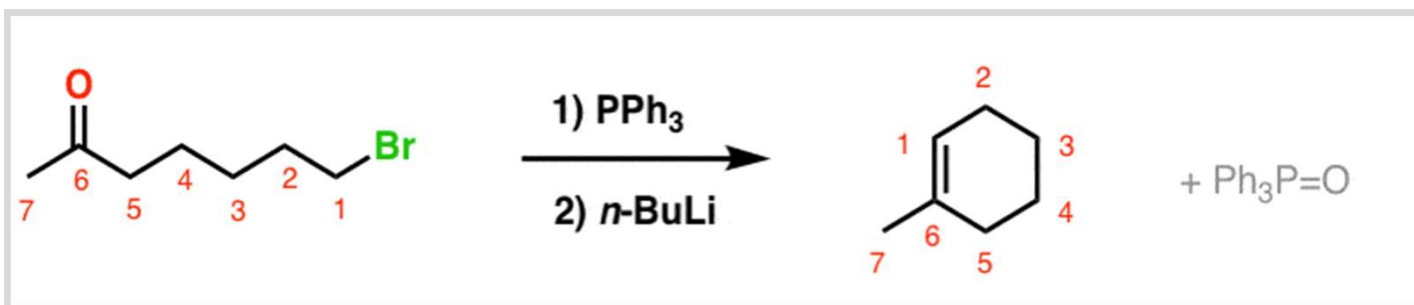
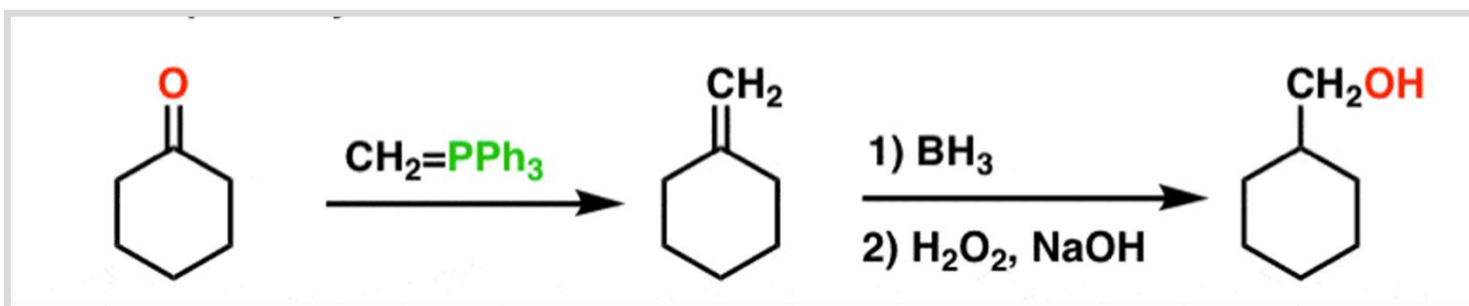
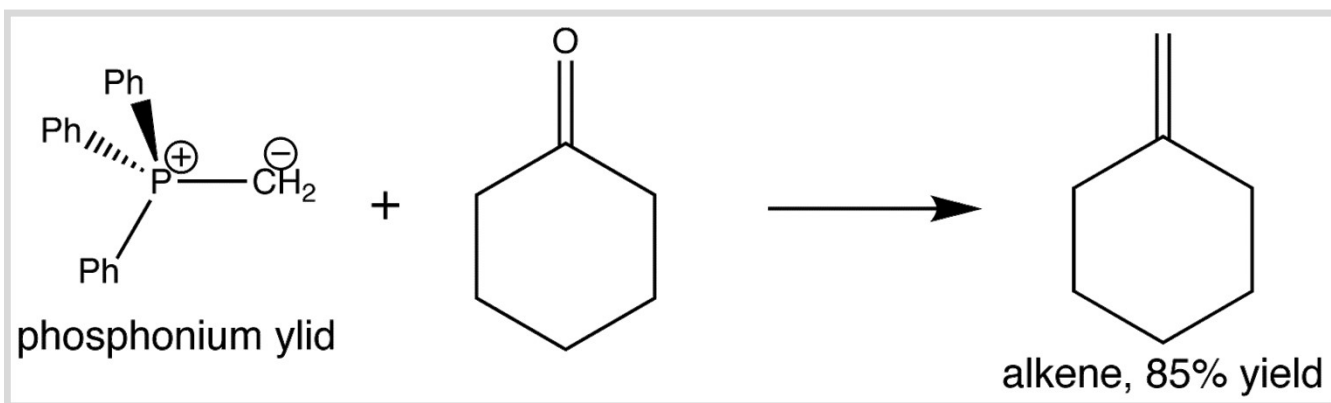


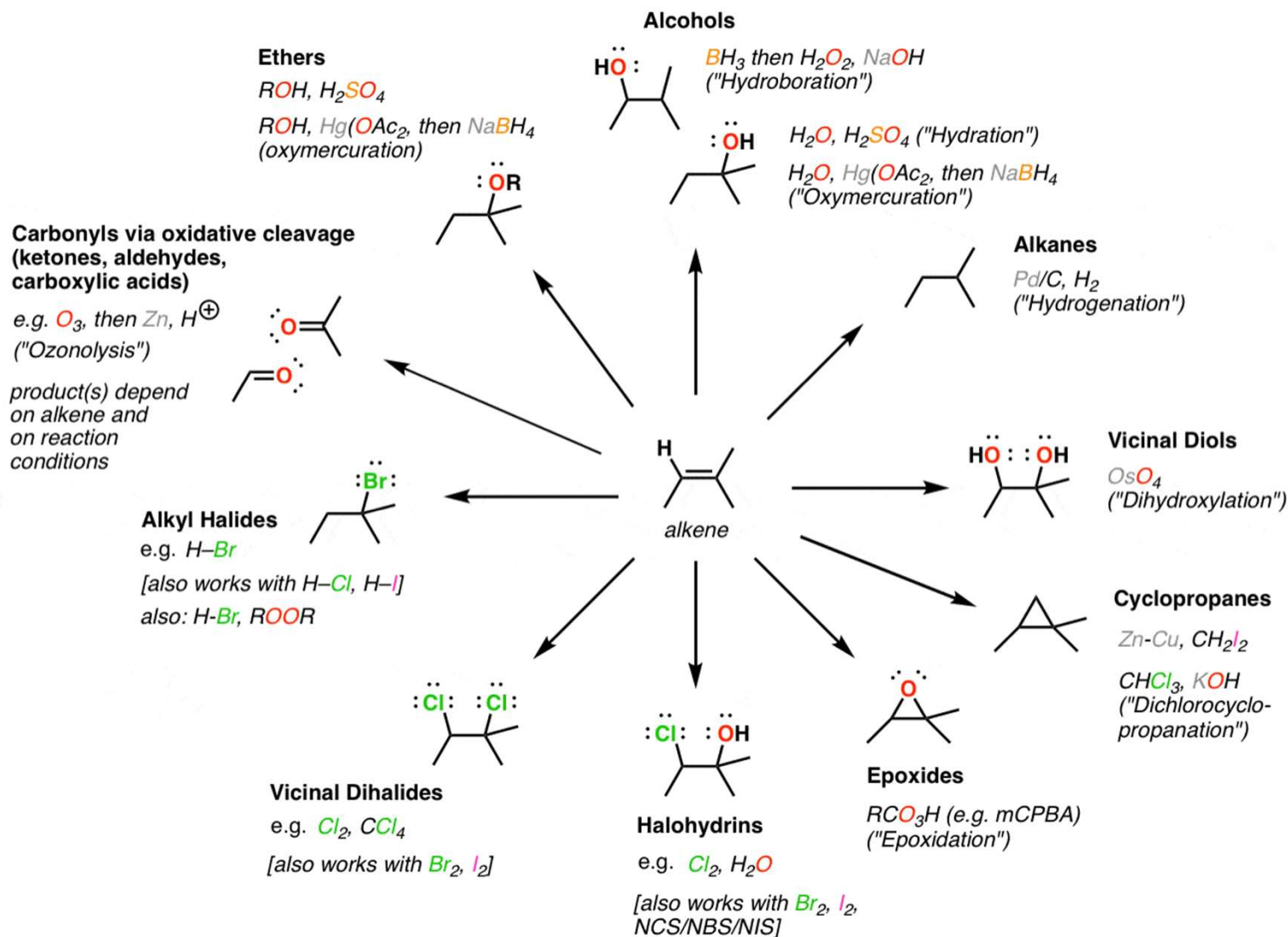
Aza-Henry Reaction

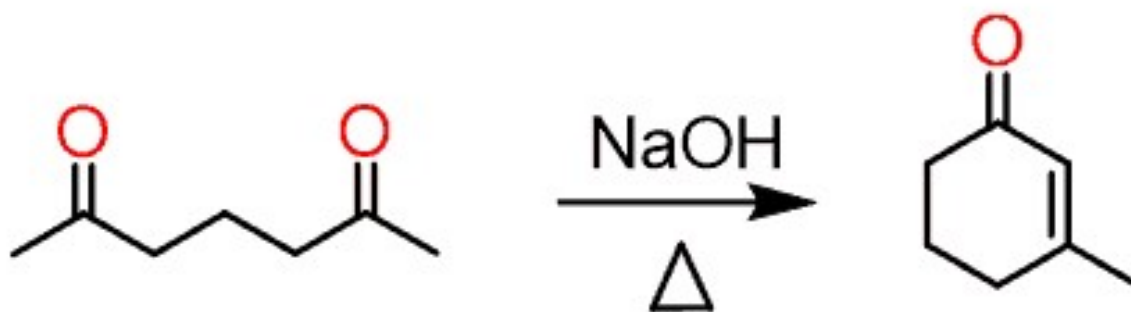
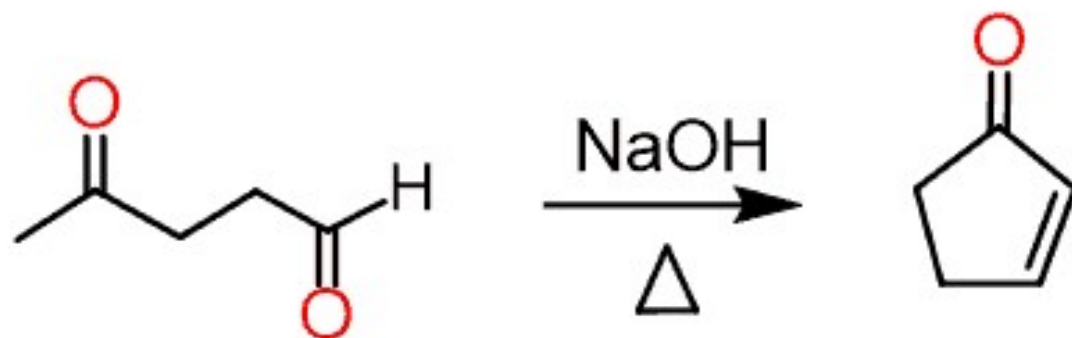


Wittig reaction

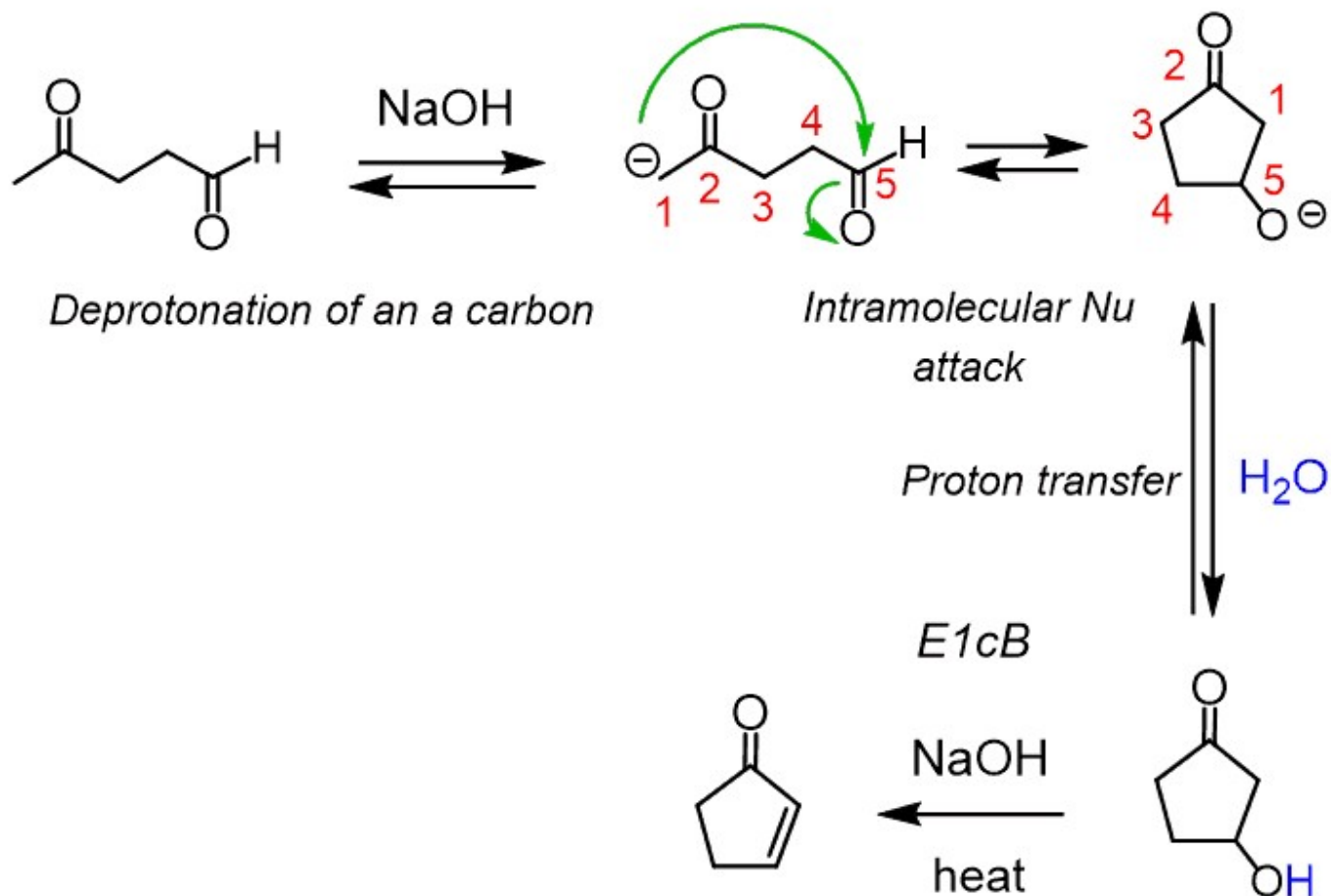




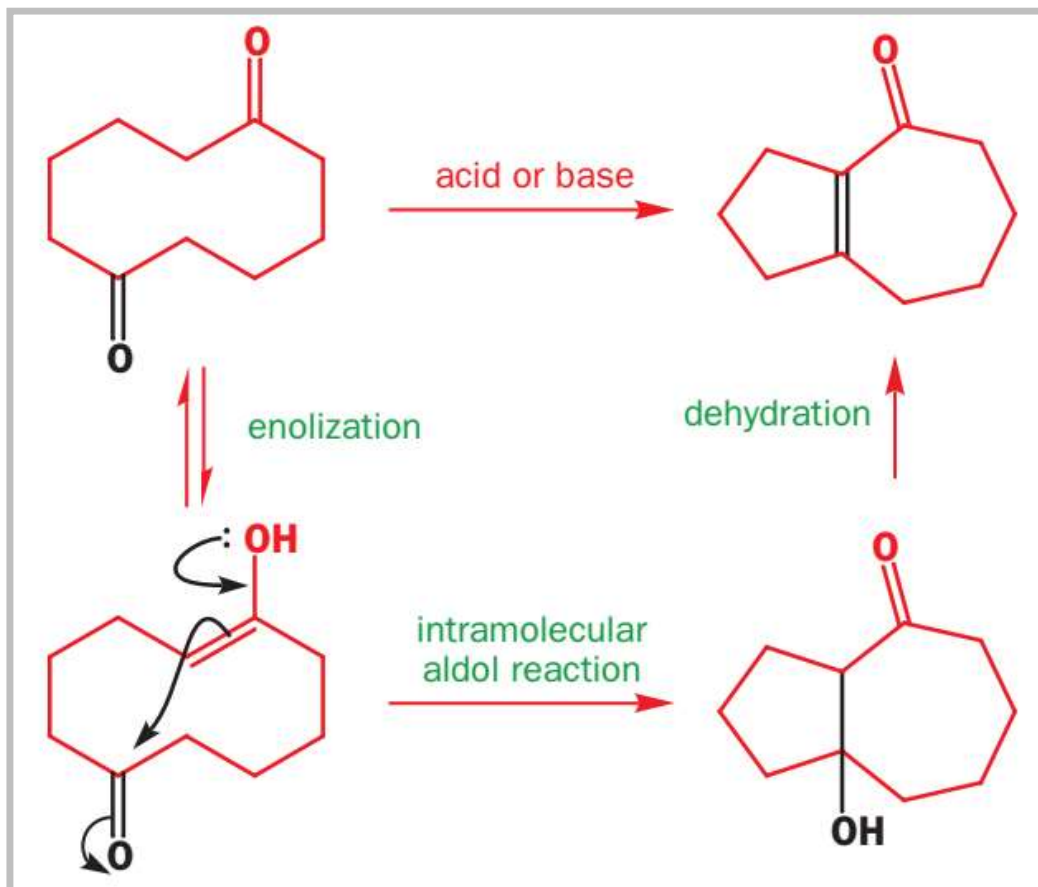


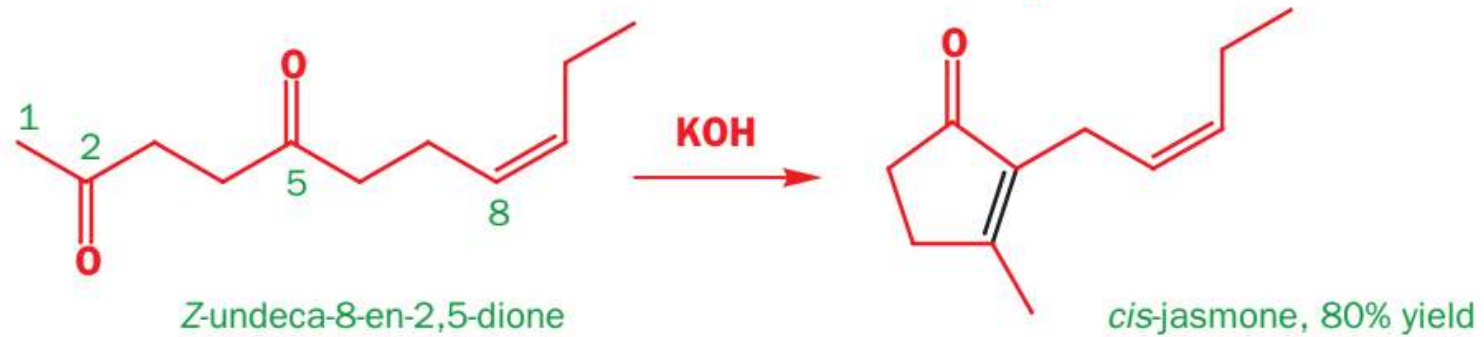
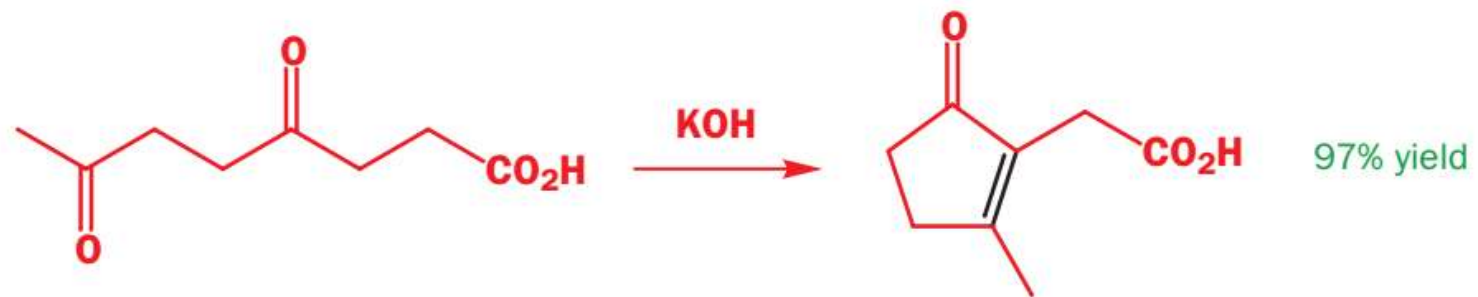


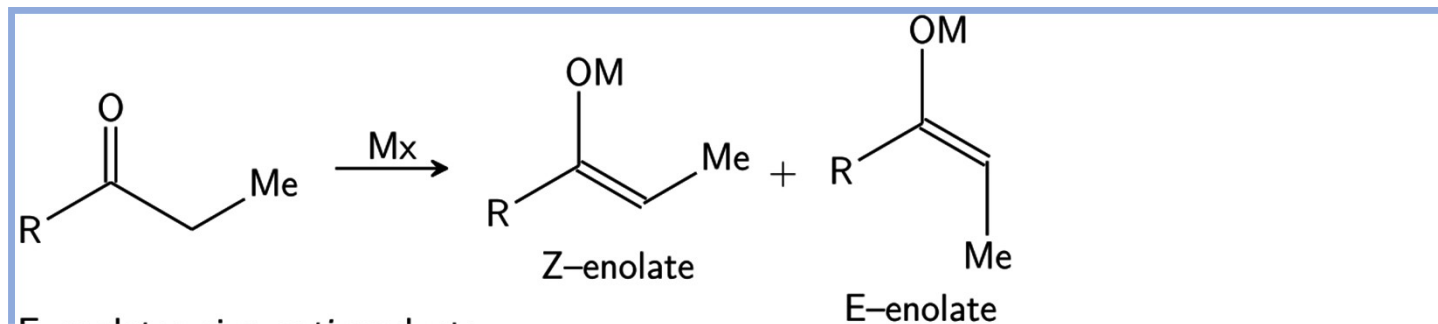
An Intramolecular Aldol Reaction



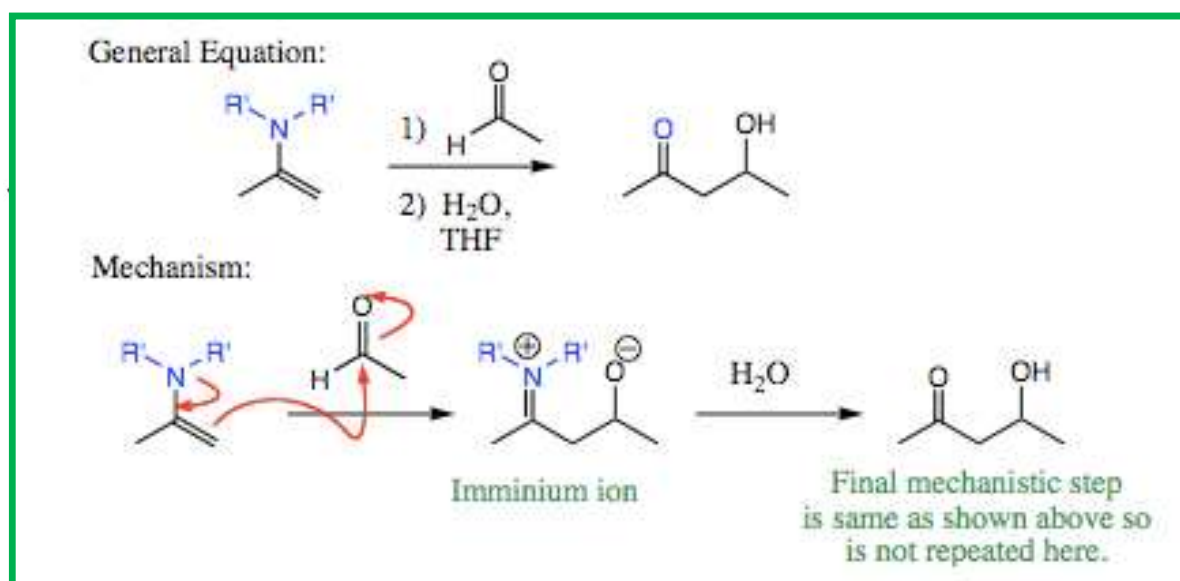
Intramolecular aldol reactions

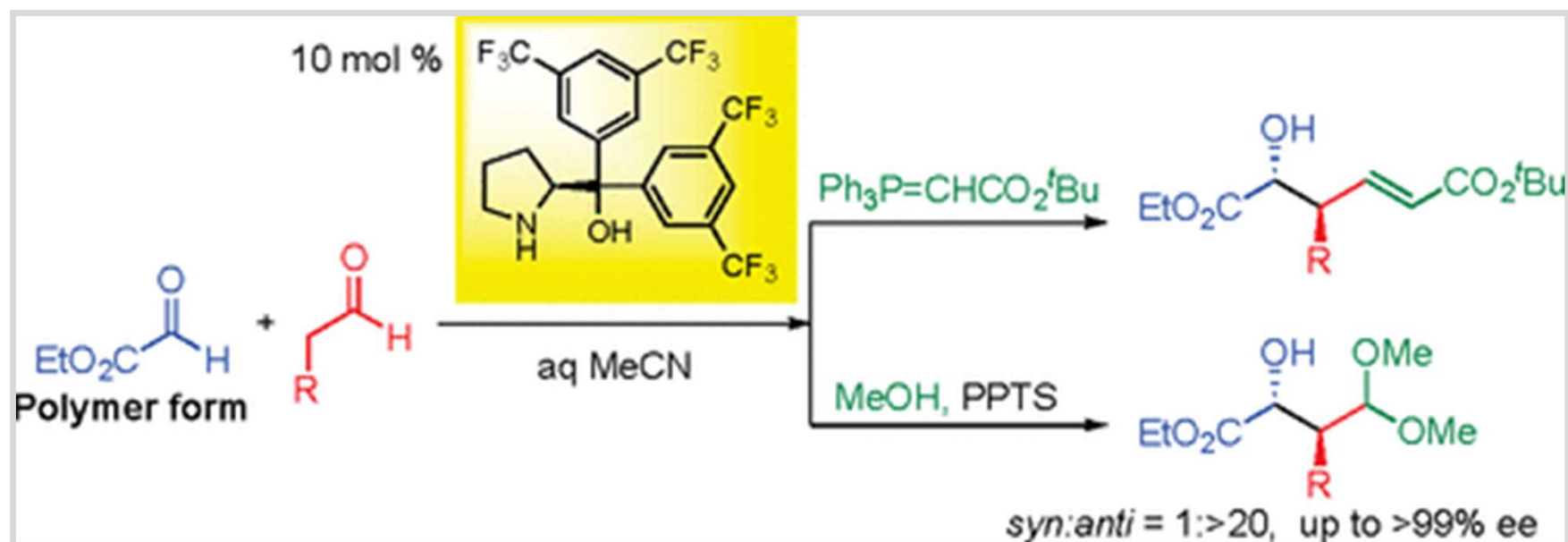


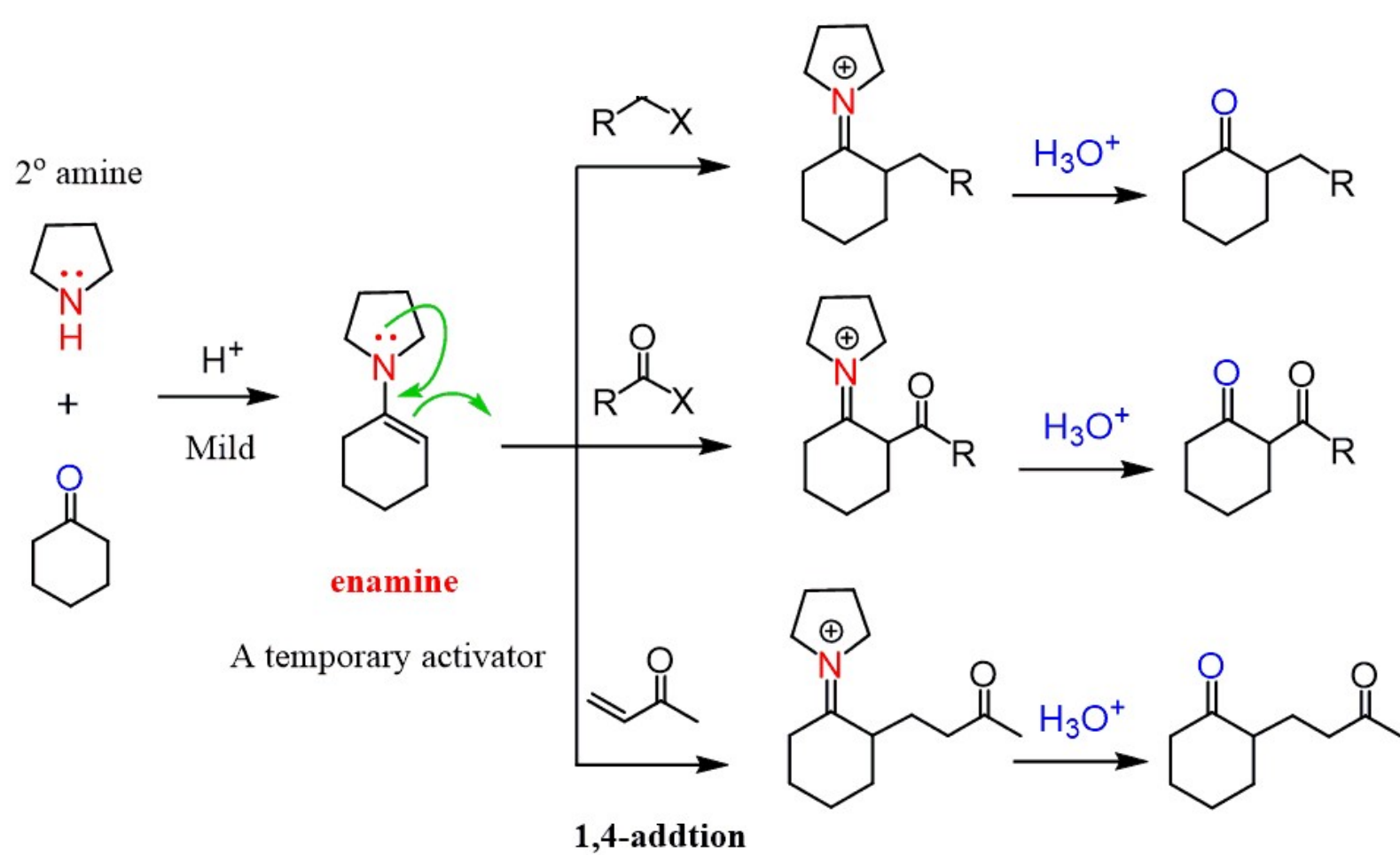


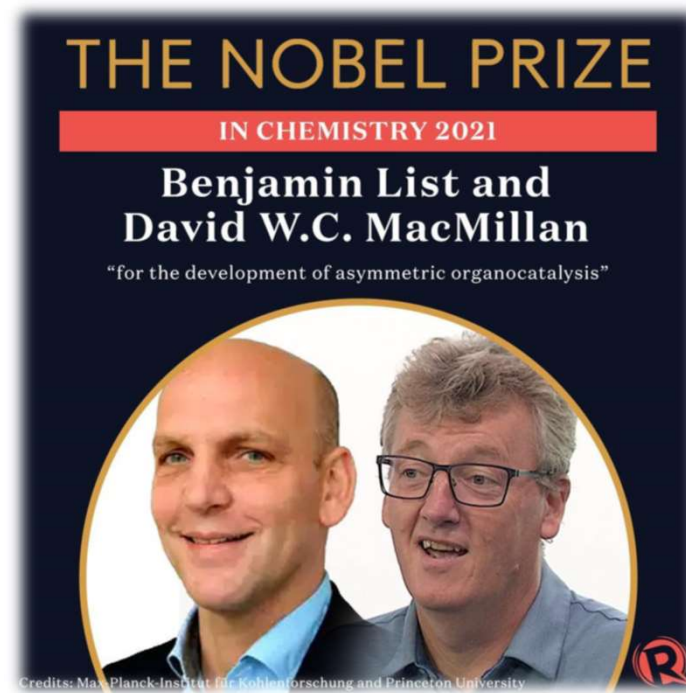
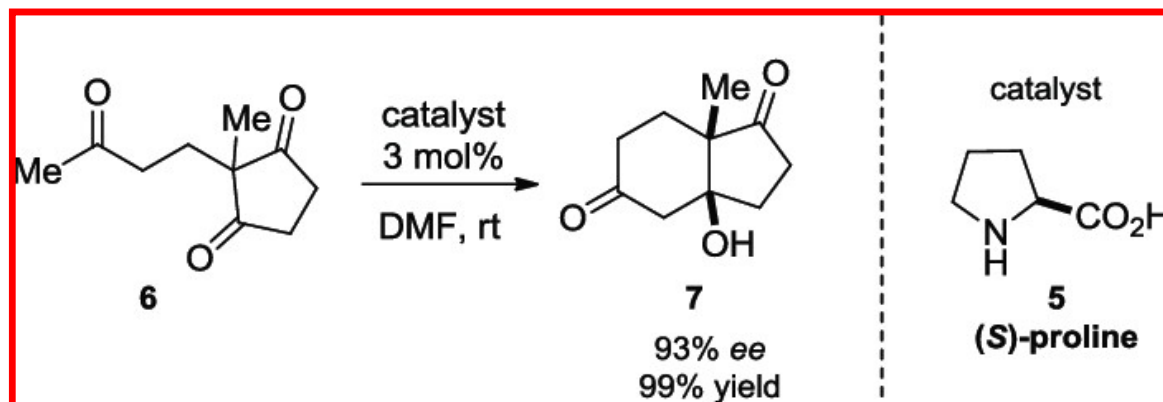
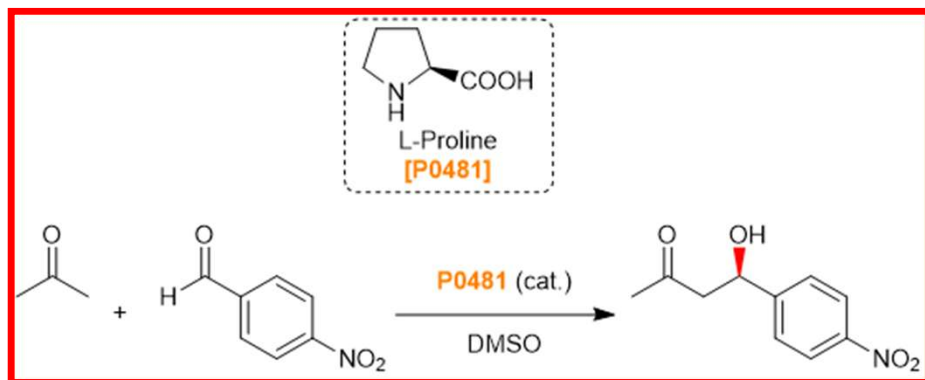


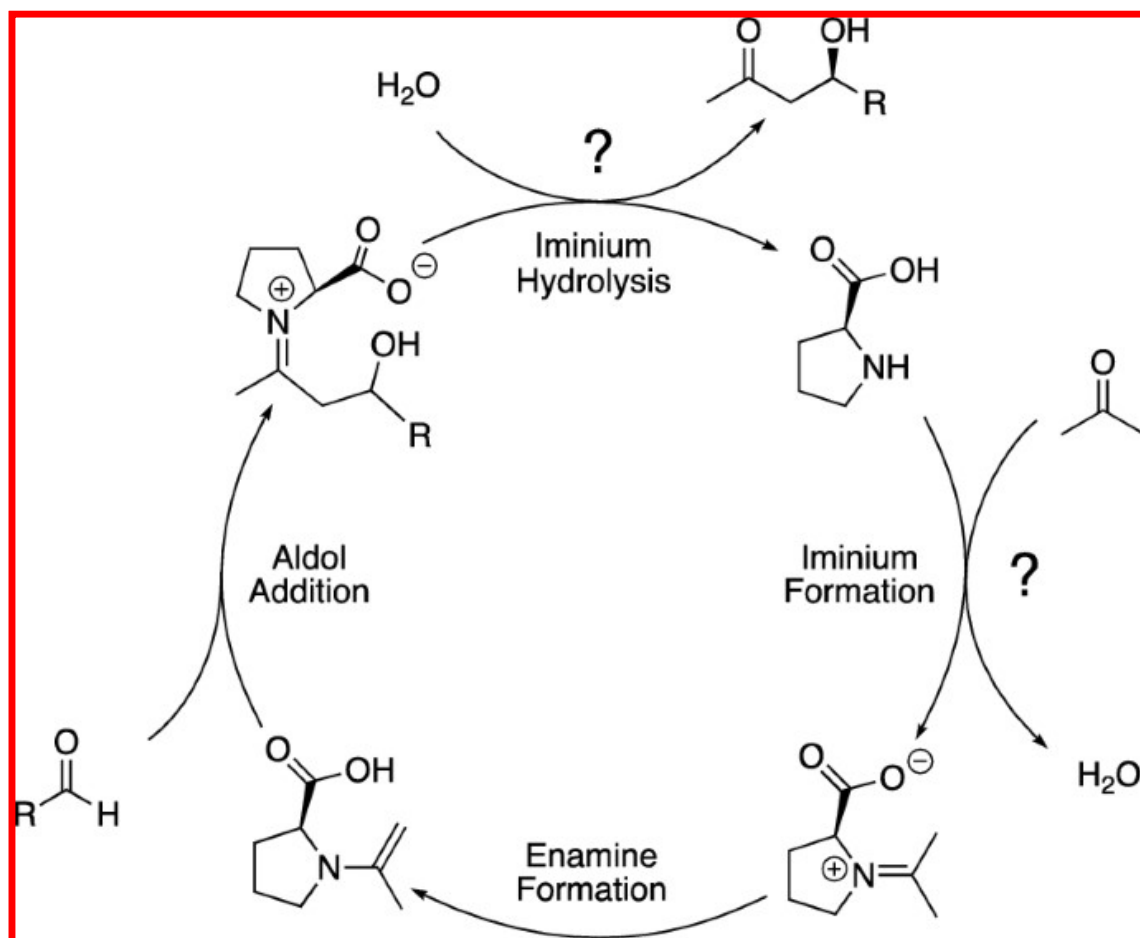
Where strong base or acid can't be used











Mechanism:
through enamine

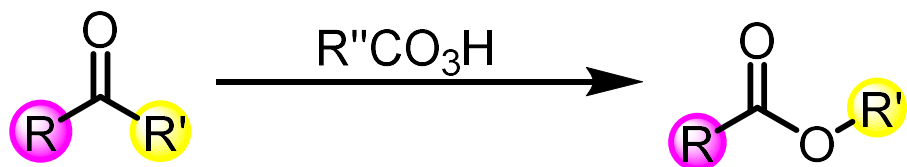
Name Reactions

Content:

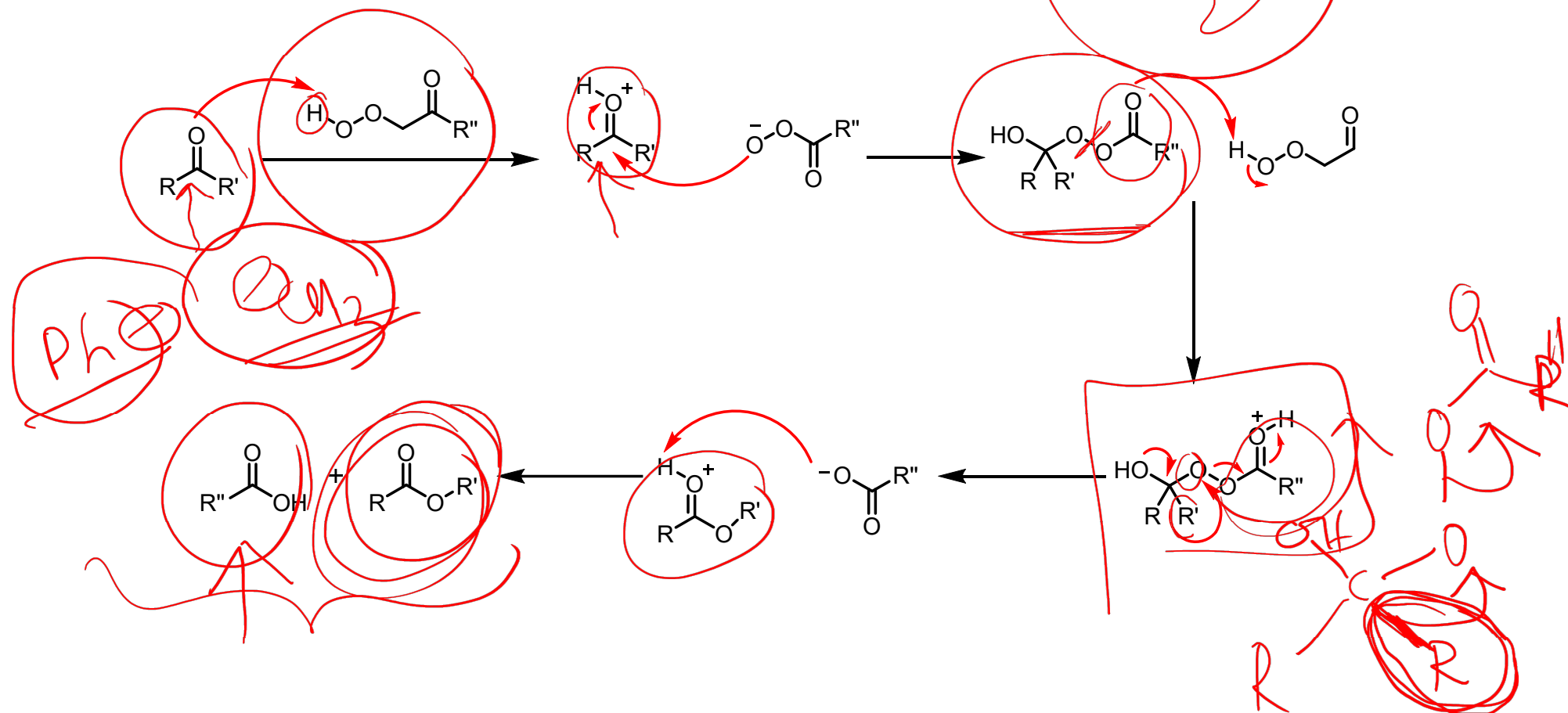
- 1. Baeyer–Villiger oxidation.**
- 2. Beckman rearrangement.**
- 3. Claisen Rearrangement.**
- 4. Ireland–Claisen rearrangement.**
- 5. Cope rearrangement reaction.**
- 6. Friedel-Craft reaction.**
- 7. Wittig Reaction.**
- 8. Fisher indole synthesis.**

Baeyer–Villiger oxidation

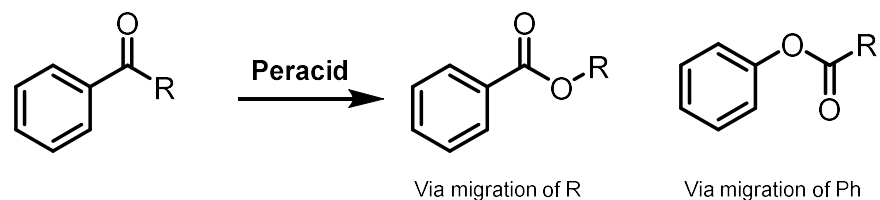
- The Baeyer-Villiger Oxidation is the oxidative cleavage of a carbon-carbon bond adjacent to a carbonyl, and insertion of an oxygen atom which converts ketones to esters and cyclic ketones to lactones
- The Baeyer-Villiger can be carried out with peracids, such as *m*-CPBA, or with peroxide and a Lewis acid.



Mechanism of Baeyer-Villiger oxidation

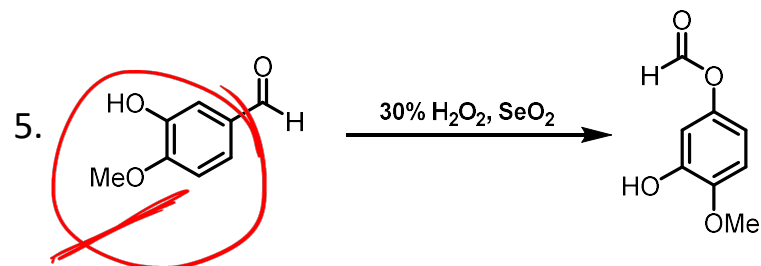
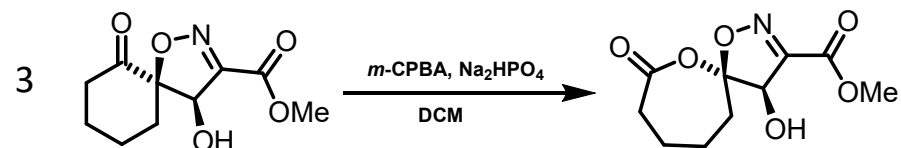
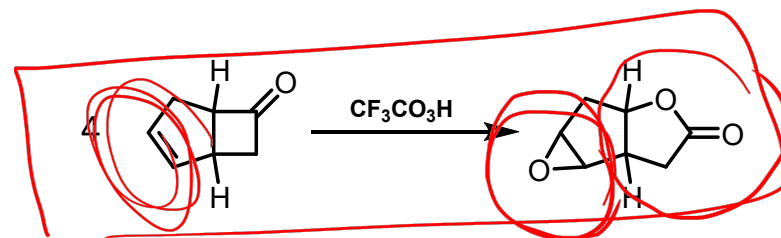
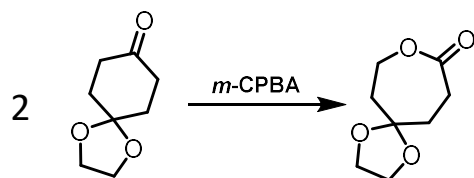
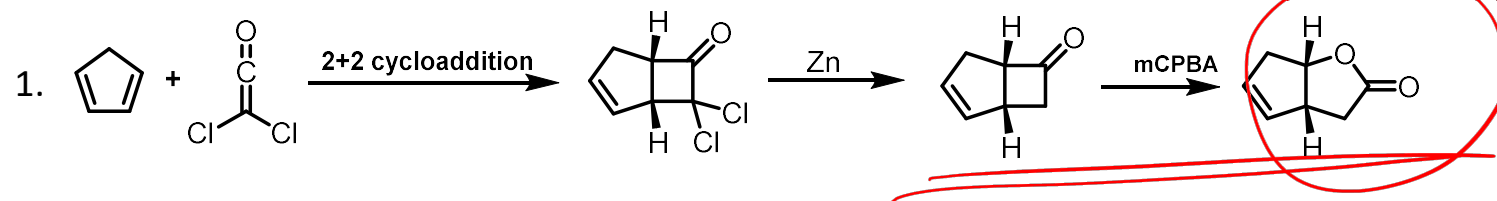


- The reaction is regiospecific in nature and depends on the relative migration ability of the substituents attached to the carbonyl group.
- In unsymmetrical ketones that group migrates which is more electron releasing.
- Migratory aptitude of alkyl groups is in the order $3^\circ > 2^\circ > 1^\circ > \text{CH}_3$
- Electron releasing substituents in the aryl group facilitate migration. The migratory order of aryl groups is $p\text{-anisyl} > p\text{-tolyl} > \text{phenyl} > p\text{-chlorophenyl} > p\text{-nitro-aryl}$.



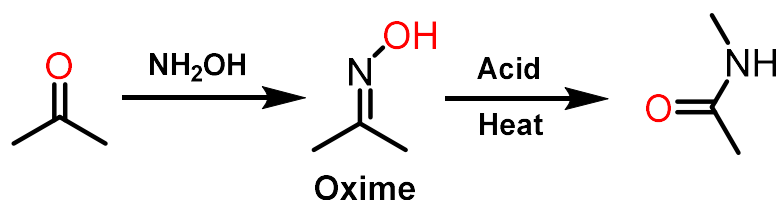
Examples:

mCPBA + NaHCO₃ ?



Beckman rearrangement

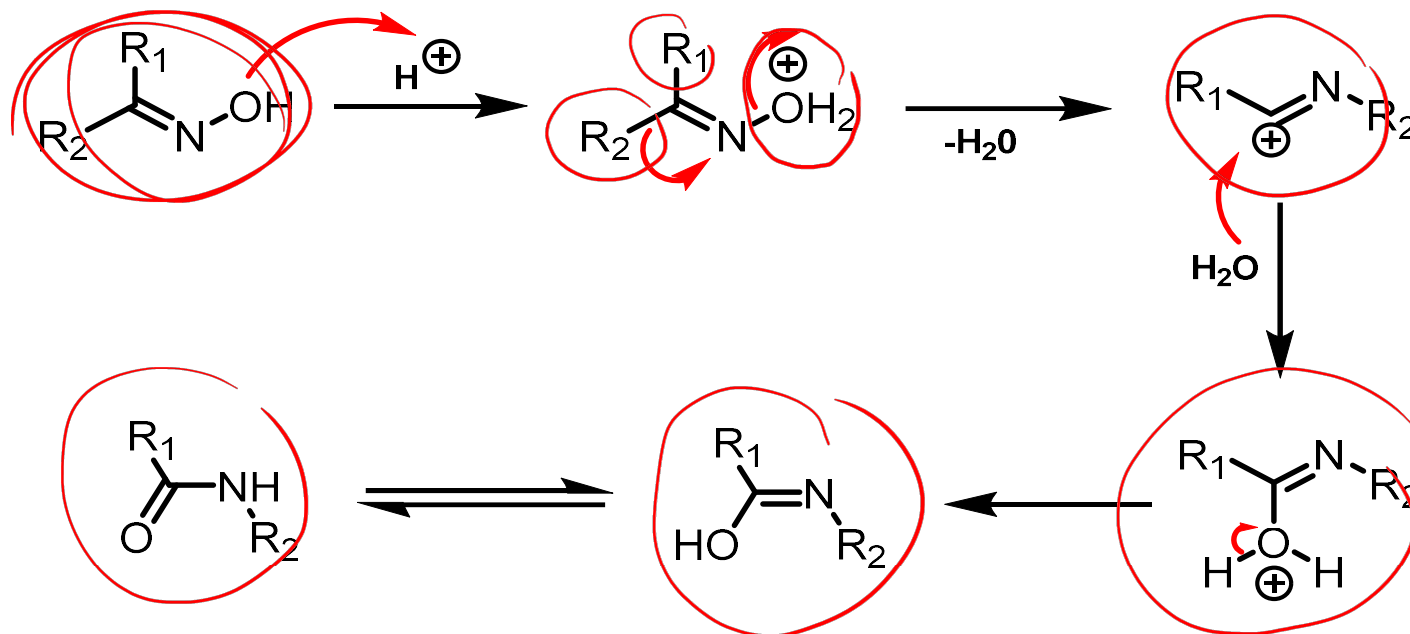
- The Beckmann rearrangement is a rearrangement of an oxime functional group to substituted amides.
- It is often catalysed by acid; however, other reagents have been known to promote the rearrangement. These include tosyl chloride, thionyl chloride, phosphorus pentachloride, etc



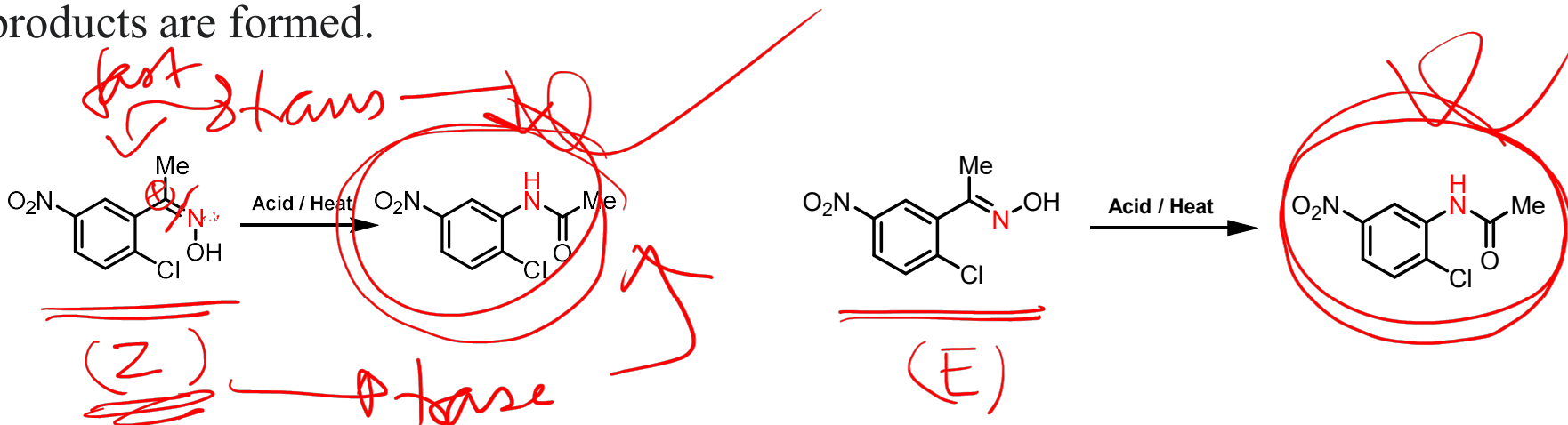
Mechanism:

Steps:

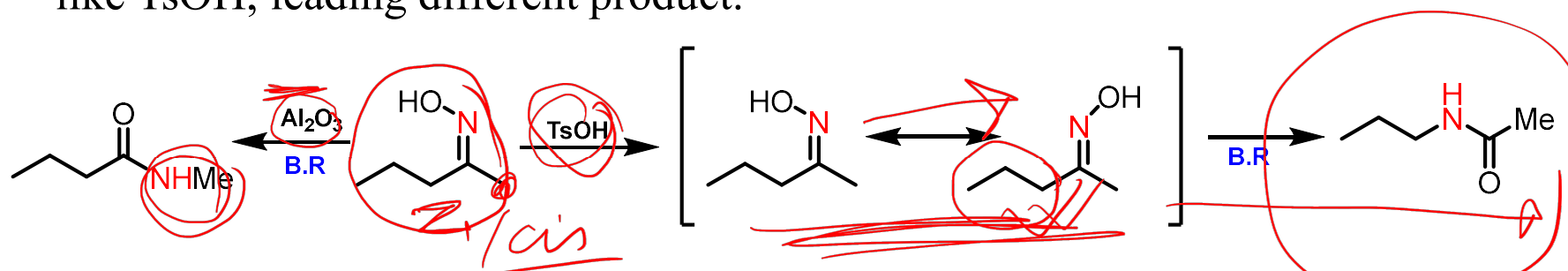
- The oxime oxygen is protonated by acid, This leads to formation of OH_2^+ , which is a much better leaving group than OH
- The C-C bond breaks, migrating to the nitrogen, forming a new C-N bond and displacing the water as a leaving group
- This forms a free carbocation, which is then attacked by water
- The positively charged oxygen is deprotonated by base to give the tautomer of the amide.
- Then tautomerism occur to give final product.



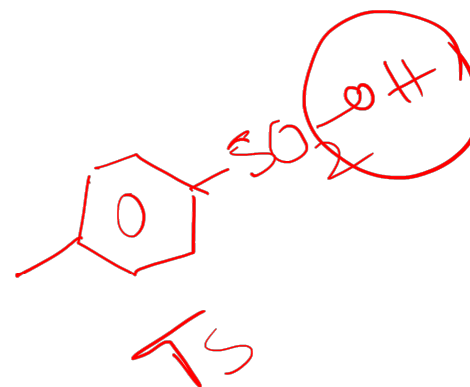
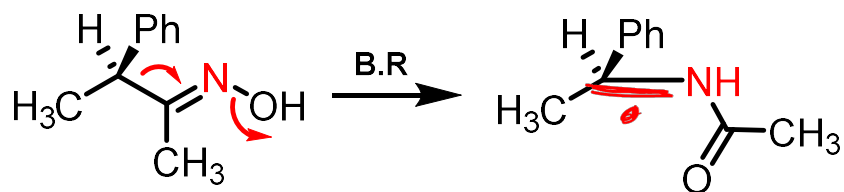
- The rearrangement occurs stereospecifically for ketoximes and N-chloro/N-fluoro imines, with the migrating group being anti-periplanar to the leaving group on the nitrogen.
- The rearrangement of aldoximes occurs with stereospecificity in the gas phase and without stereospecificity in the solution phase.
- Nitron rearrangement also occurs without stereospecificity; the regioisomer formed has the amide nitrogen substituted with the group possessing the greatest migratory aptitude.
- Rearrangement is strictly intermolecular S_N2 in nature, hence no crossover products are formed.



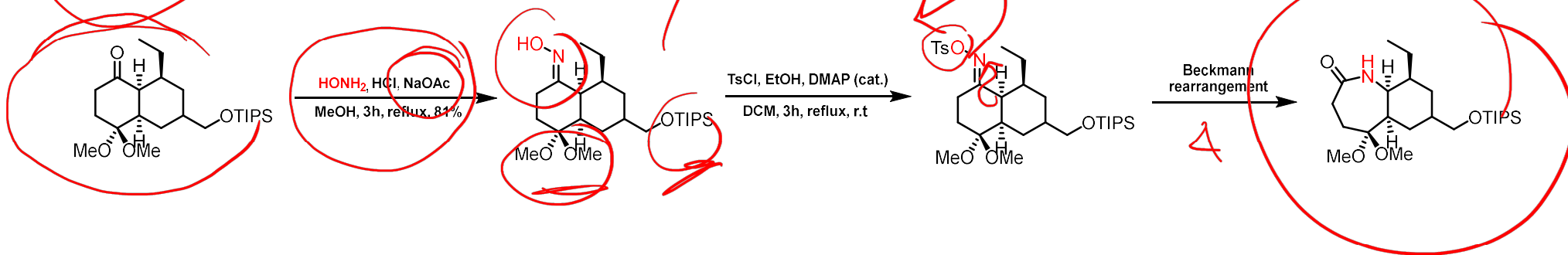
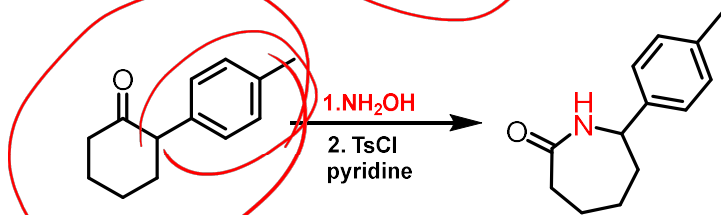
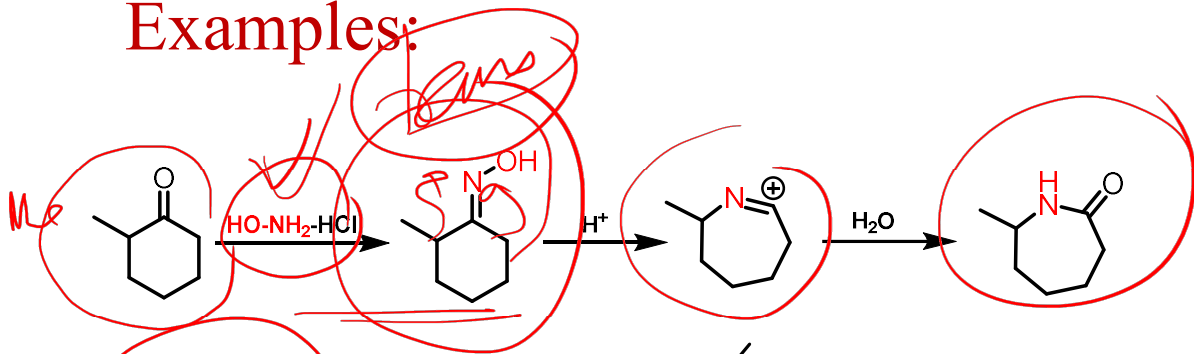
- Some reagents show oxime isomerization faster than Beckmann rearrangement like TsOH, leading to different product.



- If migrating group is chiral it will retain its stereochemistry during migration.

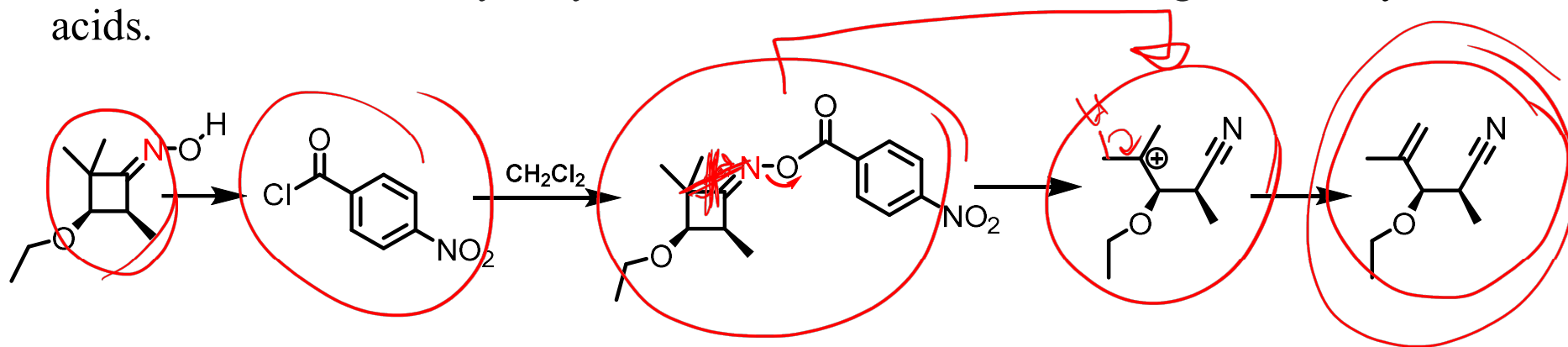


Examples:



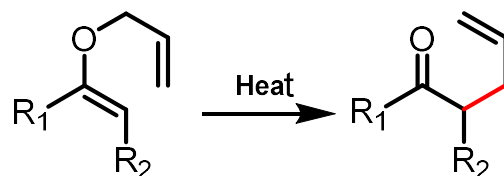
Beckmann fragmentation :

- The Beckmann fragmentation is another reaction that often competes with the rearrangement, though careful selection of promoting reagent and solvent conditions can favor the formation of one over the other, sometimes giving almost exclusively one product.
- When the group α to the oxime is capable of stabilizing carbocation formation, the fragmentation becomes a viable reaction pathway.
- The reaction generates a nitrile and a carbocation, which is quickly intercepted to form a variety of products
- The nitrile can also be hydrolyzed under reaction conditions to give carboxylic acids.

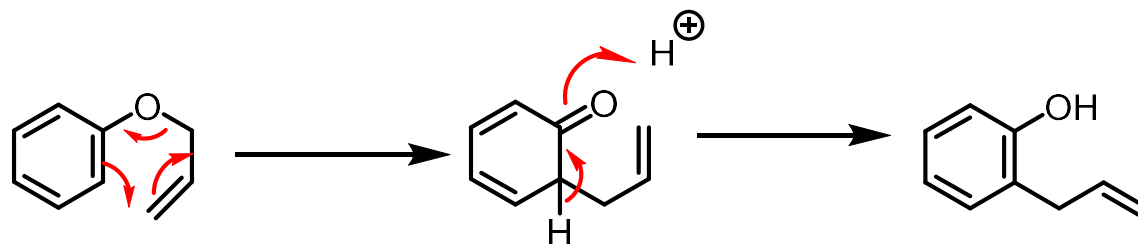


Claisen Rearrangement

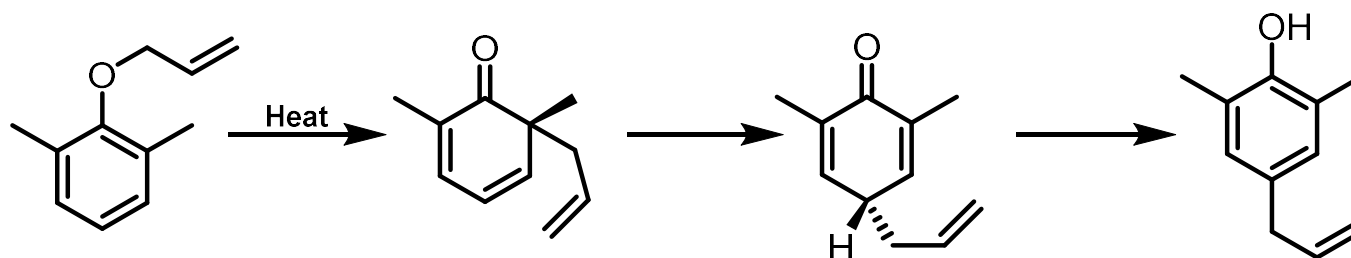
- Claisen rearrangement reaction is [3+3] sigmatropic rearrangement reaction.
- Rearrangement of allyl vinyl ethers to the corresponding γ,δ -unsaturated carbonyl compounds.



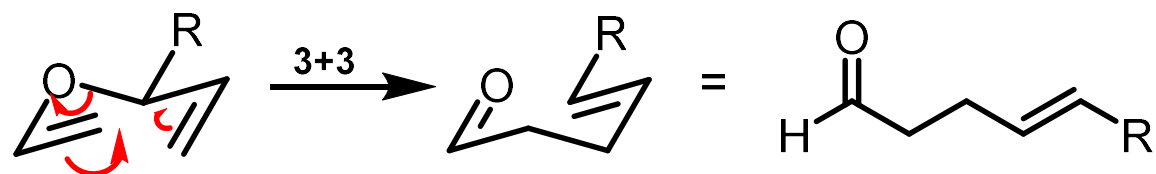
- This is a one-step mechanism without ionic intermediates or any charges.



- Allyl group also appears to attack the *para* position on the ring.

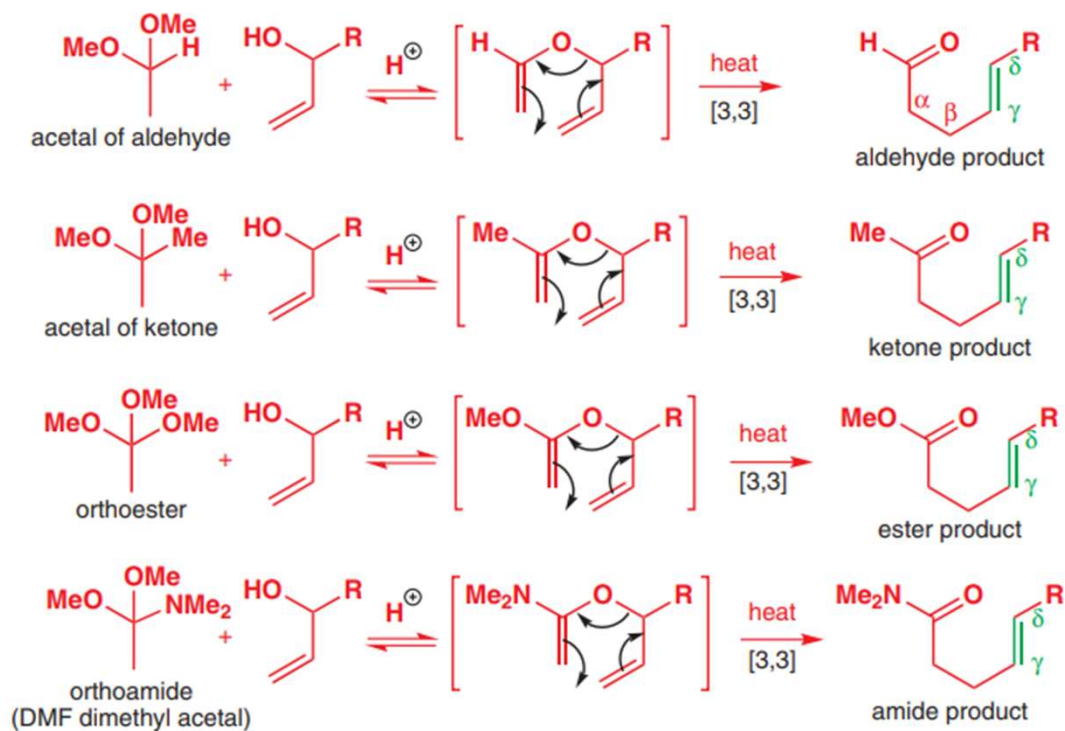


- Stereochemistry may arise if there is a substituent on the saturated carbon atom next to the oxygen atom.
- The resulting double bond strongly favors the trans (*E*) geometry
- Because substituent prefers an equatorial position on the chair transition state.

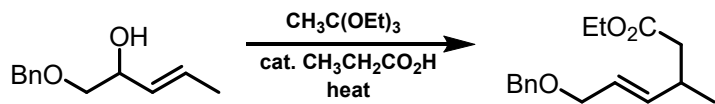
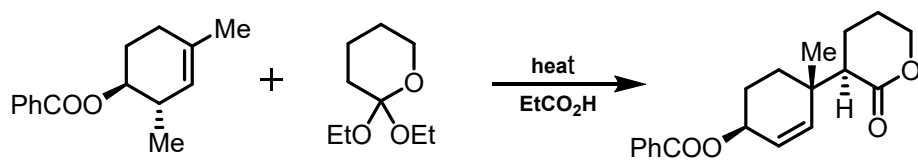
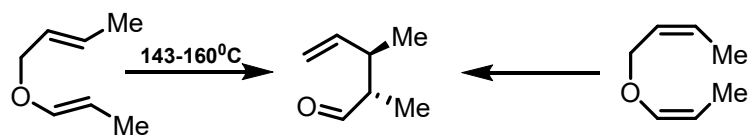
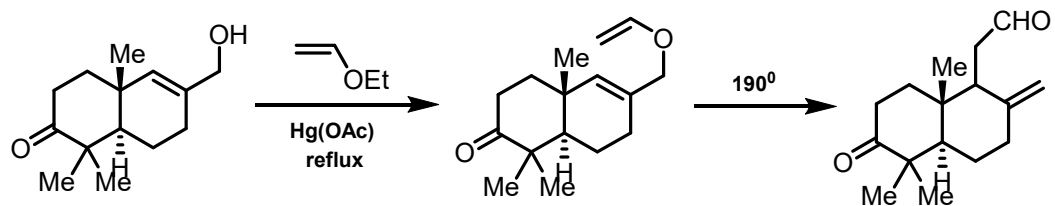


Significance:

- General synthesis of γ,δ -unsaturated carbonyl compounds.

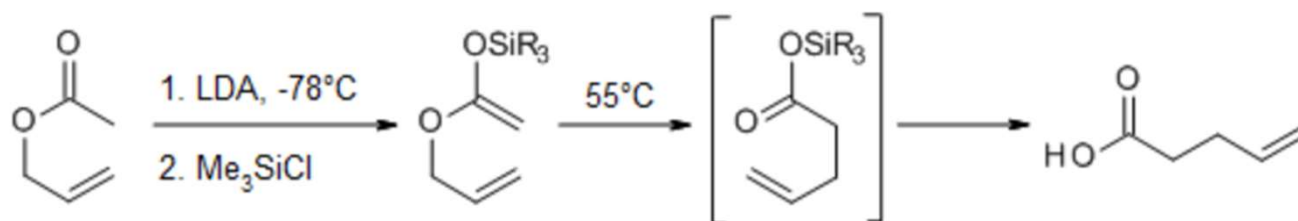


Examples:

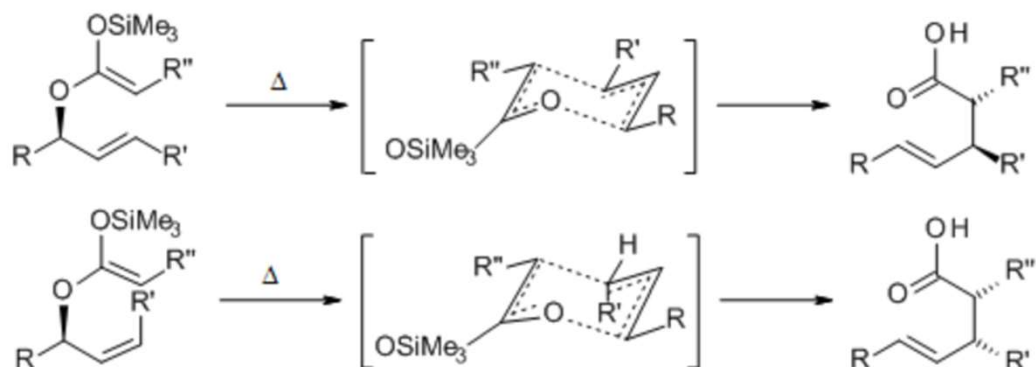


Ireland–Claisen rearrangement

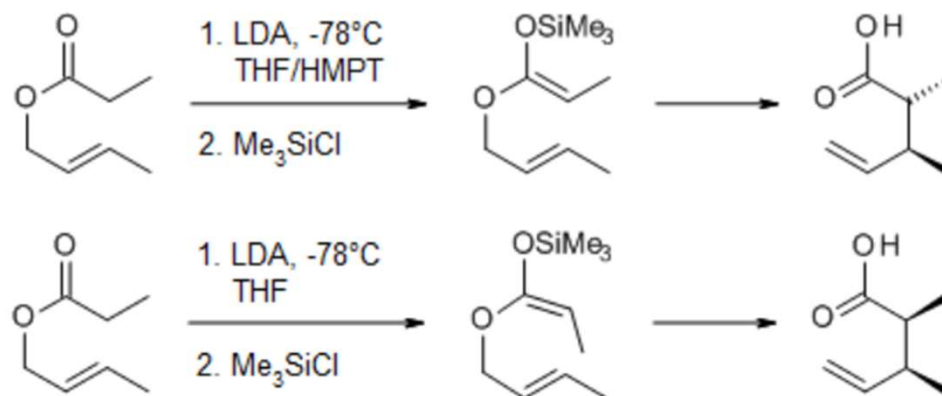
- Variant of the Claisen Rearrangement.
- The Ireland–Claisen rearrangement is the reaction of an allylic carboxylate with a strong base
- It give a γ,δ -unsaturated carboxylic acid.
- The rearrangement proceeds via silylketene acetal.



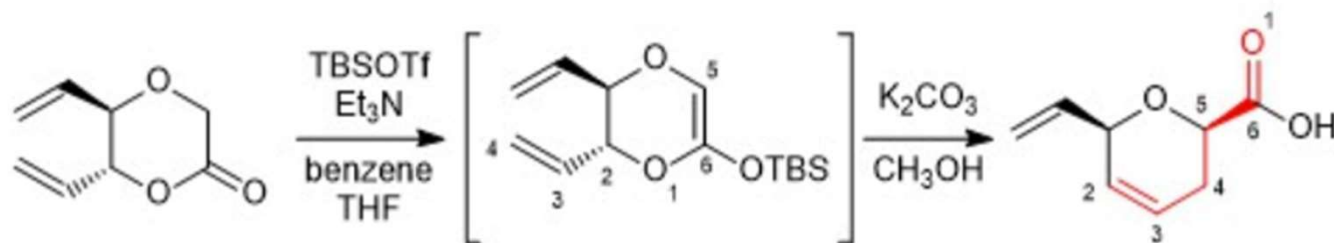
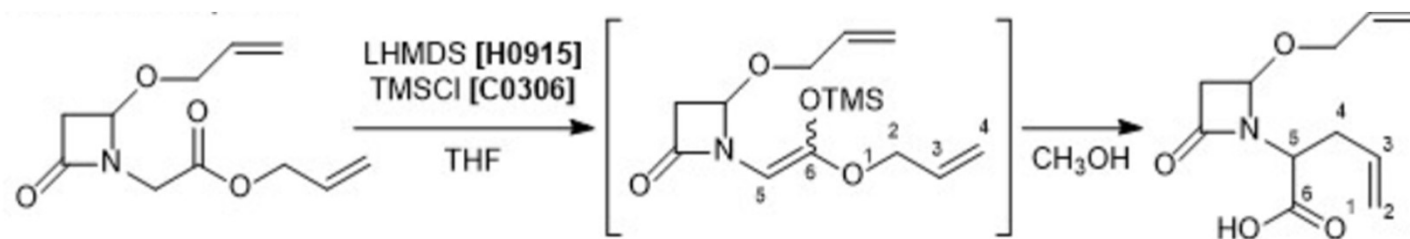
- Ireland modification proceeds with a high degree of stereoselectivity:



- An advantage of the Ireland-Claisen Rearrangement is the option of controlling the enolate geometry through the judicious choice of solvent.

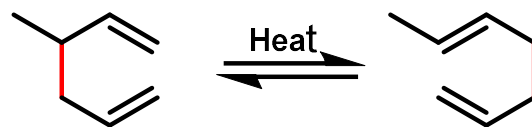


Examples:

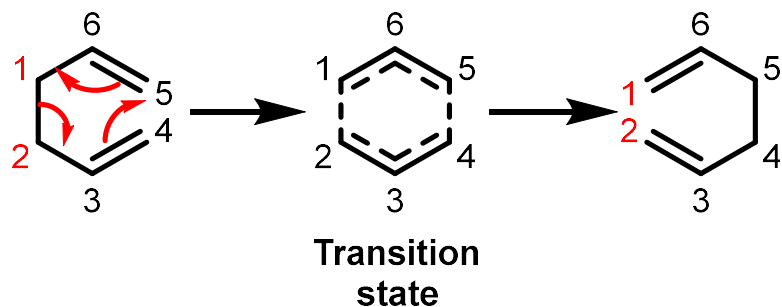


Cope rearrangement reaction

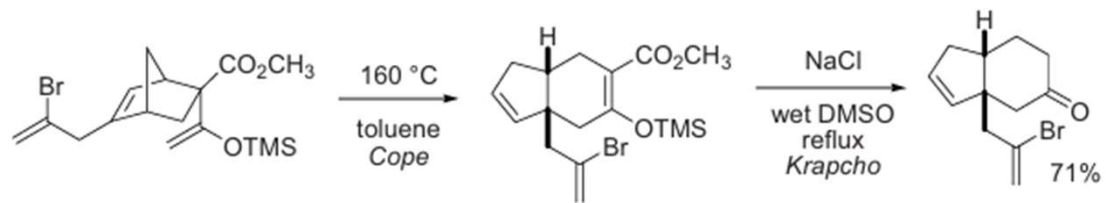
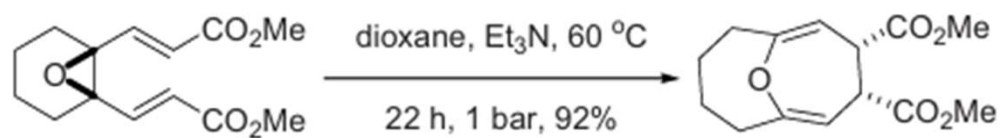
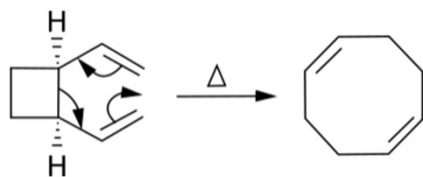
The Cope rearrangement is a [3,3]-sigmatropic rearrangement of 1,5-dienes.



The Cope rearrangement occurs in a single step with a concerted transition state.

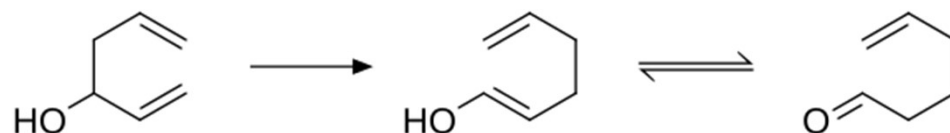


Examples:

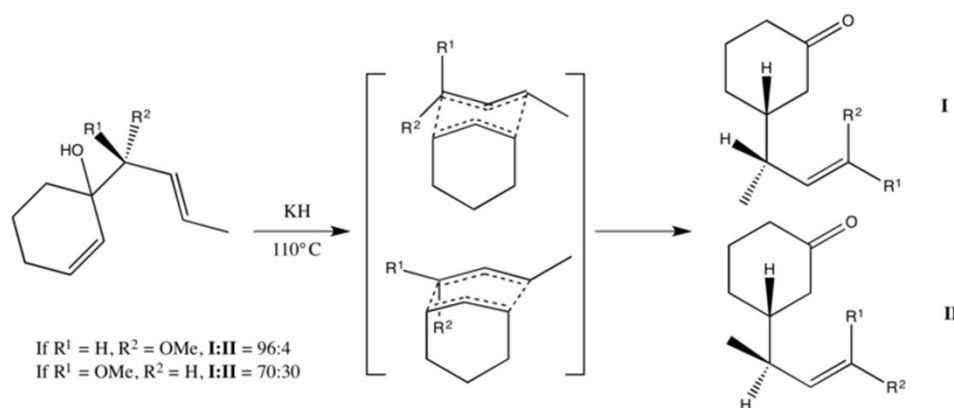


Oxy cope rearrangement

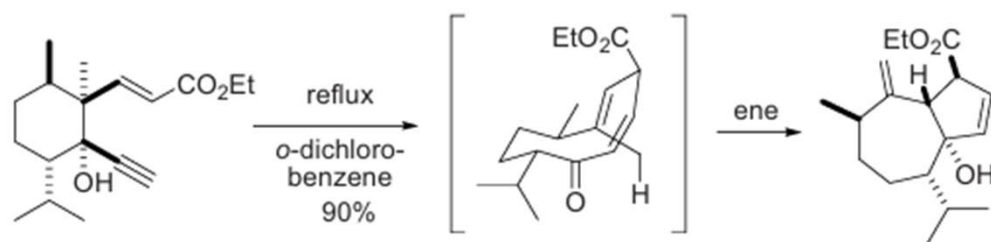
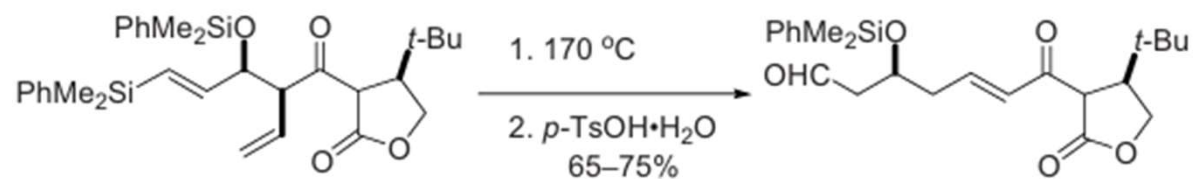
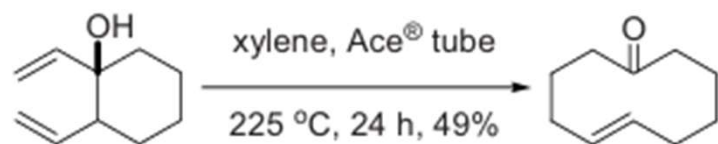
- In the oxy-Cope rearrangement, a hydroxyl group is added at C3 forming an enone after keto-enol tautomerism of the intermediate enol.



Mechanism:

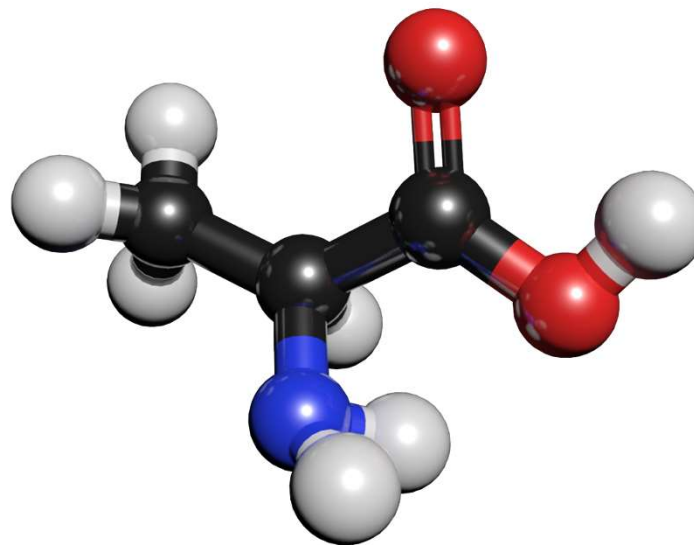
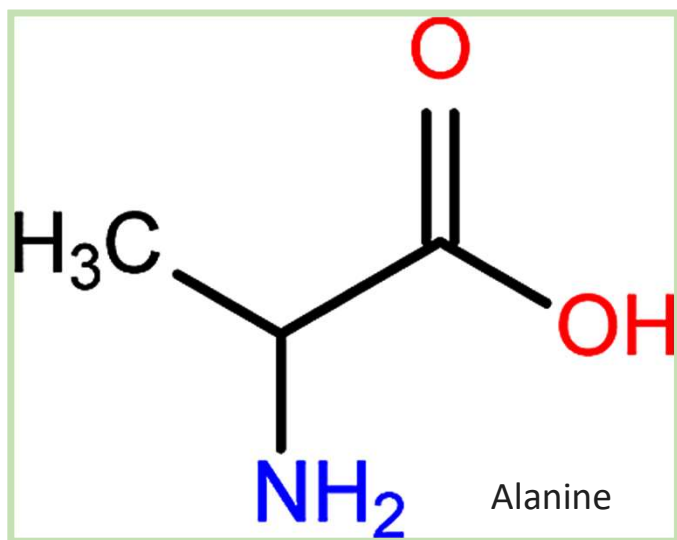


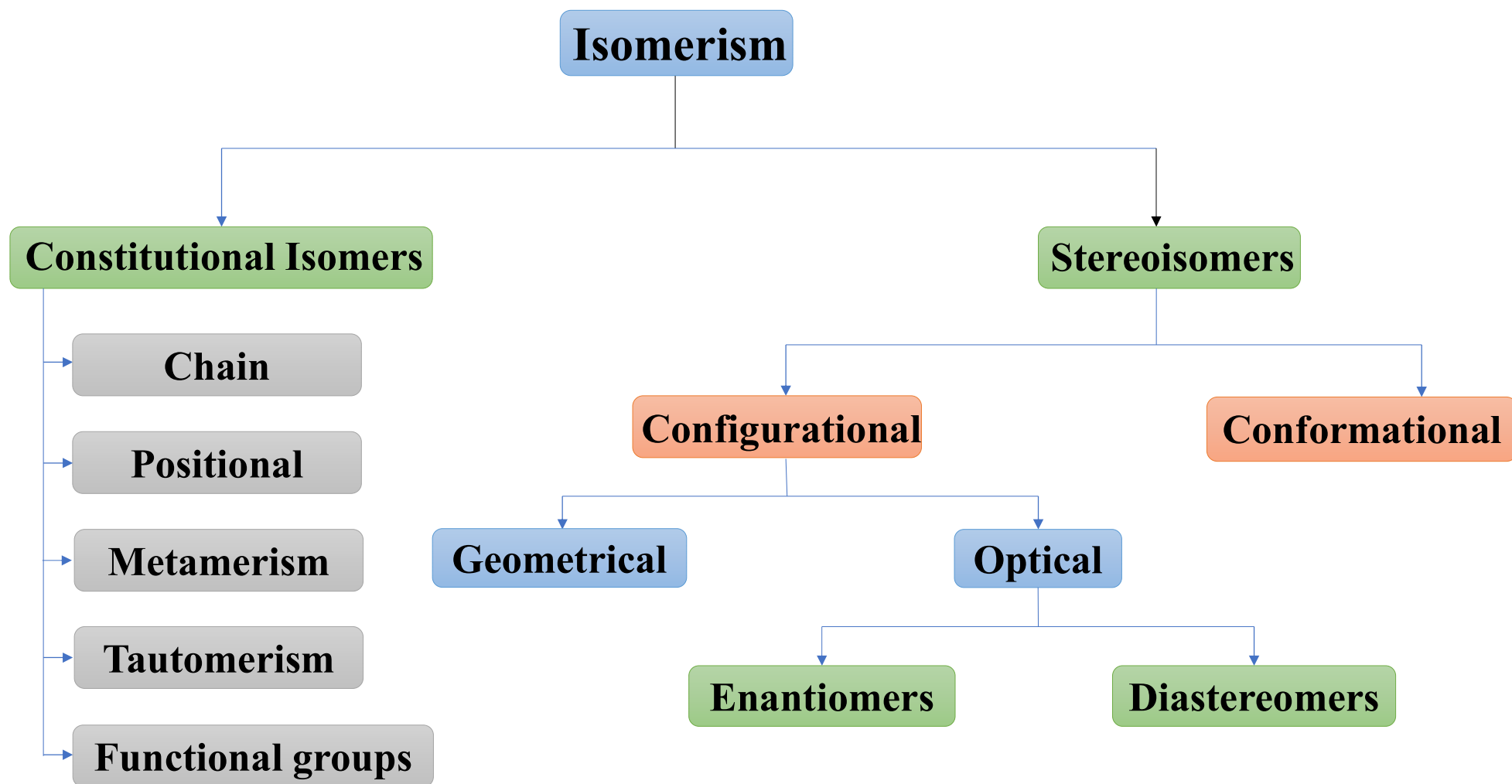
Examples:



Stereochemistry

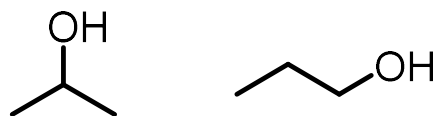
It's the study of three-dimensional structure of molecules.





Constitutional : isomers differ in the order in which the atoms are connected so they can contain different functional groups and/ or bonding patterns (*e.g.* branching)

example: 1-propanol, 2-propanol

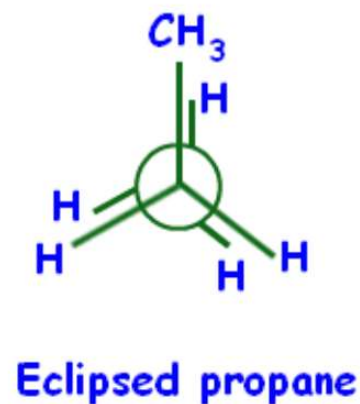
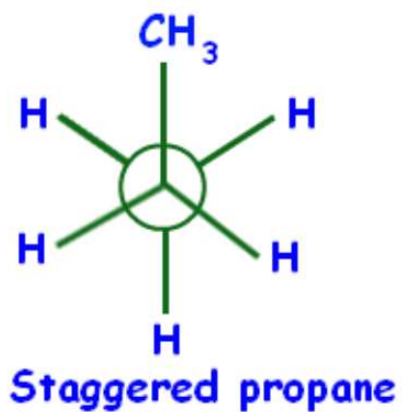


Stereoisomers: have the same functional groups and the same atoms order, they differ only in the arrangement of atoms and bonds in space.

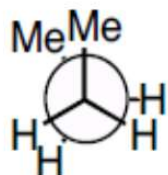


Conformational isomers

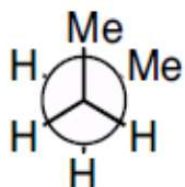
- Stereoisomers that are produced by rotation about bond (single bond), and are often rapidly interconverted at room temperature.



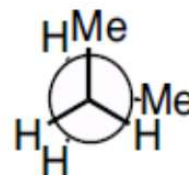
Butane conformational isomers



fully eclipsed
(eclipsed)
 $\theta_{\text{Me/Me}} = 0^\circ$



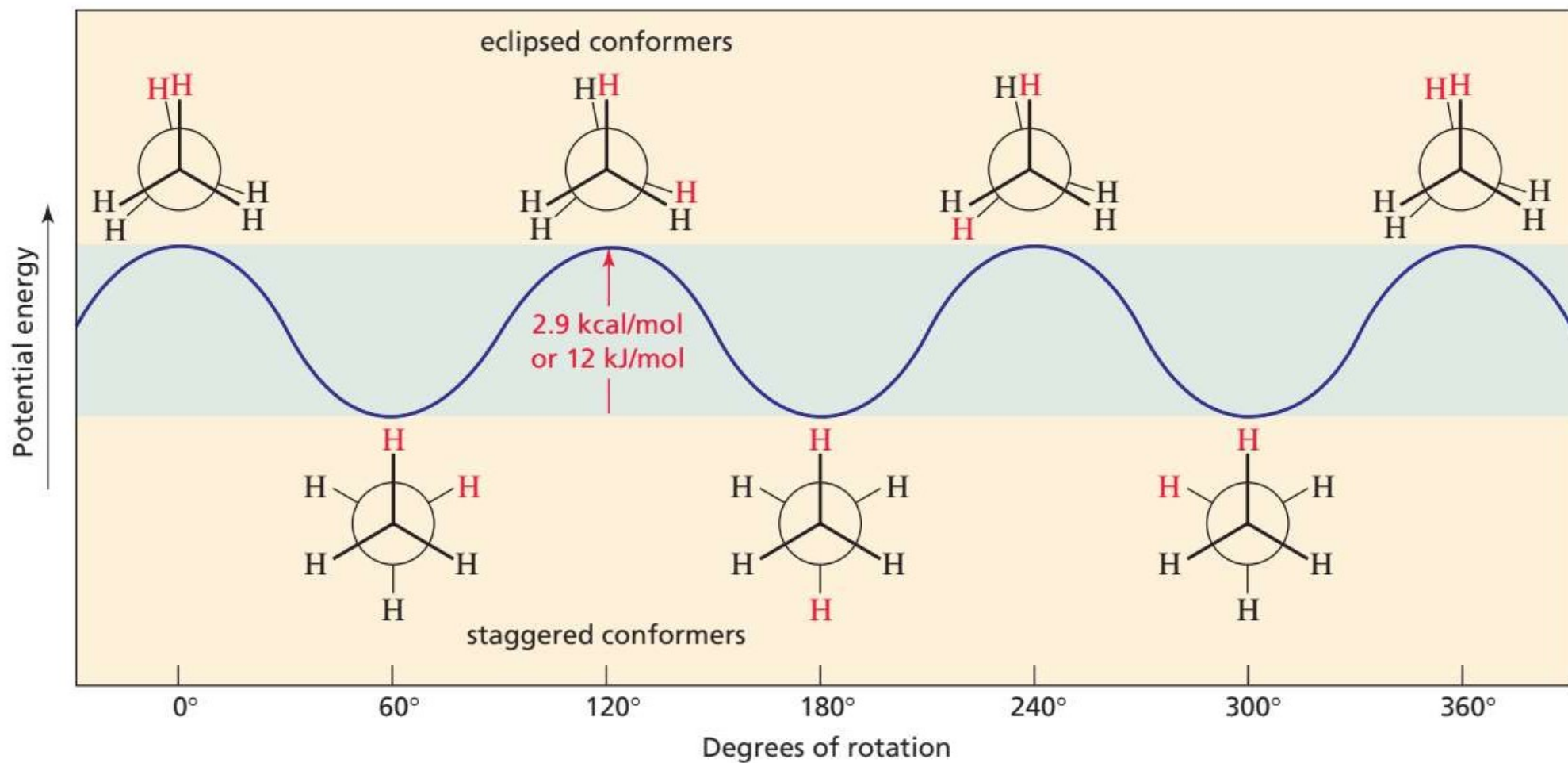
gauche
(staggered)
 $\theta_{\text{Me/Me}} = 60^\circ$



eclipsed
(eclipsed)
 $\theta_{\text{Me/Me}} = 120^\circ$



anti
(staggered)
 $\theta_{\text{Me/Me}} = 180^\circ$



▲ **Figure 2.4**

Potential energy of ethane as a function of the angle of rotation about the carbon-carbon bond.

Configurational isomers

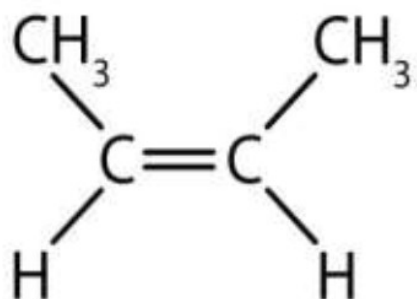
Stereoisomers that do not readily interconvert at room temperature and can be separated.

Geometric isomers

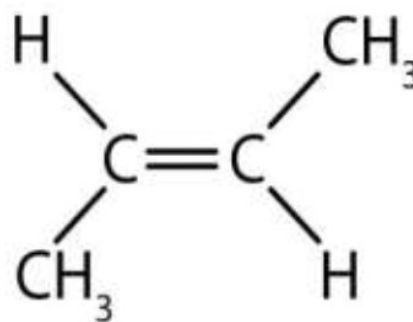
Optical isomers

Geometric isomers

These isomers occur where you have restricted rotation somewhere in a molecule (ex double bond) or across a ring system



cis-2-butene



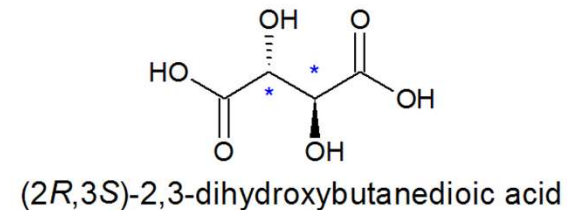
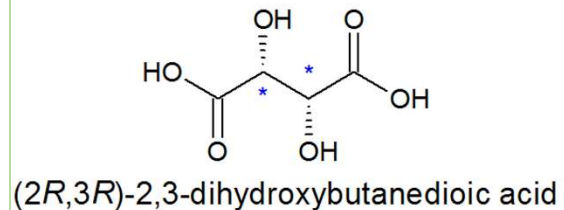
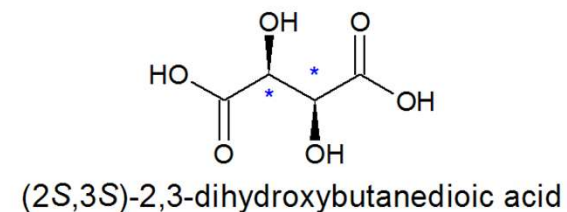
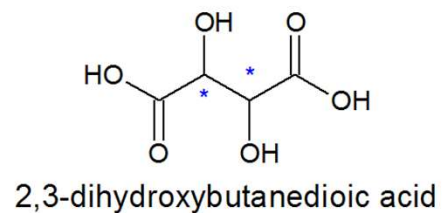
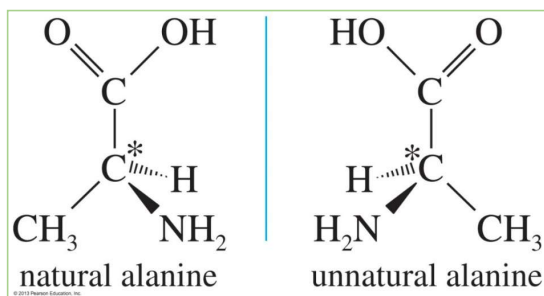
trans-2-butene

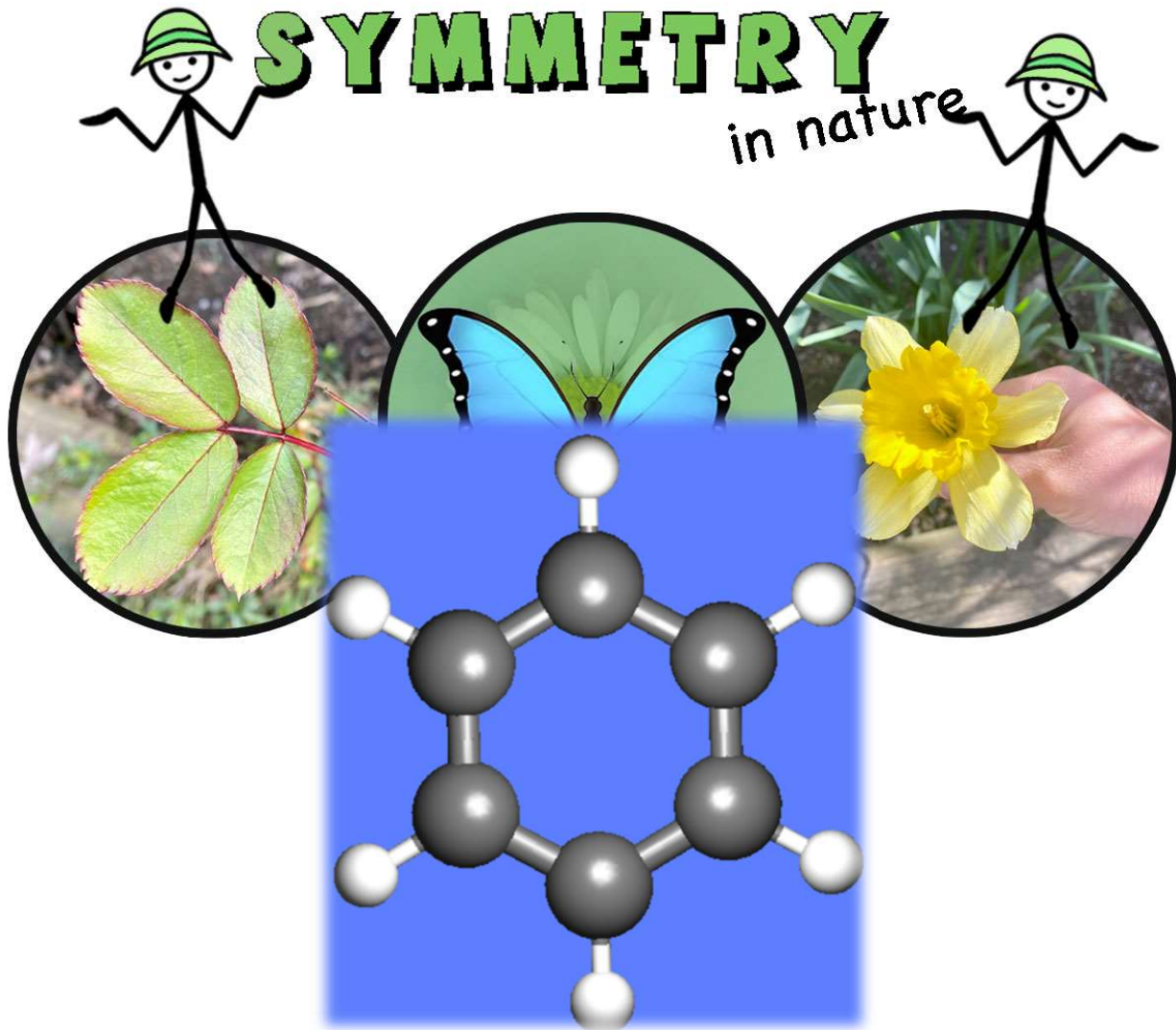
Optical isomers

Configurational isomers that differ in the 3D relationship of the substituent about one or more atoms.

Enantiomers

Diastereomers



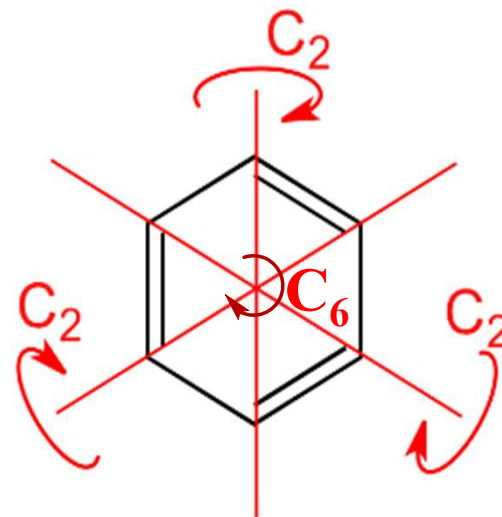
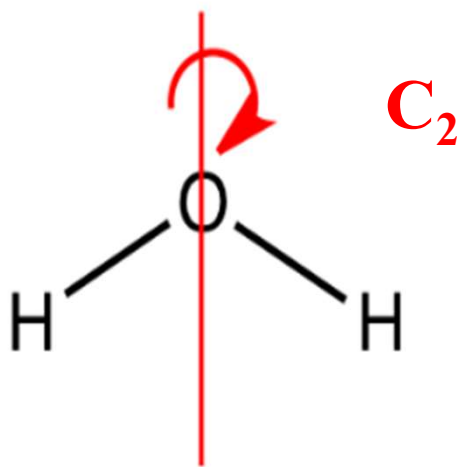
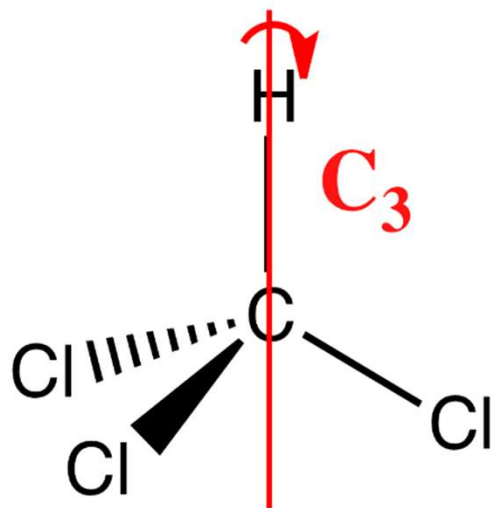


Symmetry

- **Symmetry element** is a geometrical entity such as a **plane, a line, a point** with respect to which after operation we get **superimposable on the original one**.
- There are four types of symmetry elements. These are
 - (a) **Simple or proper axis of symmetry (C_n)**
 - (b) **Plane of symmetry (σ)**
 - (c) **Centre of symmetry (i)**
 - (d) **Alternative axis of symmetry (S_n)**

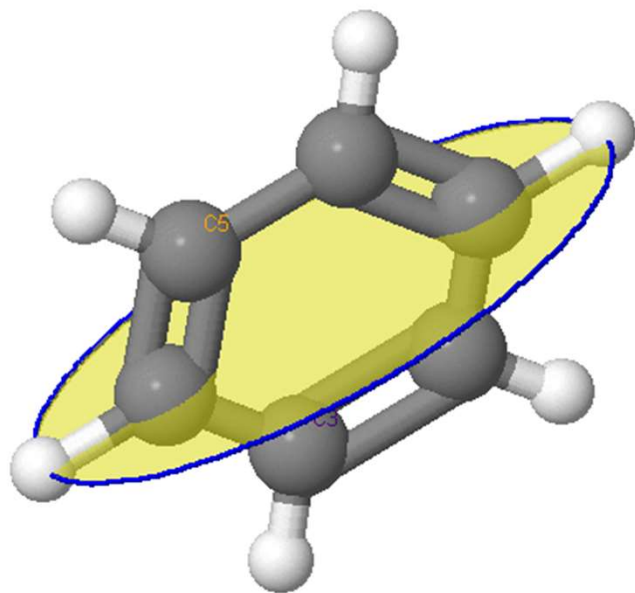
□ Simple or proper axis of symmetry (C_n)

If the molecule is **rotate by an angle of $360^\circ/n$ around an imaginary line (axis)** passing through the molecule, an indistinguishable structure from original results.

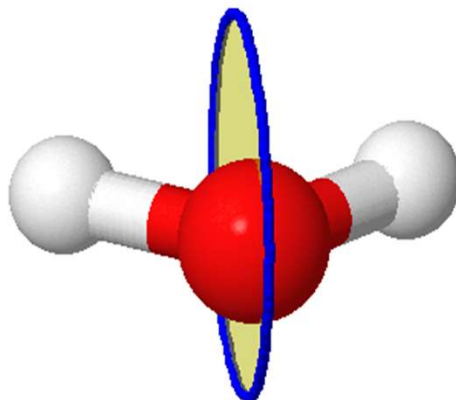


□ Plane of symmetry (σ)

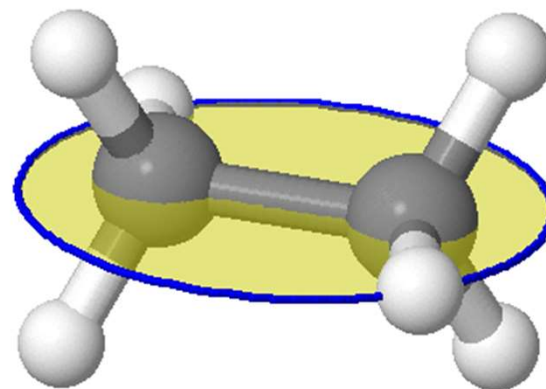
An imaginary plane that **bisects a molecule** in such a way that one half of molecule is mirror image of other half (**plane acting as mirror**)



Benzene



Water



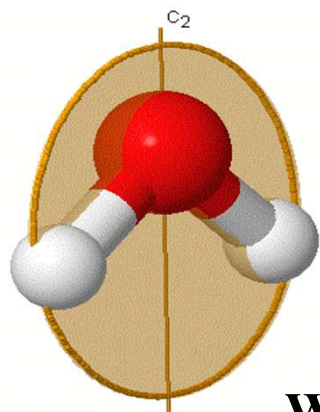
Ethane

Classification of plane of symmetry

Depending upon **the position of σ**
with respect to **principal axis**

σ_v

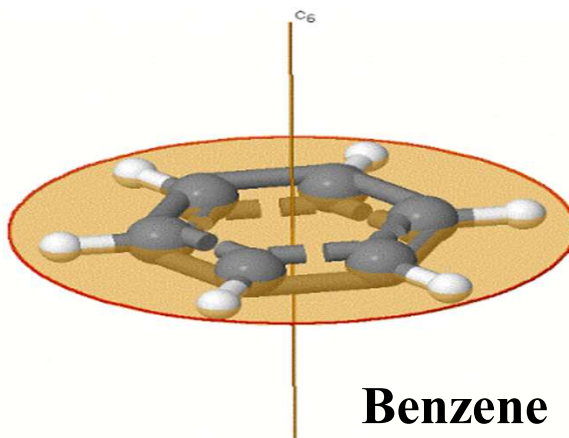
➤ Plane **containing**
principal axis



Water

σ_h

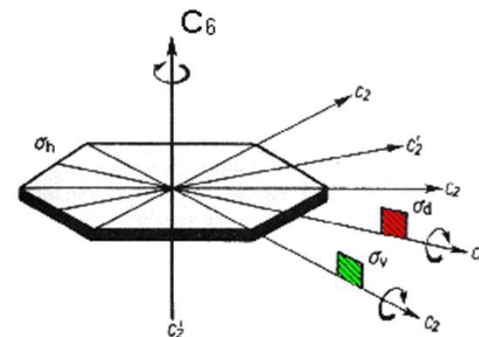
➤ Plane **perpendicular**
to principal axis



Benzene

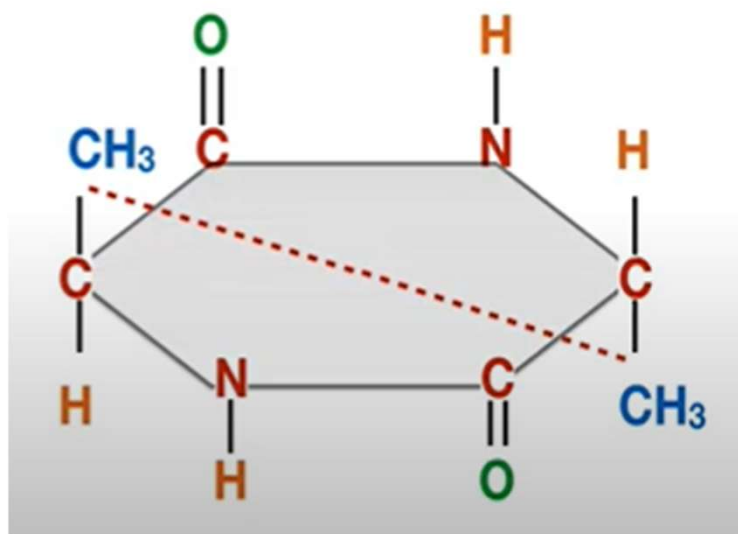
σ_d

➤ Plane which **contains**
principal axis **and bisects**
angle between two C₂



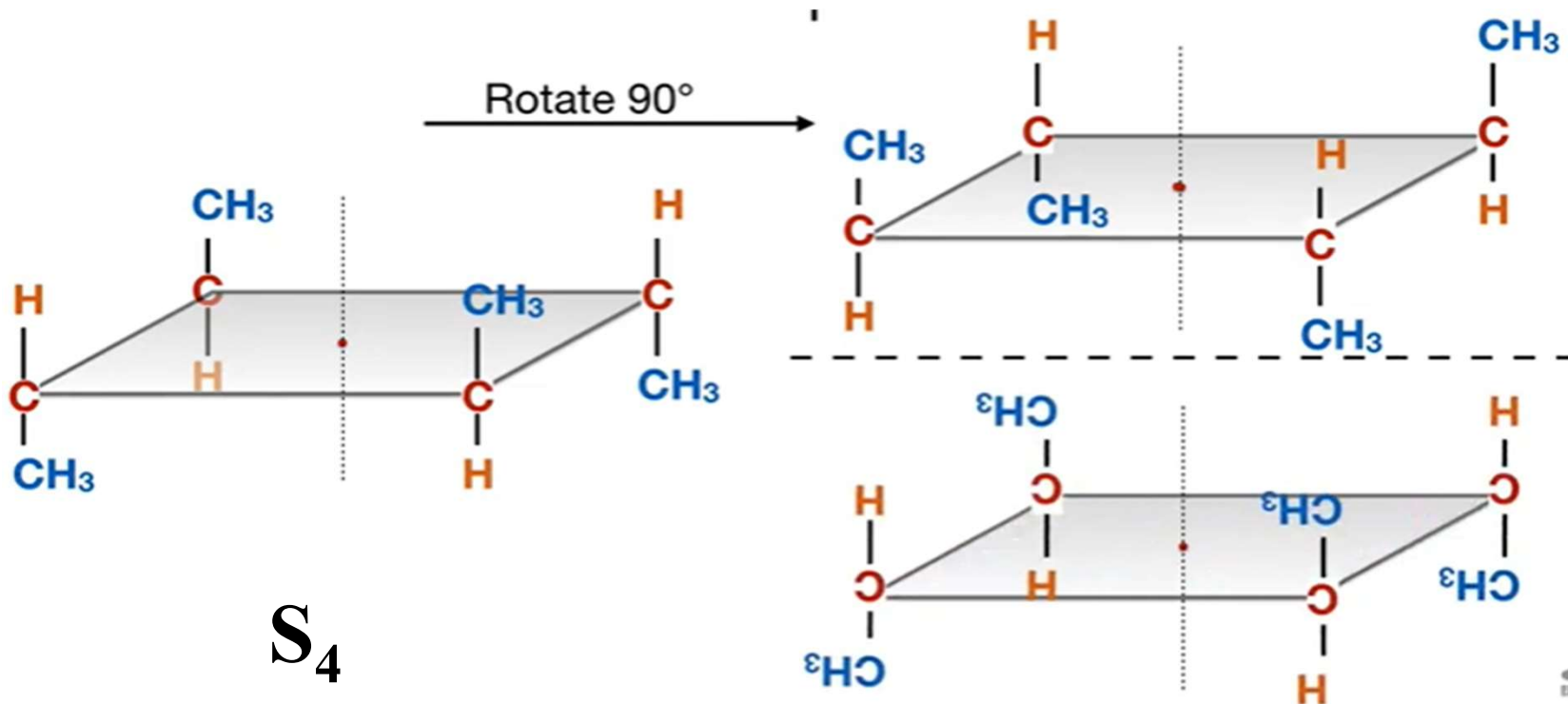
□ Centre of symmetry (i)

A point within a molecule such that if a **straight line is drawn from any atom of the molecule to the point and then extended equal distance beyond the point**, another identical atom will be found at the end of the line.

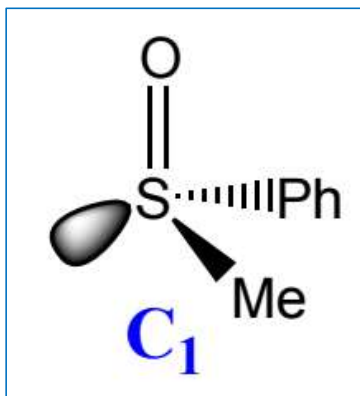


□ Alternative axis of symmetry (S_n)

An imaginary axis in molecule such that **rotation of the molecule about the axis by $360^\circ/n$ followed by reflection in a plane perpendicular to this axis** generates a indistinguishable structure to original.

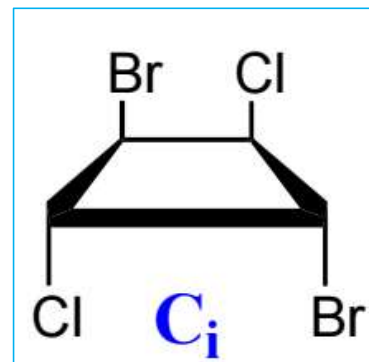


Symmetry elements and Point Group:



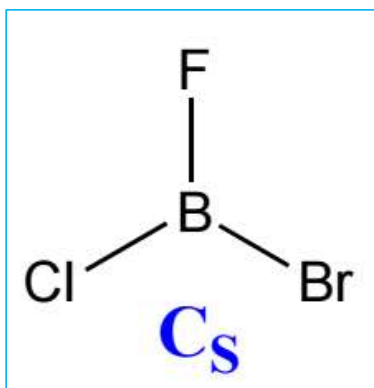
Sym elements

E

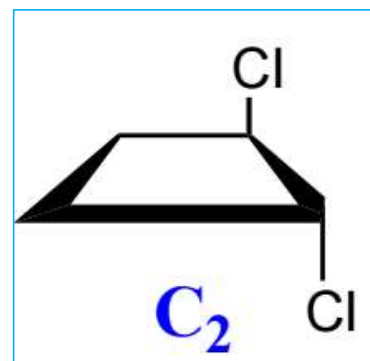


Sym elements

i



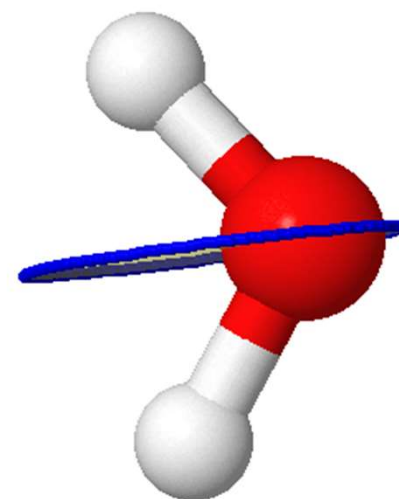
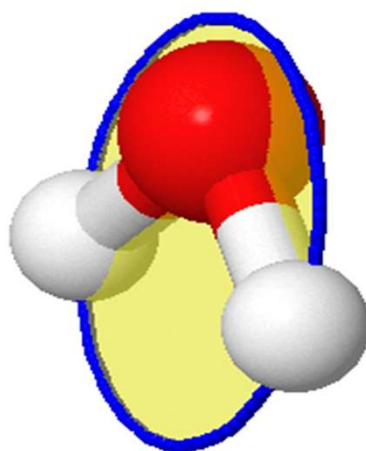
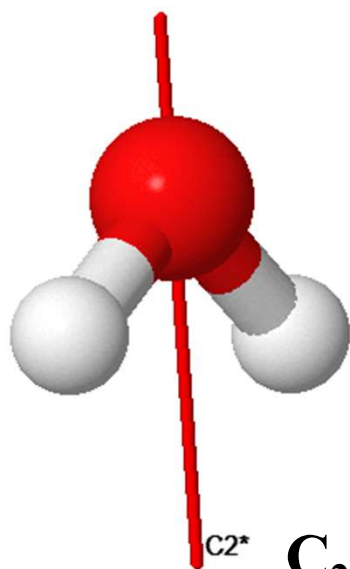
σ



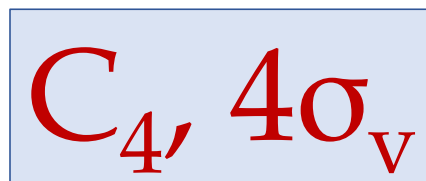
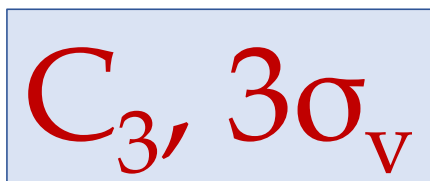
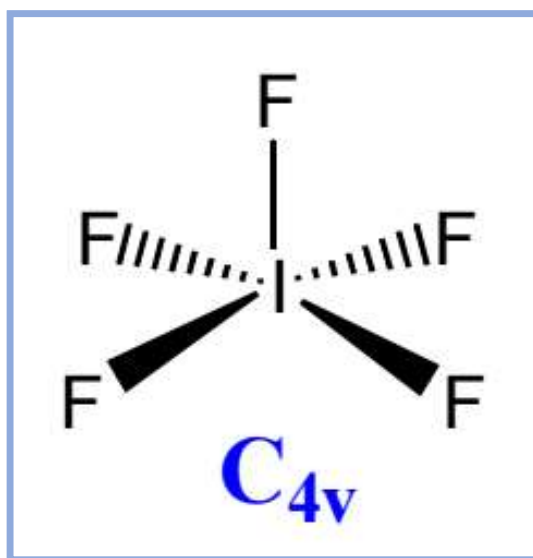
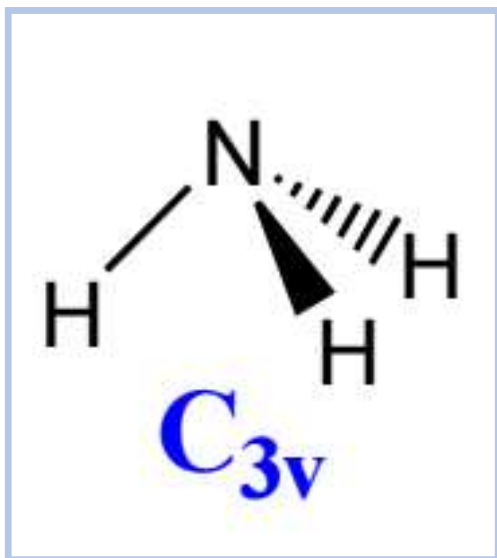
C_n

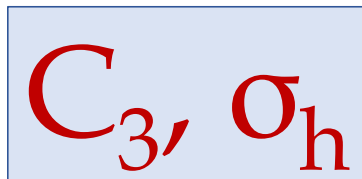
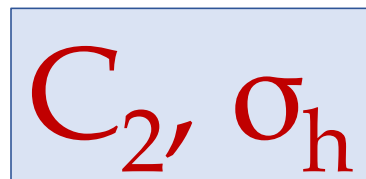
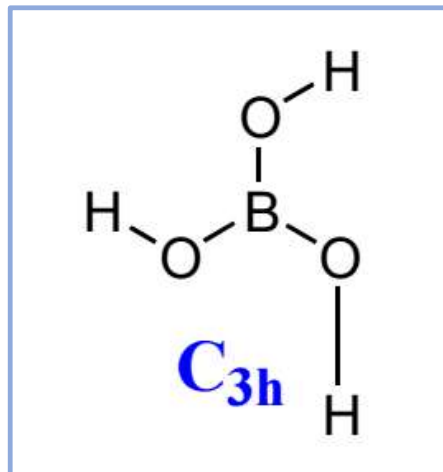
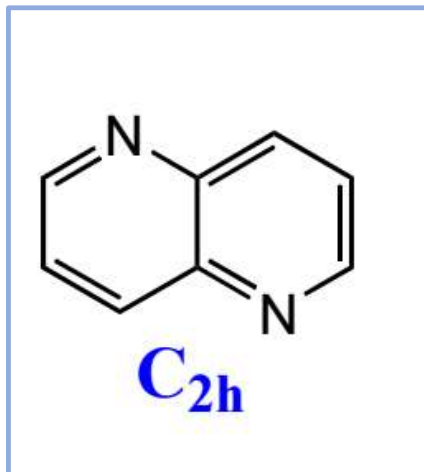
Point group of H₂O

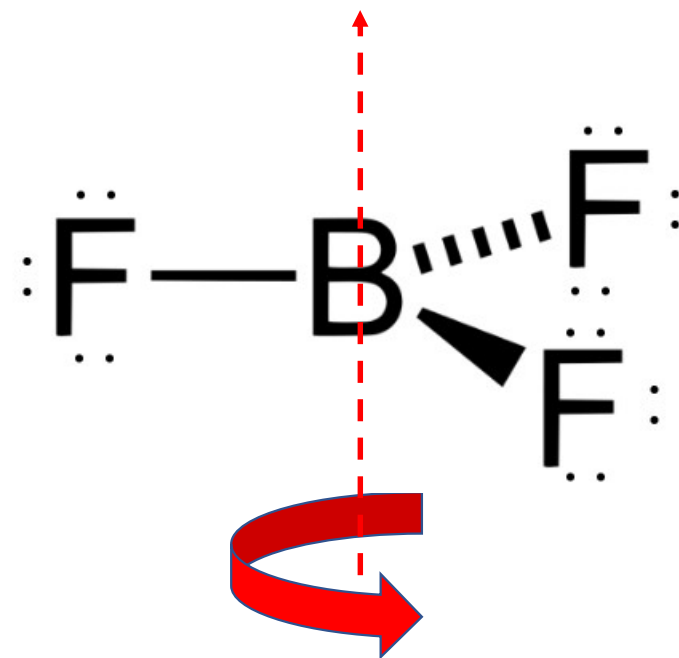
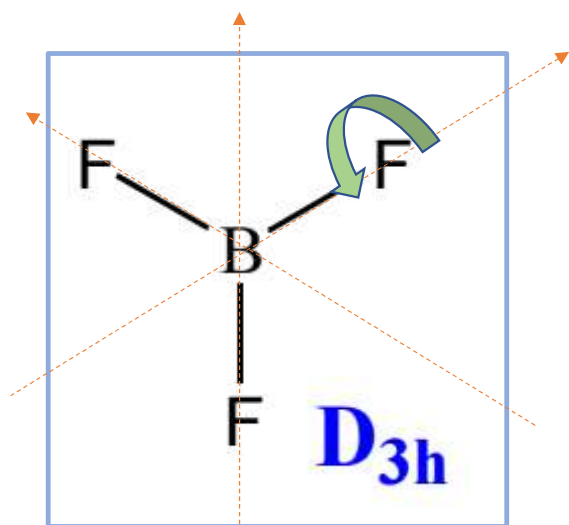
➤ Symmetry elements :



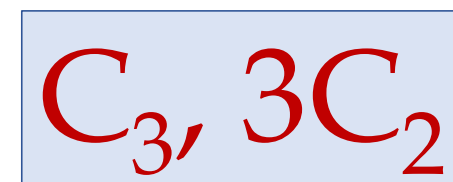
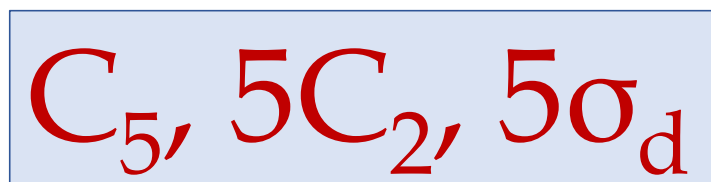
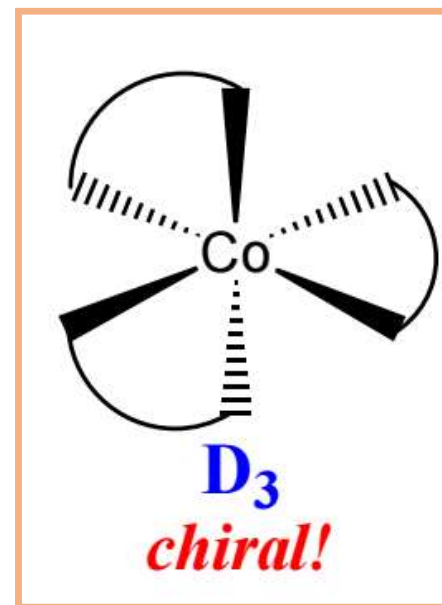
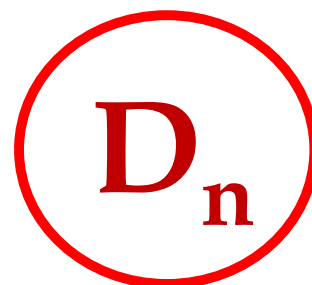
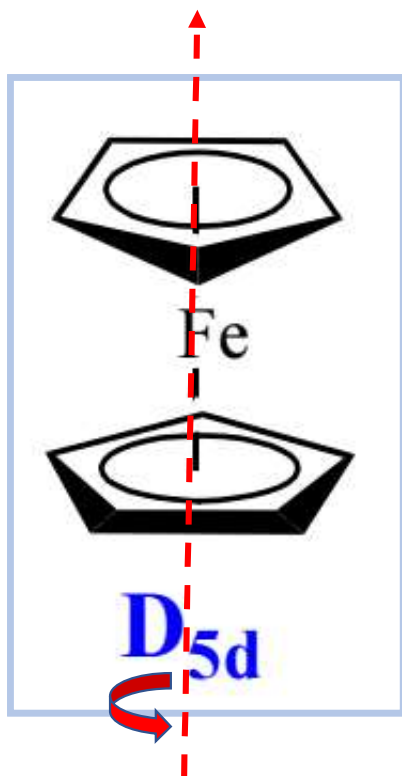
Point group is C_{2v} (C_{nv})





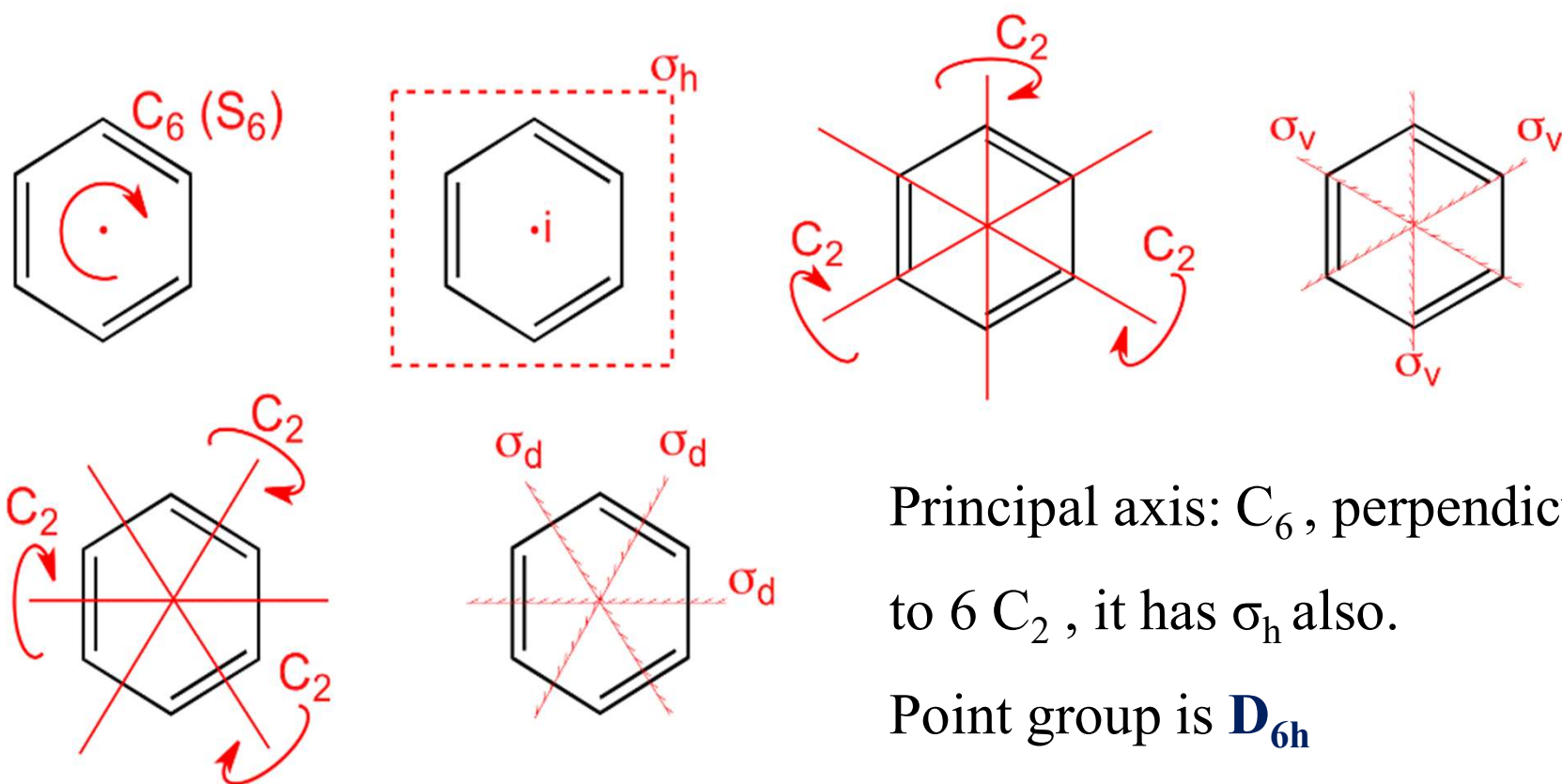


$C_3, 3C_2, 3\sigma_v$ and σ_h



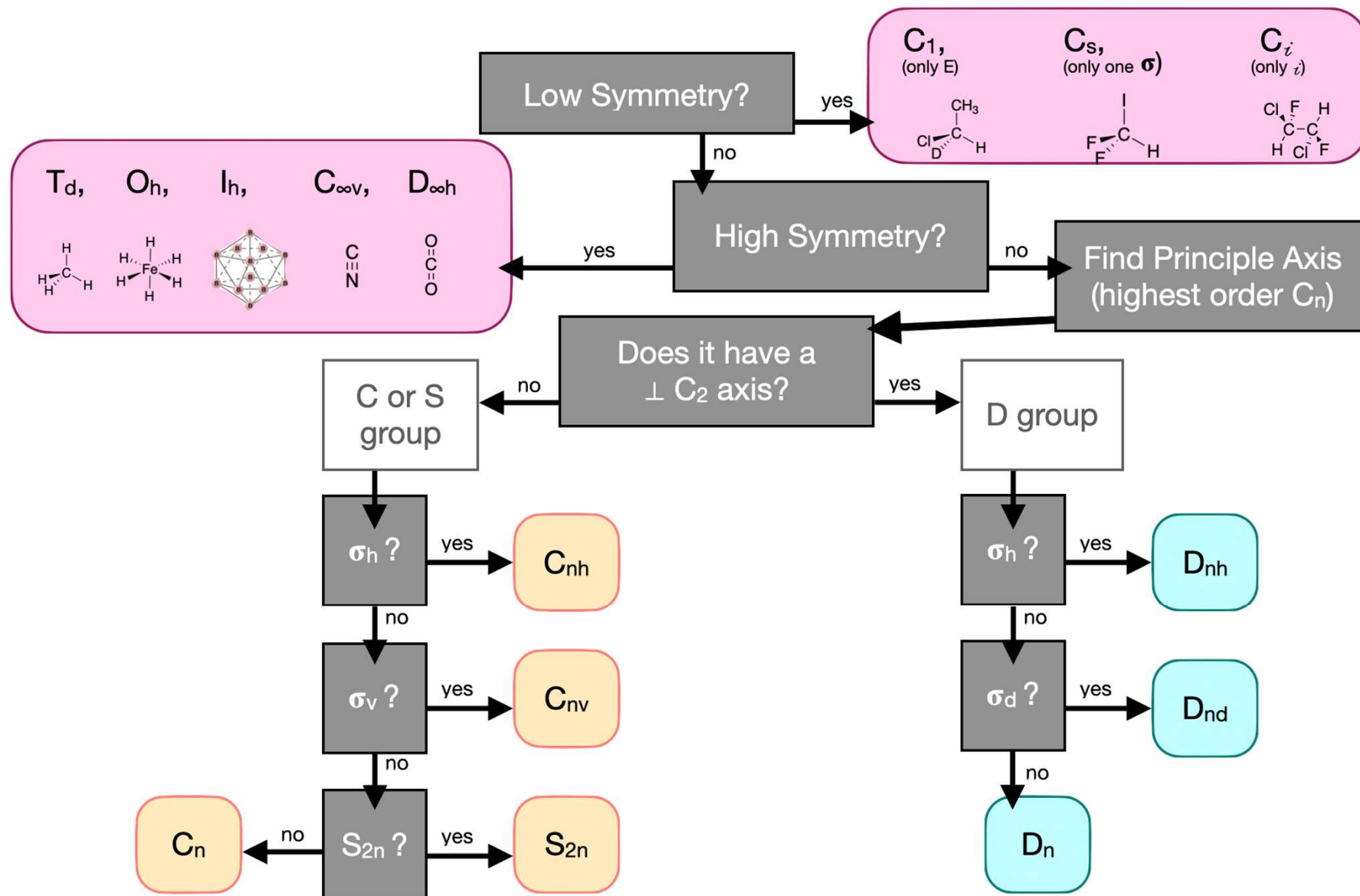
Point group of benzene (C_6H_6)

➤ Symmetry elements:



Principal axis: C_6 , perpendicular to 6 C_2 , it has σ_h also.

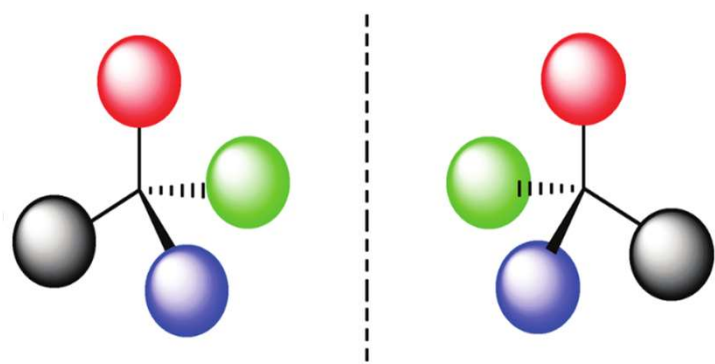
Point group is **D_{6h}**



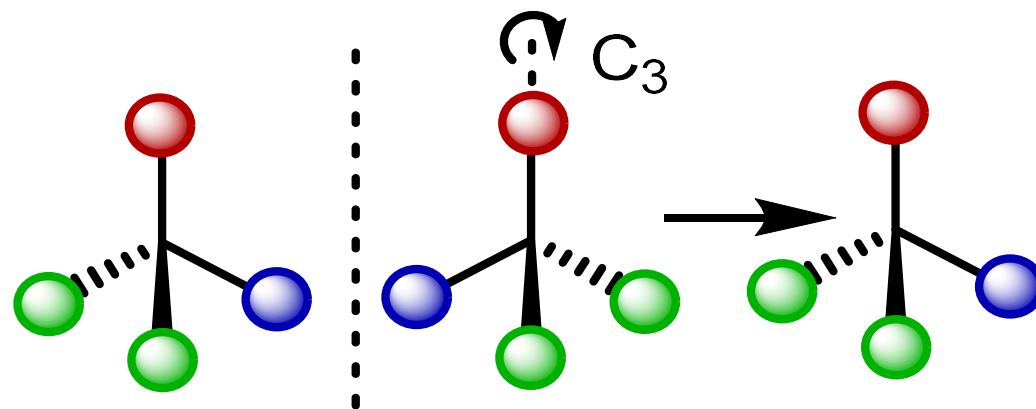
Point gr.	Sym elements*	h***	Point gr.	Sym elements*	h***
C_1	E	1	C_i	i	2
C_s	σ	2	C_n	C_n	n
S_n	S_n	n	C_{nv}	$C_n, n\sigma_v$	2n
C_{nh}	C_n, σ_h	2n	D_n	$C_n, nC_2 \perp C_n$	
D_{nd}	$C_n, nC_2 \perp C_n, n\sigma_d$	4n	D_{nh}	$C_n, nC_2 \perp C_n, \sigma_h, n\sigma_v$	4n
$C_{\infty v}$	linear no i	∞	$D_{\infty h}$	linear with i	∞

Chiral molecule:

- If the **mirror image** of a molecule is **not superimposable on the original**.
- If the molecule **doesn't have** σ plane of symmetry, inversion centre, Alternative axis of symmetry, then it is said to be chiral molecule.
- Chiral molecule **may have** C_n ($n > 1$) **axis** of symmetry.
- Chiral molecules **rotate the plane** of plane polarized light.



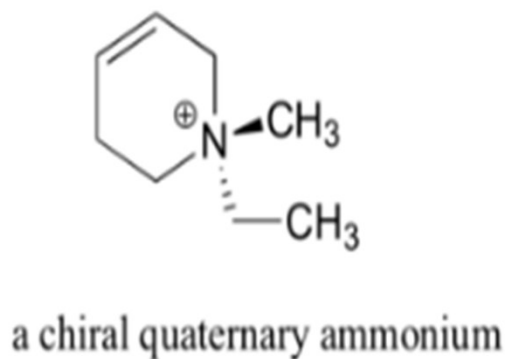
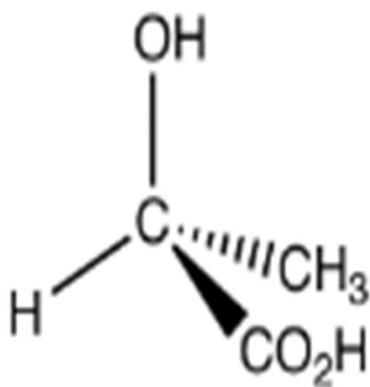
Non-superimposable mirror image

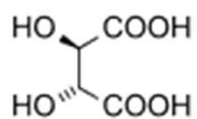
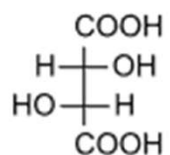


Superimposable mirror image

Chiral centre:

- Chiral centre is an atom **bonded tetrahedrally to four different atoms or groups**. Chiral centre may be **C, N, P, S, Si** etc.
- If the molecule has **one chiral centre** it **must be** chiral molecule. But if **multiple chiral centres** are present, then it **may not be** chiral molecule.

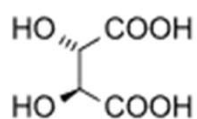
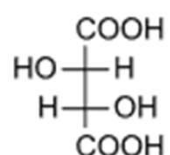




(natural) tartaric acid

L-(+)-tartaric acid

dextrotartaric acid



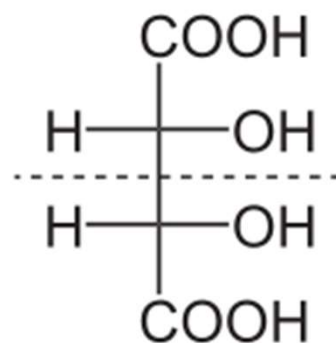
D-(-)-tartaric acid

levotartaric acid

(1:1)

DL-tartaric acid

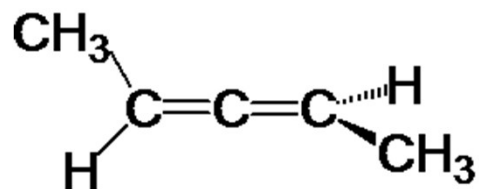
"racemic acid"



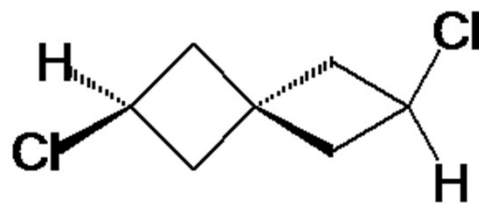
mesotartaric acid

Chiral molecule having no chiral centre:

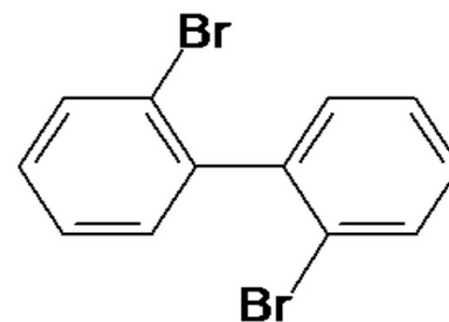
➤ Axial chirality:



Dimethylallene

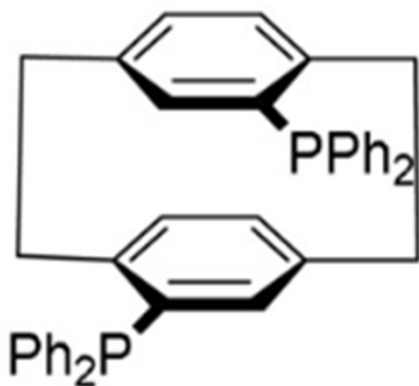


Dichlorospiroheptane

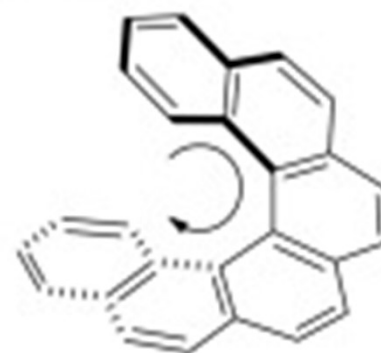
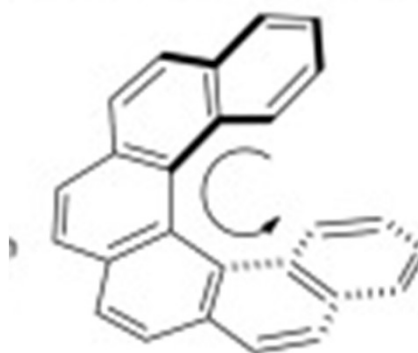


Dibromobiphenyl

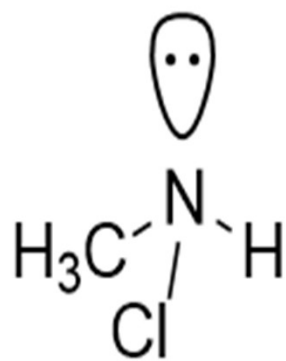
➤ Planner chirality:



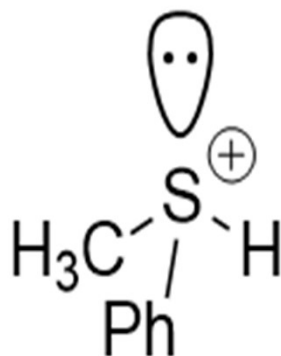
➤ Helical Chirality:



Condition of the molecule to be chiral including lone pair

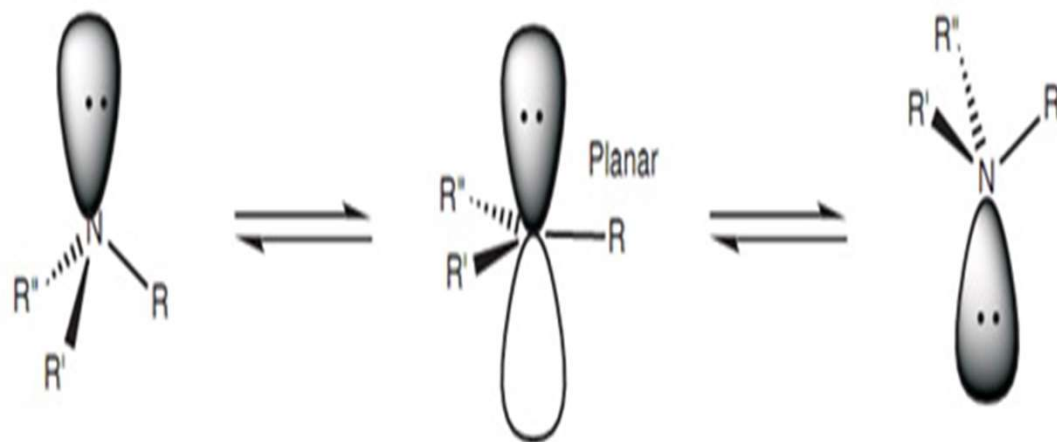


Achiral



chiral

Pyramidal inversion or umbrella inversion



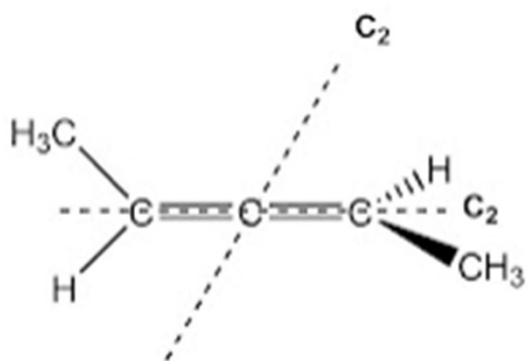
Rapid interconversion at room temperature forms racemic mixture.

Stereogenic centre or Stereocentre:

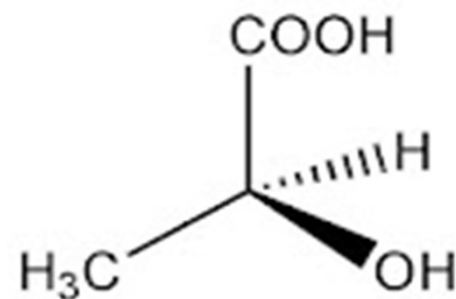
- If any two ligands attached to an atom are interchanged spatially resulting a new stereoisomer.

Asymmetric and Dissymmetric molecules:

- molecule without any C_n axis ($n > 1$) of symmetry, called **Asymmetric molecule**
- molecule with C_n axis ($n > 1$) of symmetry, called **dissymmetric molecule**.
- **All are chiral molecules.**



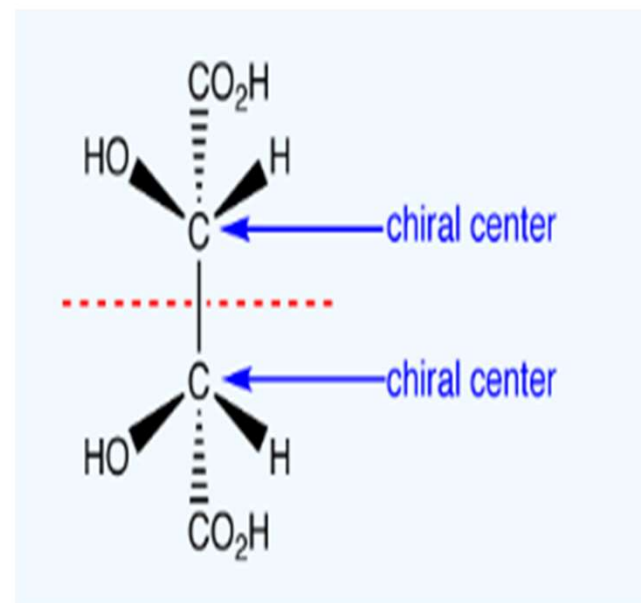
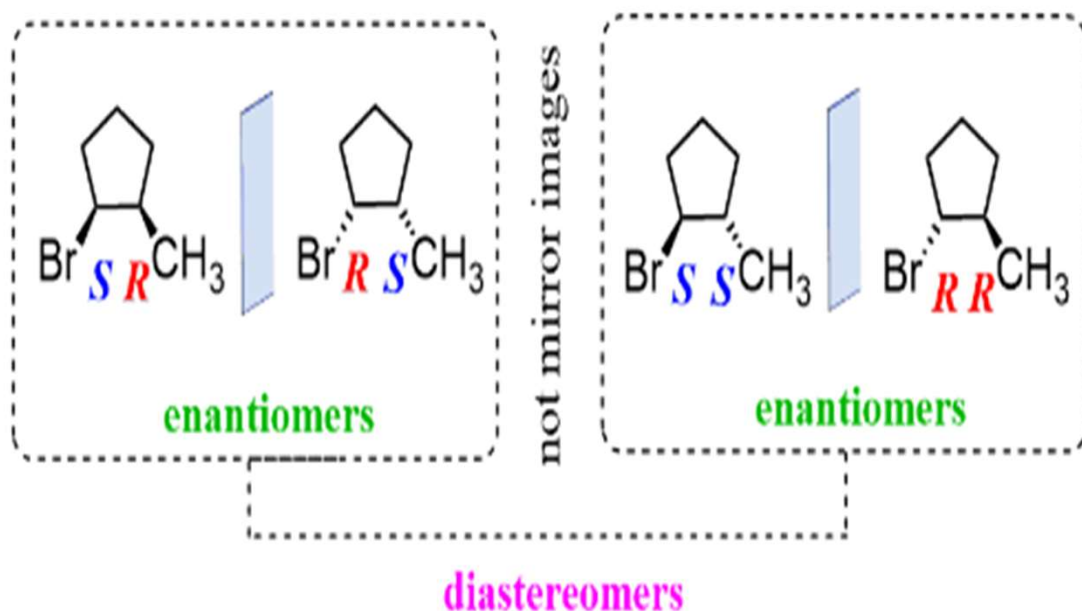
Dissymmetric molecule



Asymmetric molecule

Enantiomers, Diastereomers and Meso isomers

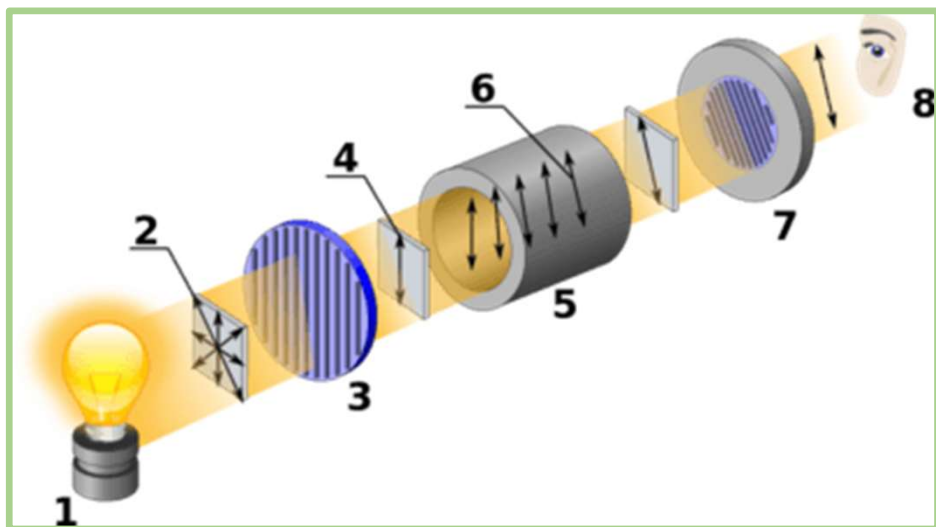
- ❑ Enantiomers: Isomers having non-superimposable mirror image relationship.
- ❑ Diastereomers: Isomers having no mirror image relationship.
- ❑ Meso Isomers: Molecules having multiple stereocentres and a plane of symmetry.



Meso Isomer

Optical Activity

Optical activity is the ability of a chiral molecule to rotate the plane of plane-polarised light



Formula for Specific Rotation

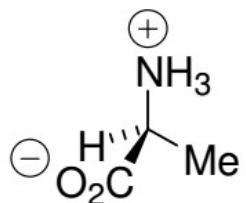
$$[\alpha] = \frac{\alpha_{\text{observed}}}{c \times l}$$

Specific rotation (in degrees*)

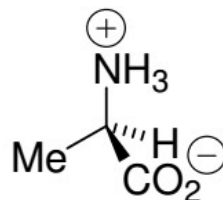
Concentration (units = g/ml)

Observed rotation (units = degrees)

Path length (units = dm)



(+)-alanine $[\alpha]_D = + 8.5^\circ$



(-)-alanine $[\alpha]_D = - 8.5^\circ$

**Racemic
50:50**

If one enantiomer is in excess?

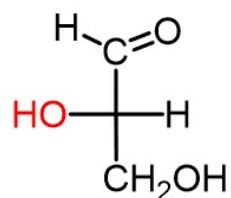
Enantiomeric Excess (ee)

$$\% \text{ ee} = 100 (\alpha_{\text{observed}} / \alpha_{\text{specific}})$$

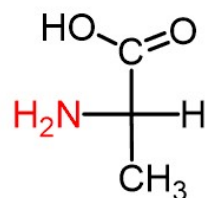
$$\begin{aligned} \% \text{ ee} &= \% \text{ of R} - \% \text{ of S} \\ &= 100 \times (\text{R} - \text{S}) / (\text{R} + \text{S}) \end{aligned}$$

R/S Nomenclature according to CIP rule

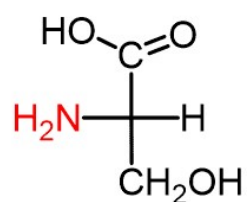
➤ Developed by three chemists, R.S. Cahn, C. Ingold, V. Prelog



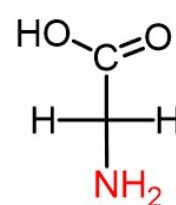
L-Glyceraldehyde



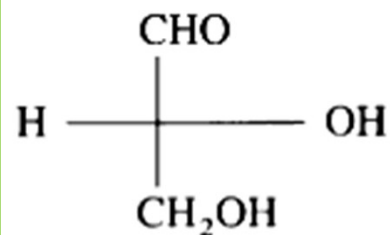
L-Alanine



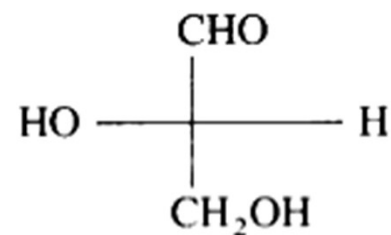
L-Serine



Glycine-achiral



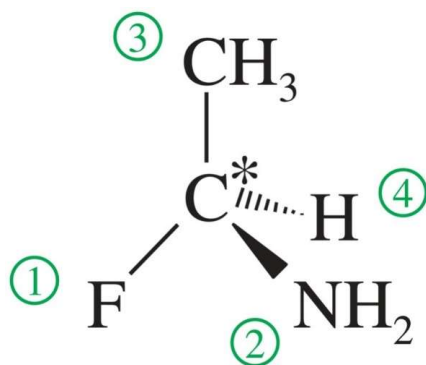
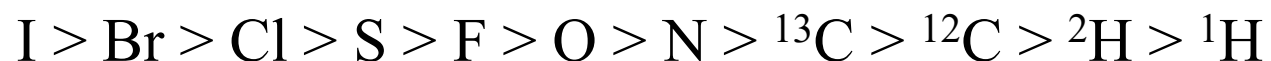
D-Glyceraldehyde
(+) Glyceraldehyde



L-Glyceraldehyde
(-) Glyceraldehyde

(*R*) and (*S*) Configuration: Step 1 Assign Priority

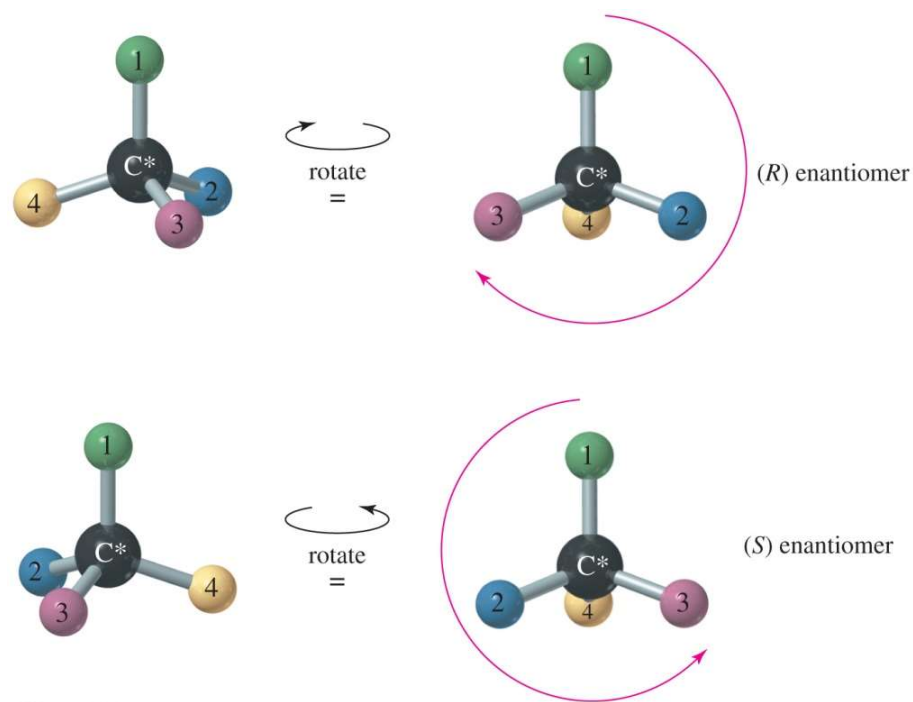
- Assign a relative “priority” to each group bonded to the asymmetric carbon.
- Atoms with higher atomic numbers receive higher priorities.

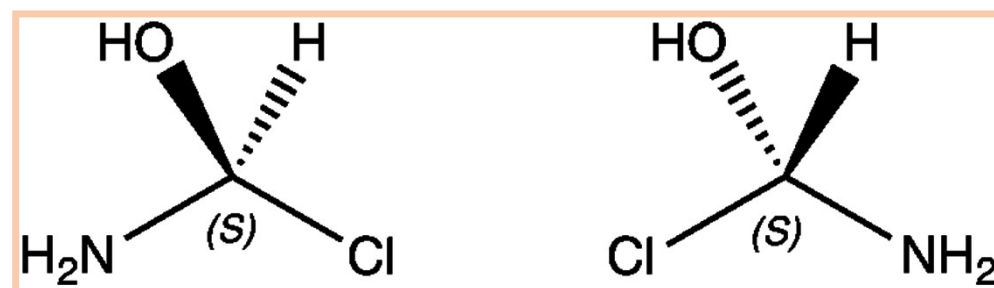
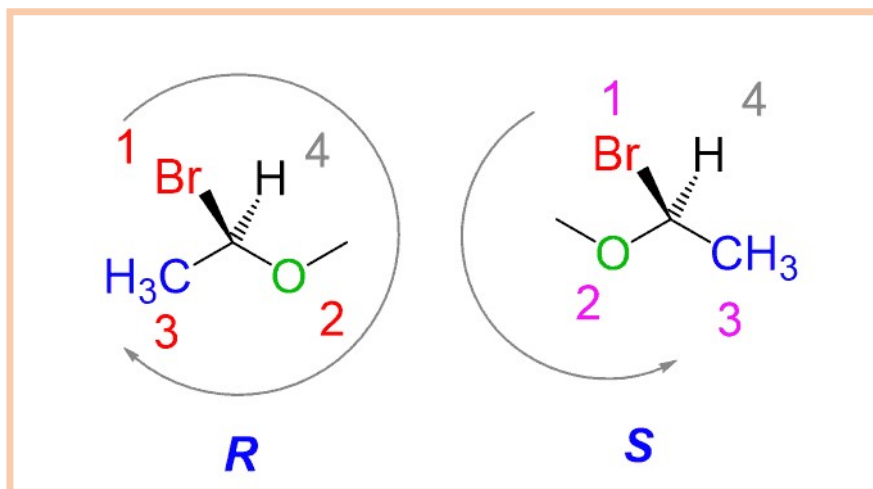
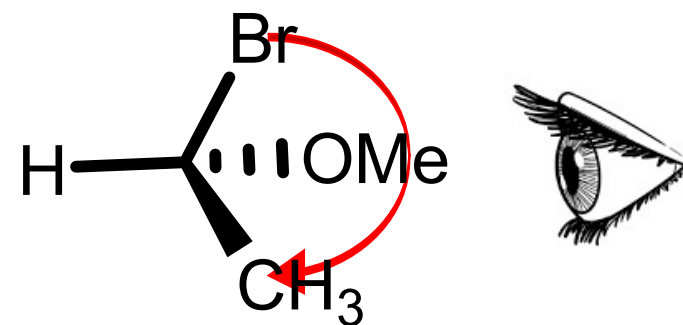
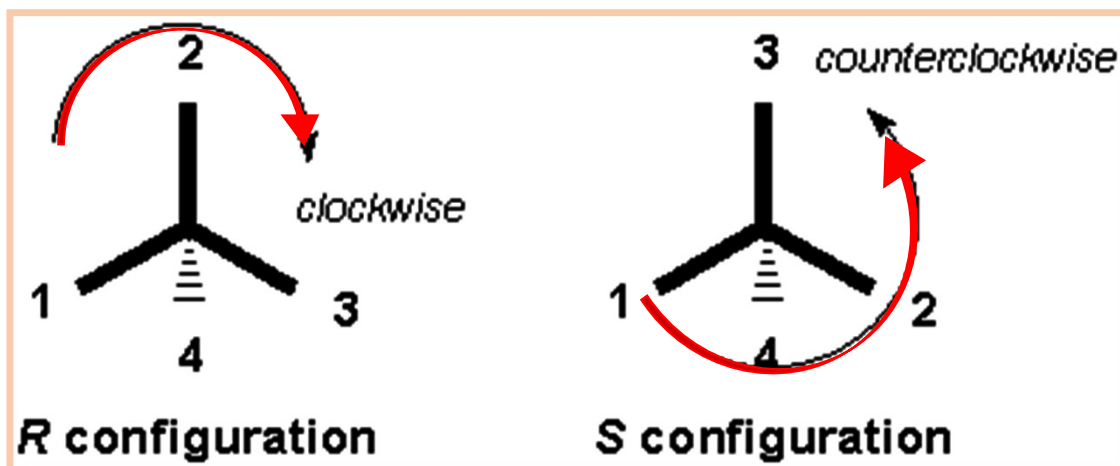


Atomic number: $\text{F} > \text{N} > \text{C} > \text{H}$

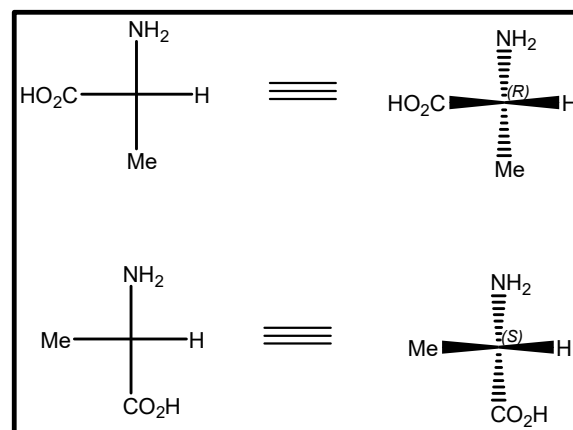
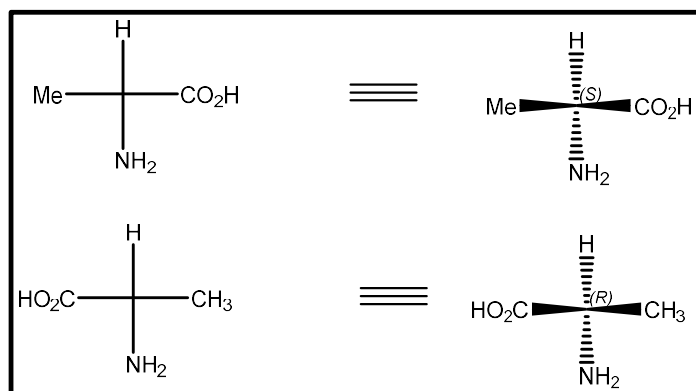
(R) and (S) Configuration: Step 2

- Working in 3-D, rotate the molecule so that the lowest priority group is in back.
- Draw an arrow from highest (1) to second highest (2) to lowest (3) priority group.
- Clockwise = (*R*), Counterclockwise = (*S*)



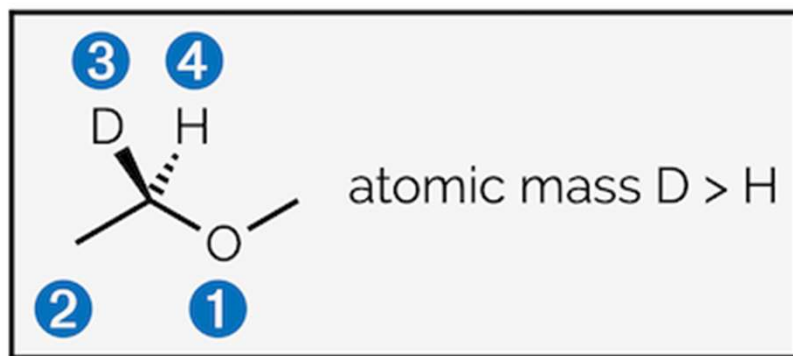


Determine R and S Configurations on a Fischer Projection

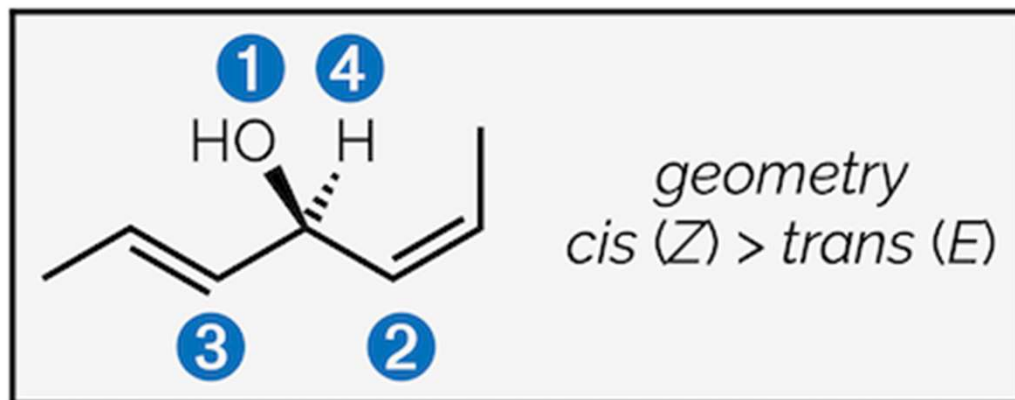
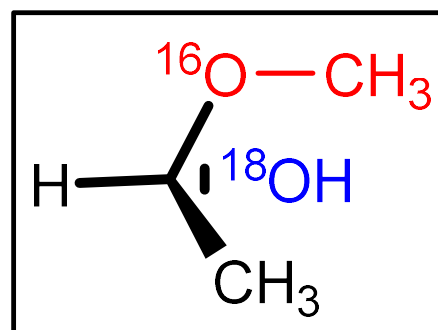


With the #4 ranked substituent at Vertical Line (pointing “in the back”), If that path is clockwise, the chiral center is *R*. If the path is counter-clockwise, the chiral center is *S*.

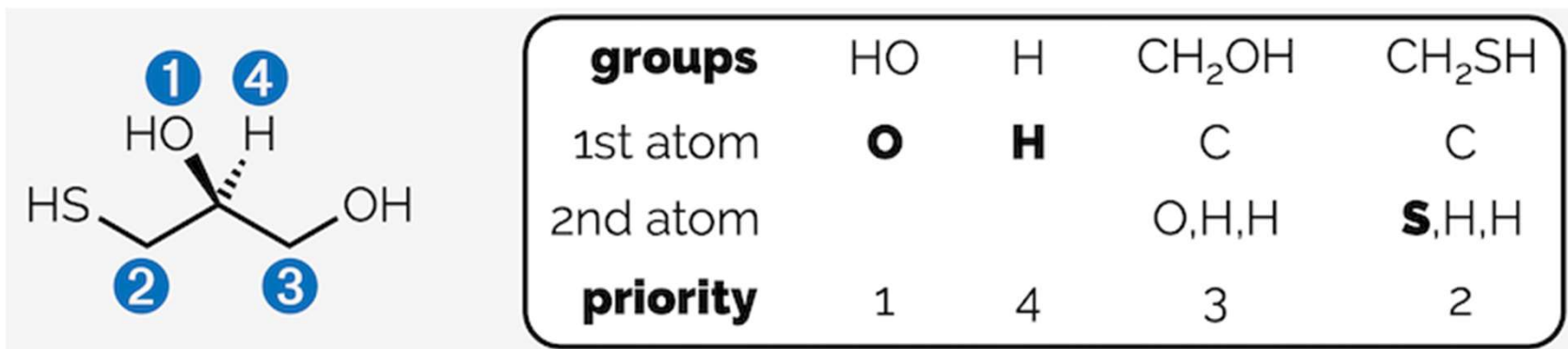
With the #4 ranked substituent at Horizontal Line (pointing “in the Front”), If that path is clockwise, the chiral center is *S*. If the path is counter-clockwise, the chiral center is *R*.



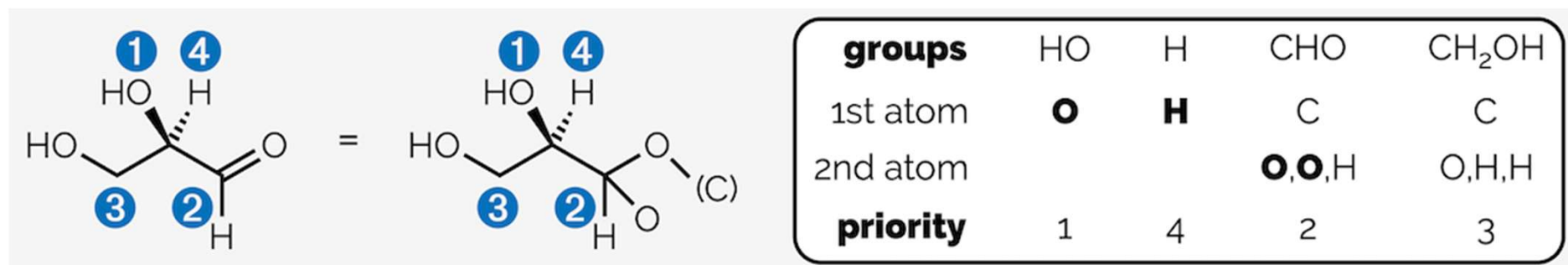
Higher atomic mass precedes
lower atomic mass (isotopes)



Cis (Z) has higher priority
than *trans* (E)



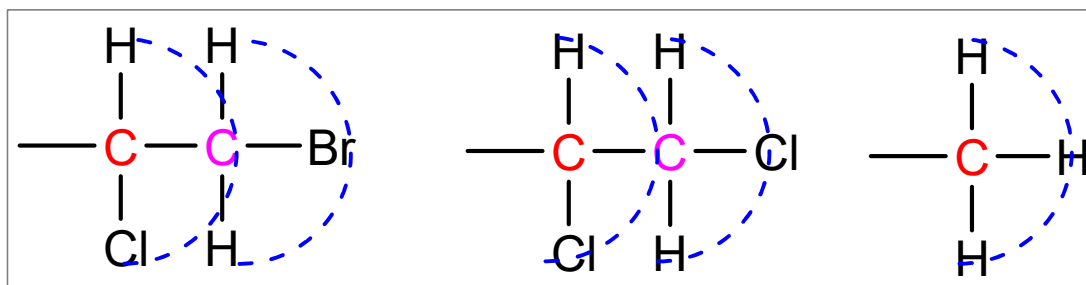
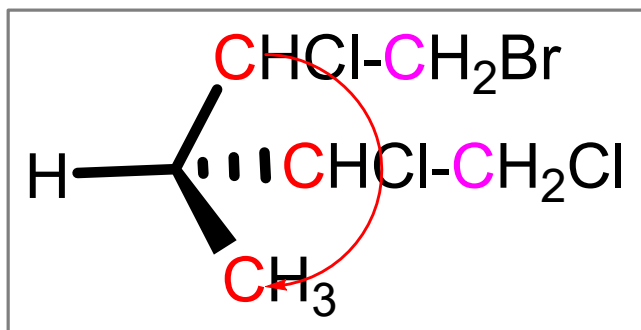
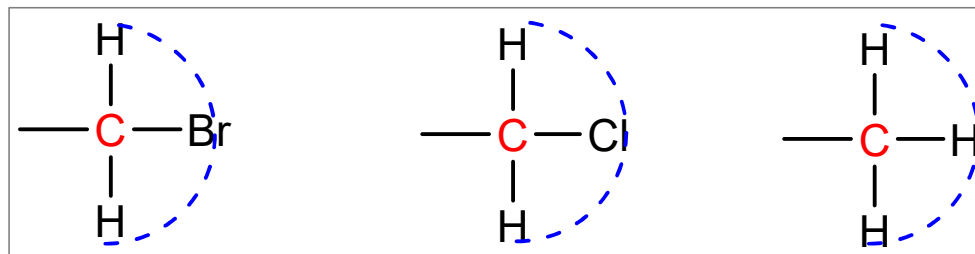
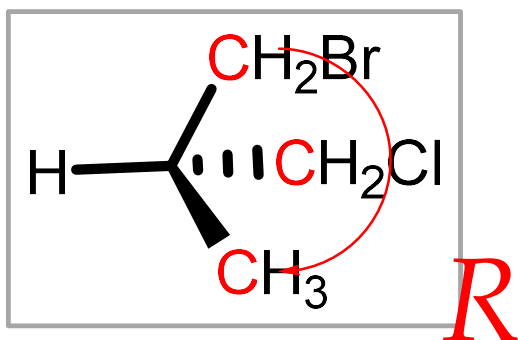
If the atoms directly attached to the chiral center are the same, then compare the next atom in the chain until a difference is found

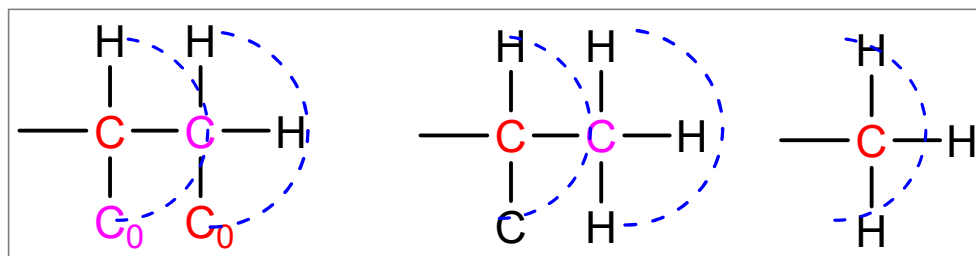
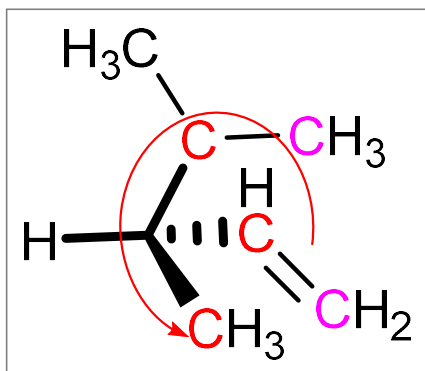


Multiple bonds have the same number of single bonds...Breaking rules

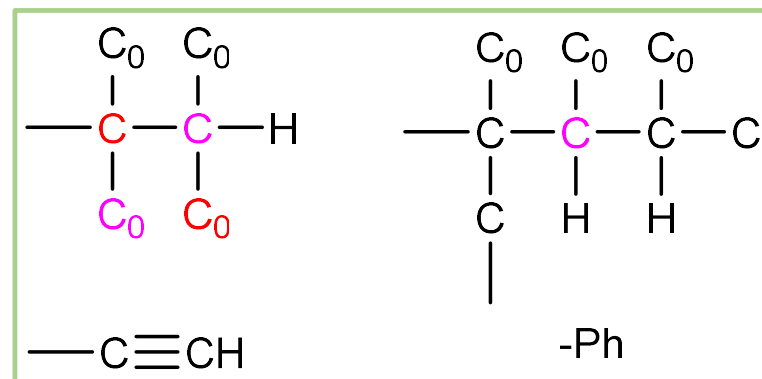
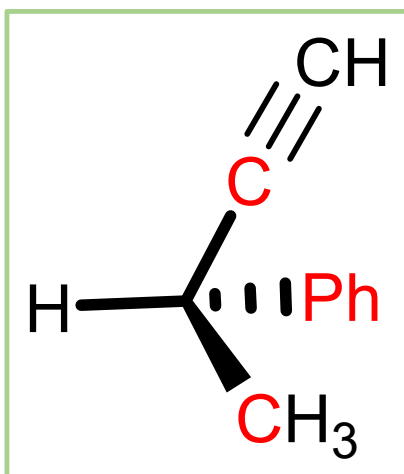
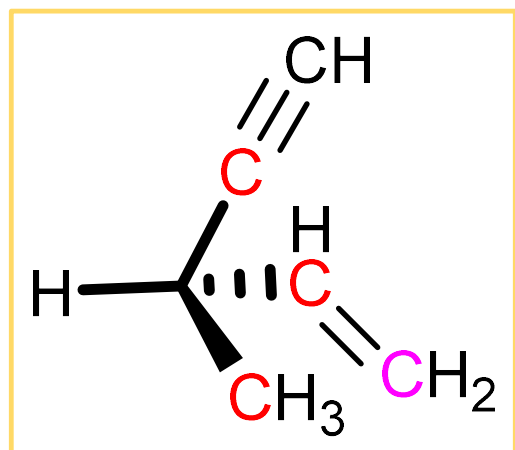
(*R*) and (*S*) Configuration: Assign Priority (Breaking Ties)

In case of ties, use the next atoms along the chain of each group as tiebreakers.

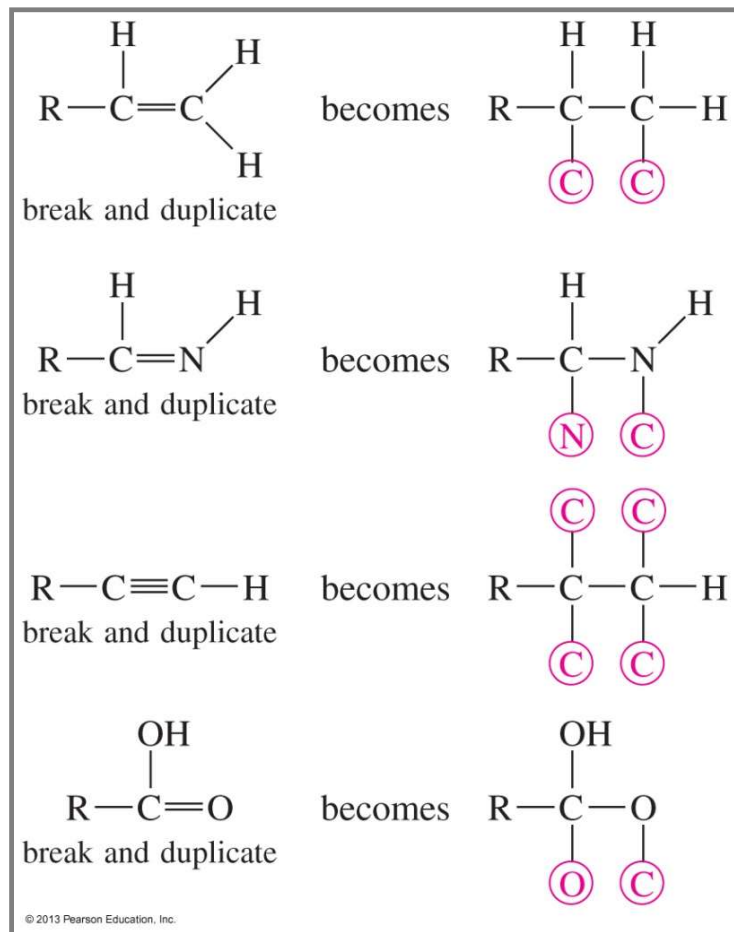




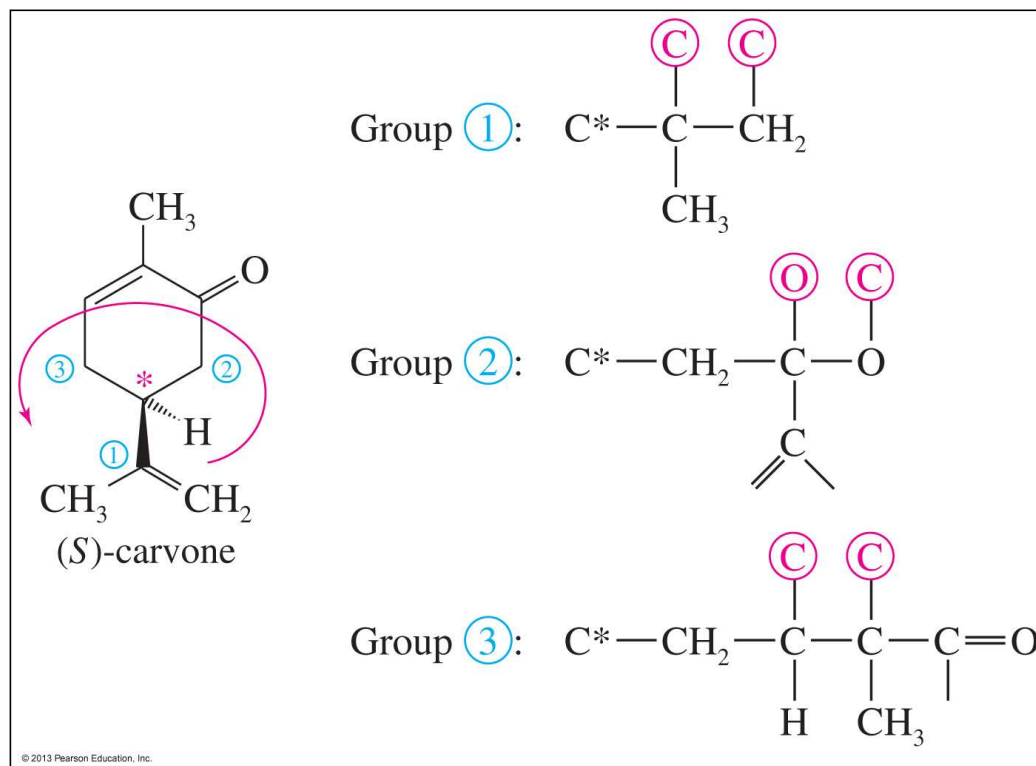
S

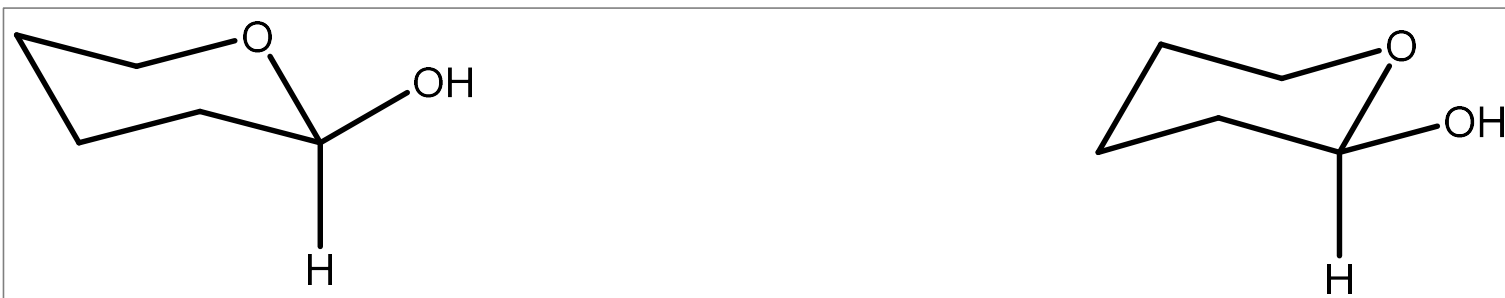


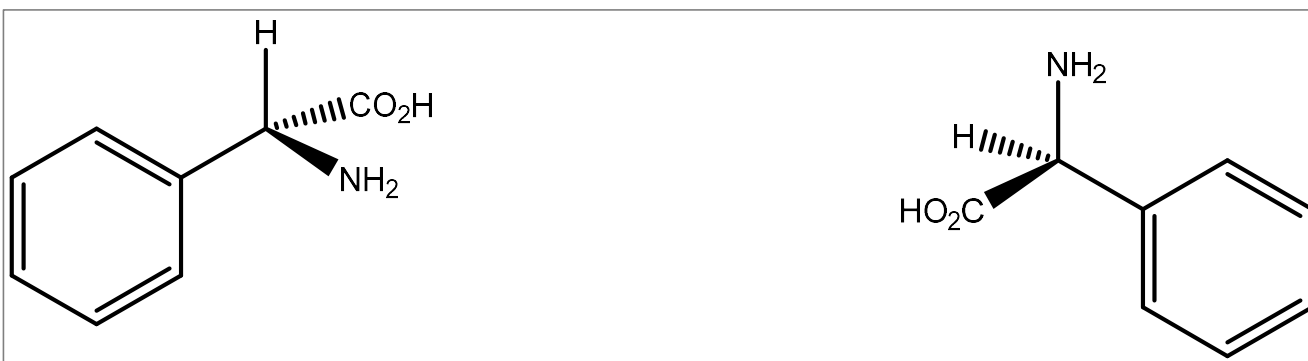
(R) and *(S)* Configuration: Step 1 **Assign Priority** (Multiple Bonds)

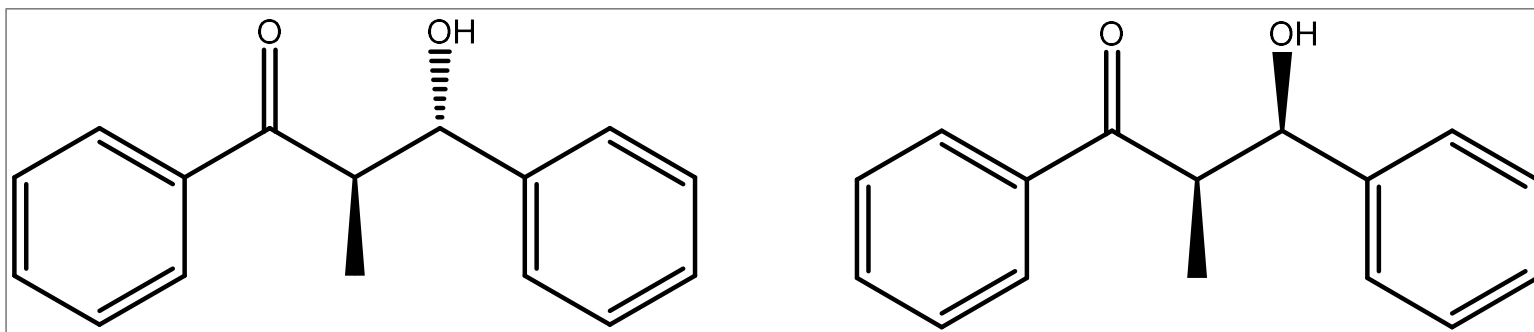


Configuration in Cyclic Compounds

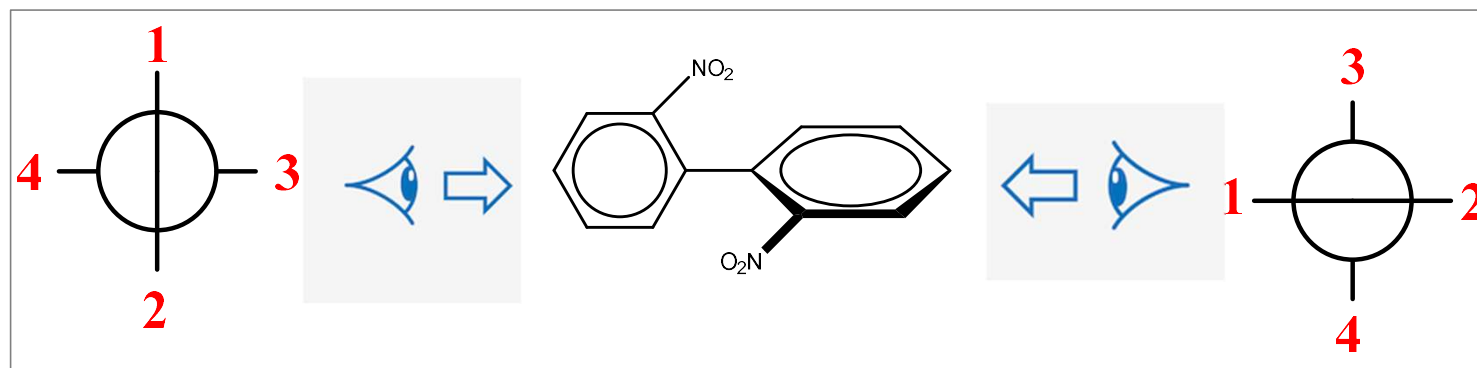
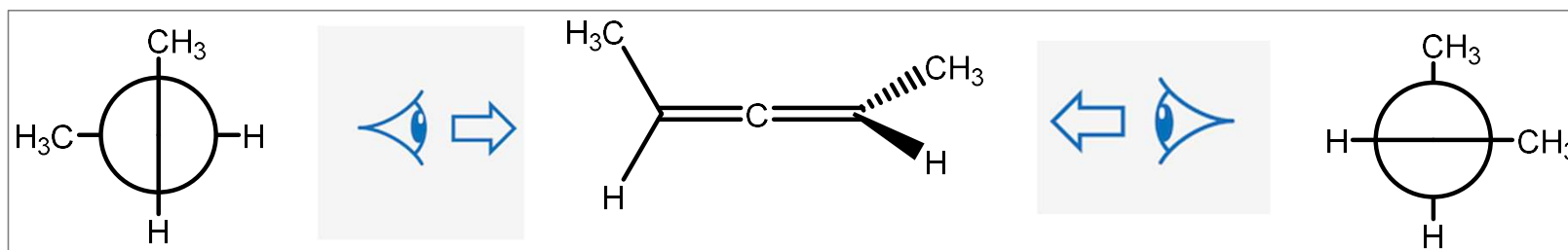








axial chirality: Nomenclature



Importance of chiral chemistry

- All proteins, enzymes, amino acids, carbohydrates, nucleosides and a number of alkaloids and hormones are chiral compounds.
- 56% of the drugs currently in use are chiral products and 88% of the last ones are marketed as racemates consisting of an equimolar mixture of two enantiomers.
- Cells are mostly composed of chiral molecules.

